HYPER STRONG



User Manual

10Ft Battery Energy Storage System

HSL3G01-03200EU/ HSL3A01-03008EU HSL3G0U-06400EU/ HSL3A0U-06017EU

HSL3G0U-09600EU/ HSL3A0U-09025EU HSL3G0U-12800EU/ HSL3A0U-12034EU

HSL3G0U-16001EU/ HSL3A0U-15042EU HSL3G0U-19201EU/ HSL3A0U-18051EU

Copyright © Beijing HyperStrong Technology Co., Ltd. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Beijing HyperStrong Technology Co., Ltd.

Trademarks and Permissions

HYPER STRONG and other HyperStrong trademarks are trademarks of Beijing HyperStrong Technology Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between HyperStrong and the customer. Portions of the products, services and features described in this document may not be within the purchase or the usage scope. Unless explicitly specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without any warranties, guarantees or representations, either express or implied.

This document is subject to change without notice. While every effort has been made in the preparation of this document to ensure accuracy of the contents, the statements, information, and recommendations herein do not constitute any form of warranty, whether express or implied.

Beijing HyperStrong Technology Co., Ltd.

Address: Building C-2, Building C
Courtyard 9, Fenghao East Road, Haidian District
Beijing, 100080
People's Republic of China

Website: http://www.hyperstrong.com.cn

PREFACE

Manual Application

This manual is prepared according to the specifications of the initial edition of the liquid-cooled container models:

HSL3G01-03200EU/ HSL3A01-03008EU

HSL3G0U-06400EU/ HSL3A0U-06017EU

HSL3G0U-09600EU/ HSL3A0U-09025EU

HSL3G0U-12800EU/ HSL3A0U-12034EU

HSL3G0U-16001EU/ HSL3A0U-15042EU

HSL3G0U-19201EU/ HSL3A0U-18051EU

The technical specifications of this product are subject to updates, and this manual may be revised without prior notice. This manual applies to the energy storage battery system, focusing on product operation and functionality during transportation, assembly, installation, and commissioning. Pictures, data, and instructions herein are for reference purposes, shall not be considered as the basis for order acceptance.

Target Audience

This manual is intended for operators (customers), qualified electrical engineers, operations and after-sales personnel. Please follow the instructions provided in this manual to operate the system. Failure to do so may cause equipment damage, personnel injury, property loss.

Retention

This manual contains crucial information. Please

1. Read it carefully before operating and maintaining the system.
2. Operate the system in accordance with the instructions to avoid the risk of equipment damage, injury to personnel, or property loss.
3. Ensure the manual is properly maintained and accessible by operators and maintenance personnel.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Revision History

Document Revision History

Version	Description of Revision	Author	Auditor	Approver	Date
V1.0	New Release	Members of Product Team	Guangxin Zhang Wenli Zeng Heng Yang	Zhenghua Yang	2025/12/25
V1.1	Updated figures and layout	Kai Zhang and Hao Li	Guangxin Zhang Wenli Zeng Heng Yang	Zhenghua Yang	2026/1/30
V1.2	Optimize some pictures and FSS contents	Shuang Li Xiaokang Sun	Guangxin Zhang Wenli Zeng Heng Yang	Zhenghua Yang	2026/2/11

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Content

1. SYMBOL CONVENTIONS 1

2. ACRONYMS AND ABBREVIATIONS 2

3. SAFETY INSTRUCTIONS 3

3.1 Safety Signs 3

3.2 General Safety Precautions 4

3.3 Personal Safety 7

3.4 Electrical Safety 7

3.5 Operational Safety 8

3.6 Electrostatic Protection 8

3.7 Lifting and Transportation Safety 9

3.8 Installation and Wiring Safety 10

3.9 Operation and Maintenance Safety 10

4. TRANSPORTATION AND STORAGE 12

4.1 Cautions 12

4.2 Type of Transportation 12

4.3 Transportation Requirements 13

4.4 Storage Requirements 14

5. ESS PRODUCT 15

5.1 Product Introduction 15

5.2 Structural Appearance and Internal Layout of ESS 19

5.3 Primary Topology of ESS 24

5.4 Power Supply Topology of ESS 25

 Copyright © Beijing HyperStrong Technology Co., Ltd.

5.5 Communication Topology of the ESS 27
5.6 Rack 28
5.7 Pack 30
5.8 Power Distribution Unit (PDU) 32
5.9 System Control and Distribution Cabinet (SCDC) 33
5.10 Fire Suppression System (FSS) 36
5.11 Thermal Management System (TMS) 43
6. EQUIPMENT INSTALLATION 54
6.1 ESS Lifting 54
6.2 Mechanical Installation 58
7. PRODUCT ELECTRICAL INSTALLATION 65
7.1 Setup Tools Preparation 65
7.2 Door Requirements of Boxes and Equipment Cabinets 66
7.3 Cable Harness Preparation 67
7.4 DC Cable Connections 68
7.5 Cable Access Port 68
7.6 Container Grounding 69
7.7 SCDC External Interfaces 70
7.8 Container External Interfaces 76
7.9 Inter-Container Cable Connection Instructions 78
7.10 UPS Cautions 83
7.11 Power on/off Process 84
8. HUMAN MACHINE INTERFACE (HMI) 87
8.1 Touchscreen Content and Structure 87

 Copyright © Beijing HyperStrong Technology Co., Ltd.

8.2 Touchscreen Program 87

8.3 Basic Environment Requirements for Touchscreen 87

8.4 Touchscreen Basic Functions 88

8.5 Communications 98

9. EMERGENCY OPERATION AND ACCIDENT HANDLING.....99

9.1 Emergency Handling 99

9.2 Emergency Operations 100

10. PRODUCT LABELS 102

11. CONTACT INFORMATION 104

 Copyright © Beijing HyperStrong Technology Co., Ltd.

1. Symbol Conventions

The symbols that may be found in this document are defined as follows:

Symbol	Description
DANGER	Indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
WARNING	Indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.
CAUTION	Indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.
NOTICE	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
NOTE	Supplements the important information in the main text. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

2. Acronyms and Abbreviations

Complete Designation	Abbreviations
System Control and Distribution Cabinet	SCDC
Battery Array Management Unit	BAMU
Battery Cluster Management Unit	BCMU
Battery Energy Storage System	BESS
Battery Pack Management Unit	BMU
DC/DC Converter	DC/DC
Energy Management System	EMS
Energy Storage System	ESS
Fire Suppression System	FSS
Human Machine Interface	HMI
Heating, Ventilating and Air Conditioning	HVAC
Local Controller or Control room	LC
Battery Pack	PACK
Power Conversion System	PCS
Power Distribution Unit	PDU
Battery Rack/Cluster	RACK
SCADA (Supervisory Control and Data Acquisition) System	SCADA
Switching Mode Power Supply	SMPS
State Of Charge	SOC
State Of Energy	SOE
State Of Health	SOH

 Copyright © Beijing HyperStrong Technology Co., Ltd.

3. Safety Instructions

3.1 Safety Signs

Safety signs on the energy storage system equipment contain essential information for safe operations. Manual removal or damage is strictly prohibited.

Table 3.1 shows common safety signs.

Signs	Explanation
	It indicates that the area contains fire-hazardous materials, and pyrotechnics are strictly prohibited in this location.
	It indicates that the equipment in the area shall not be disassembled.
	It indicates that the area contains fire-hazardous materials, and smoking is prohibited in this location.
	It indicates that the device poses a danger of entrapment of body parts and that inserting body parts into the machine is prohibited.
	It indicates that the area prohibits the stacking of items.
	It indicates that all personnel operating the equipment shall wear protective gloves to prevent electric shock or static electricity.
	It indicates that the equipment needs to be grounded. The equipment shall be grounded through the grounding point, ensuring regular and safe operation.
	It indicates an immediate hazardous situation that, if not avoided, will result in serious personal injury or death.
	It indicates a flammable substance that, if not avoided, could result in severe injury or death.
	It is used to convey safety warning information about equipment or the environment. It may lead to equipment damage, data loss, performance degradation, or other unpredictable results if not avoided.
	It indicates that cables have been placed in this area and you need to be aware of the cables underneath

 Copyright © Beijing HyperStrong Technology Co., Ltd.

	your feet when servicing or passing through the site to prevent tripping.
	It indicates a risk of electric shock from this equipment or device. Operators should take care while operating this equipment or device to avoid electrocution.
	It indicates that the machinery of the equipment may pose a risk of injury and that care should be taken when operating the equipment to prevent injury from the machinery.
	Do not dispose of this product as household waste.
	Read the instructions before performing any operation on the product.
	It is required to wear goggles.
	Ear protection shall be worn.
	In case of contact with the eyes, rinse the eyes immediately with running water or normal saline and seek medical help in time.
	Live parts! Please do not touch them until 5 minutes after disconnecting from the power sources.
	Emergency Exit: It indicates an emergency safety exit in the direction of the arrow of movement and that you can escape in the order given in the event of an emergency danger.

3.2 General Safety Precautions

The safety precautions outlined in this document provide essential guidance for selecting measuring and test equipment during the installation, operation, and maintenance of the equipment.

Statement

Before transporting, storing, installing, operating, using, or maintaining the equipment, carefully read this document, adhere strictly to the instructions provided, and follow all the safety instructions indicated on the equipment and within this document.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

For the purpose of this document:

"Equipment" refers to the products, software, components, spare parts, and/or services related to this document.

"The Company" refers to the manufacturer (producer), seller, and/or service provider of the equipment.

"You" refers to the entity that transports, stores, installs, operates, uses, and/or maintains the equipment. The **Danger**, **Warning**, **Caution**, and **Notice** statements included in this document are supplementary to, but do not encompass, all necessary safety precautions. Operators shall also comply with relevant international, national, and regional safety standards.

The Company shall not be liable for any consequences resulting from violations of safety requirements or safety standards concerning the design, production, and usage of the equipment.

The equipment shall be operated in an environment that meets the design specifications. Use in non-compliant environments may result in faulty, malfunctioning, or damaged, which are not covered under warranty. The Company shall not be liable for any property loss, personal injury, or even death caused thereby.

Ensure compliance with all applicable laws, regulations, standards, and specifications during transportation, storage, installation, operation, use, and maintenance.

Do not perform reverse engineering, decomplication, disassembly, adaptation, implantation, or other derivative operations on the equipment software.

Do not study the internal implementation logic of the equipment, obtain the source code of the equipment software, violate intellectual property rights, or disclose any of the performance test results of the equipment software.

During the storage phase of the energy storage system, the customer should implement safety measures to mitigate unforeseen circumstances.

The Company shall not be liable for any of the following circumstances or their consequences:

- Equipment damage caused force majeure, including but not limited to earthquakes, floods, volcanic eruptions, debris flows, lightning strikes, fires, wars, armed conflicts, typhoons, hurricanes, tornadoes, and other extreme weather conditions.
- Operation of equipment beyond the conditions specified in this document.
- Installation or usage in environments that do not comply with international, national, or regional standards.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- Failure to follow the operation instructions, operation and maintenance manual, or safety precautions on the product and in the document.
- Unauthorized removal, modification of the product, or alteration of software code.
- Equipment damage resulting from improper storage conditions that deviate from the requirements specified in the product document.
- Failure to prepare materials and tools that comply with local laws, regulations, and related standards.
- Negligence, gross negligence, intentional breach, improper operations, or other third-party actions causing equipment damage.
- Unauthorized change to the application of battery energy storage system.
- Battery stored beyond the maximum storage period.
- Failure to charge promptly, resulting in capacity loss or irreversible damage to the battery.
- Installation or operation by unqualified personnel.
- Mixing the batteries supplied by the Company with other batteries, including mixing with different brands or batteries with incompatible capacities, causing accelerated capacity degradation.

Safety Precautions

- During installation, overhaul, or maintenance, place clear warning signs or safety caution tape near the operating area. Prominent signs shall also be displayed at the front and rear switches of the energy storage system to prevent accidental closures.
- Technicians responsible for installation and maintenance shall be professionally trained, fully understand all safety precautions, be familiar with the composition and working principle, and master correct operation methods.
- Only qualified professionals are authorized to perform equipment removal and overhaul operations.
- Replacement and modification of equipment or components (including software) shall be carried out by personnel authorized by HyperStrong.
- Ensure the escape routes remain fully accessible during the maintenance and overhaul activities.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- After completing maintenance, overhaul, and other operations, remove all tools and restore the site to its original state for safe and regular equipment operation.
- During the storage, shall establish safety measures to prevent unforeseen incidents.

3.3 Personal Safety

DANGER	Please wear protection gear before operations to prevent electric shocks or burns.
DANGER	Equipment operates in high voltage. Any accidental contact may result in a fatal shock hazard. It is essential to have at least two personnel presents during maintenance or operation for safety.
DANGER	When the equipment is powered on, it is strictly prohibited to move, open, or dismantle the electric equipment container door or touch any internal components. Before performing maintenance operations, ensure the equipment has been safely powered off.
DANGER	Avoid contact cords or operating equipment during storms.

3.4 Electrical Safety

CAUTION	After performing AC and DC double-side disconnect operations on the energy storage system, wait at least 15 minutes before opening the front door to perform maintenance, overhaul, or other functions on the energy storage system.
NOTICE	When the energy storage system is electrically connected for a test run or other operations, use high-quality measuring equipment with a range and accuracy that meets the site requirements to ensure that the electrical parameters meet the requirements. Ensure that measuring equipment is correctly connected and used according to standards to prevent arcs and other hazards.
DANGER	Even if all external circuits are disconnected, high voltage may still exist between the positive and negative terminals of the energy storage battery pack. If measurements shall be conducted under charged conditions, use dedicated instruments,

 Copyright © Beijing HyperStrong Technology Co., Ltd.

	and ensure that only qualified professionals perform the tests.
WARNING	If the energy storage system operates abnormally or requires maintenance or an, shut down the equipment and confirm that the energy storage system completely stops running before proceeding with any maintenance or repair work.

Grounding

The following requirements apply to equipment that shall be grounded:

- During installation, it shall be grounded first.
- During removal, the ground wire shall be the last item disconnected.
- Before operations, check the electrical connections to ensure that the equipment is reliably grounded. Operate the equipment without a grounding conductor is strictly prohibited.
- The equipment shall remain permanently connected to a protective earth. Destruction of the grounding conductor is prohibited. (HyperStrong uses TN-S protective earthing system).

3.5 Operational Safety

NOTE	Read all equipment information and instructions carefully and completely before operation.
NOTICE	Before operation, please make sure that all cables are correctly connected. During operation, please pay attention to the normal operation of the equipment fan. Do not block the vent.
DANGER	Do not place small metal devices such as crank pins, screws, etc. next to connectors, slots, sockets, or circuits to prevent short circuits.
CAUTION	If the equipment gets damaged, please do not repair it yourself. Contact HyperStrong to assign professional personnel to repair. Do not modify the internal equipment of the battery system privately to avoid danger.

3.6 Electrostatic Protection

Accumulation of static electricity can result in electric shocks, fire, explosion, failure and damage of electronic devices, and other safety hazards. The energy storage system contains circuit boards and other static-sensitive components. To mitigate the risks associated with static electricity, operators shall implement static protection measures, minimize static electricity generation, accelerate the dissipation of static electricity, and neutralize static electricity.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

NOTICE	The prevention methods include but are not limited to: • During replacement, all equipment devices that have not yet been installed should be retained in bags with anti-static shielding, and temporarily removed equipment devices should be placed on foam pads with anti-static features. • Do not touch soldering points, pins, or exposed circuits.

EMC Related Cautions

The system's emission characteristics make it suitable for use in industrial areas, meaning Class A electromagnetic environment. If used in a residential environment, where Class B is typically required, this equipment may not provide adequate protection to radio-frequency communication services. User may need to implement mitigation measures, such as relocating or re-orienting the equipment.

Mains power quality should conform to the standards of a specialized industrial distribution network. For users requiring uninterrupted operation during or after power mains interruptions, it is recommended to connect the system's information storage components to an uninterruptible power supply (UPS) or a backup power source.

Some portable RF communications equipment such as high-power walkie-talkie may degrade the electromagnetic performance of the system. If the essential system performance is lost or impaired due to Electromagnetic disturbances, maintain a safe distance between high-power transmitters and the system, and contact customer service to address the issue.

3.7 Lifting and Transportation Safety

- When walking on top of the equipment, strictly follow the procedures for working at heights.
- Wear appropriate personal protective equipment (PPE) as required, including safety helmets, work clothing, safety shoes, and gloves.
- When using lifting machinery, adhere to all relevant safety operation guidelines. Do not engage in improper lifting practices.
- Protect the equipment during handling and storage. Prevent it from falling, tumbling, or being subjected to excessive pressure.
- Batteries transported shall be securely fastened to prevent severe vibration, impact, or compression.
- Drivers and transporters shall undergo appropriate training. For those handling hazardous materials, training should include

 Copyright © Beijing HyperStrong Technology Co., Ltd.

emergency response procedures for fire prevention, spill containment, and emergency contact protocols. They shall be familiar with the items being transported and understand the correct handling procedures.

3.8 Installation and Wiring Safety

- During the entire mechanical and electrical installation process, strictly adhere to the relevant standards and requirements of the project location.
- Electrical wiring for the system shall only be operational after inspection and verification. Ensure warning signs are prominently displayed during the process, such as "Personnel Working: Do Not Close the Gate" or "High-Voltage Danger.".
- Ensure that electrical wiring equipment perforations are sealed properly after installation. The sealing shall be tight, secure, uniform in thickness, and have a neat appearance.
- Operate carefully at the installation site to prevent electric shock battery short circuit.
- Wear appropriate protective equipment during installation, including helmets, work clothes, shoes, gloves.
- Pay attention to the positive and negative direction of the battery to avoid reverse installation which can lead to battery leakage, overheating, and other issues.
- Before wiring, disconnect the power supply to eliminate the risk of electric shock.
- When wiring, ensure obstacles are avoided to prevent damage to equipment or interference with operations

3.9 Operation and Maintenance Safety

- Schedule maintenance activities during favorable weather conditions. Avoid conducting maintenance in rain, snow, typhoons, or other adverse weather.
- Personnel shall wear appropriate Personal Protective Equipment (PPE) before beginning operations, including helmets, insulated gloves, long-sleeved coveralls, and insulated shoes.
- Keep an up-to-date emergency contact list on site and ensure communication devices are operational.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- During the operation and maintenance, at least two staff members should be present on site. Confirm the maintenance tasks in writing beforehand and strictly follow the outlined operational procedures.
- After completing maintenance or overhaul, securely store the tools and restore the site to its original condition to ensure the equipment can operate as intended.
- If the ambient temperature of the battery cabin is lower than 0°C, the air conditioner will automatically switch to heating mode to preheat the cabin. In case of abnormalities, contact HyperStrong professional team for assistance.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

4. Transportation and Storage

4.1 Cautions

⚠ CAUTION

- If the ESS fails to be transported and stored in accordance with the requirements specified in this manual, the warranty may be voided.

4.2 Type of Transportation

HyperBlock M is a modularized energy storage product, with a basic structure of one SCDC and several ESSs which are assembled on site. Each ESS is a 10-feet standard container with a weight of 24t. If an ESS is equipped with a control cabinet (SCDC), the weight will reach 25t. The figure presents two combined containers (one of them is equipped with a control cabinet), which is for reference only. The client is able to make flexible transportation plans in accordance with local laws and regulations.

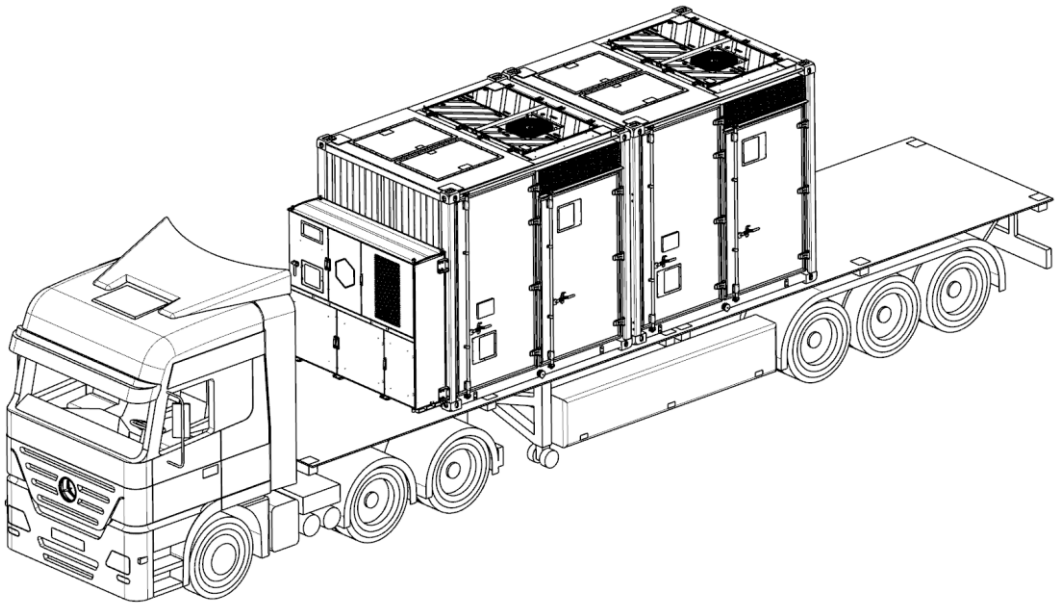


Figure 4-1 Schematic Diagram of Battery Container Transportation

 Copyright © Beijing HyperStrong Technology Co., Ltd.

4.3 Transportation Requirements

The equipment in the ESS has been installed and securely fixed inside the container before shipment, allowing the container to be lifted and transported as a single unit.

- The applicable national safety procedures and regulations shall be complied with during the operating processes of loading, unloading, and transportation for ESS containers in the country where the project is located.
- The ESS containers and all handling equipment used in the operating processes shall be properly maintained
- All personnel involved in loading, unloading and bolting shall be trained accordingly, especially safety training.
- The mechanical parameters such as external dimension, gross weight of each battery prefabrication pod shall be clarified and verified during the entire operating processes of loading, unloading, and transportation.

The following requirement shall be fulfilled prior to the transportation of ESS containers:

- All cabinet doors within the battery container shall be securely locked.
- A suitable crane or lifting device shall be selected based on site-specific conditions with adequate load capacity, boom length, and radius of rotation.
- When lifting the ESS container, it is recommended to use a single crane.
- Additional towing or traction equipment may be required when moving across inclined surfaces or slopes.
- The intended route shall be cleared of obstructions such as overhanging trees, power lines, or other impediments.
- The transportation and movement of ESS containers should be conducted under favourable weather conditions whenever practicable.
- Warning signs or cordons shall be set up and formed to restrict access to the lifting and transportation area and prevent accidents.

When transported by land, ropes shall be used to secure the corner fittings on the top of the ESS container to the vehicle and ensure that the ESS container remains

 Copyright © Beijing HyperStrong Technology Co., Ltd.

stable and the inclination angle is minimized during transportation.

4.4 Storage Requirements

- In order to prevent possible condensation or the bottom soaking from rainwater in the rainy season. The ESS should be stored on higher place.
- It is necessary to raise the container base to acclimatize site-specific conditions. The specific height should be reasonably determined based on the geological and meteorological conditions of the site.
- Store the ESS on dry, flat, and stable ground with sufficient load-bearing capacity and free of vegetation.
- Ensure the container doors and all internal equipment are securely locked.
- Storage environment temperature: -30~60℃.
- Long-term storage is not recommended because it may cause a decrease in battery capacity. Even when stored at the recommended storage temperature, irreversible capacity fading will still occur over time. The longer the system is stored, the greater of capacity will fade. Refer to the technical protocol for specific capacity fade rates.
- The relative humidity should be between 0~95%, without condensation.
- Protect the air inlet and outlet of the ESS to prevent rain water, sand, and dust from entering the container.
- Perform periodic inspections.
  - Check the container and the inner equipment for damage at least once per month.
  - For containers stored for more than six months, open the door and inspect the interior to ensure there is no visible condensation. Verify that the container and internal equipment are undamaged.
  - Inspect the system after powering it on and initiating startup procedures.
- If required, request professional assistance for testing before installation.
- For BESS systems stored for more than six months under the above conditions, a charge and discharge cycle shall be performed to bring the system SOC to 30%-40%. The storage period begins from the date of departure from the HyperStrong factory.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

5. ESS Product

5.1 Product Introduction

This product introduction is applicable for model HyperBlock M-DC-0.25P. According to the standard ESS of HSL3G01-03200EU/HSL3A01-03008EU and System Control and Distribution Cabinet (SCDC), the client can rapidly deploy energy storage system with a maximum capacity of 19.2MWh (Note: maximum 6 10-feet ESSs and 1 SCDC within 1 HyperBlock M unit), while flexibly adjusting capacity up or down as needed.

A battery cell rated 3.2V/601Ah(565Ah) is configured into a 1P104S pack. Four packs are combined to form a rack, configured as 1P416S. Each ESS can accommodate up to 4 sets of racks, 1 set of thermal management system (TMS). Meanwhile, each container is also equipped with a power distribution unit that is used for protection of auxiliary power supply loop and condition monitoring in the ESS.

HyperBlock M is composed of several ESSs and 1 set of System Control and Distribution Cabinet (SCDC) which consists of 1 set of fire suppression system (FSS), external DC power line copper conductor rail, external auxiliary power supply interface, and external communication interface. The SCDC is connected with each ESS by cables, which achieve the interaction of energy and information, which forms a set of complete energy storage unit.

The specifications of HSL3G01-03200EU/HSL3A01-03008EU is shown as follow.

Table 5-1 Product Parameters

Model	HSL3G01-03200EU	HSL3A01-03008EU
DC Side		
Rated Energy	3200.204 kWh	3008.512 kWh
Battery Configuration	416S4P	416S4P
Cell Type	3.2V/601Ah	3.2V/565Ah
Nominal Voltage Range	1123.2V - 1497.6V	1123.2V - 1497.6V
Rated Charge/Discharge Rate	0.25P (800kW)	0.25P (752kW)
Auxiliary Power Supply		
Max Auxiliary Power	3L/N/PE, 400/230V,50Hz,13.5kW L/N/PE,230V,50Hz,0.5kW (from UPS)	
System Parameter		
Dimensions(L*W*H)	2991mm (0, -6) *2438mm (0, -5) *2896mm (0, -5)	
Weight	24000kg	
Applicable Scope	Power grid side, new energy side, industry and commerce	
Degree of	IP55	

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Protection	
Operating Humidity	0 - 95%RH (No condensation)
Operating Temperature	-30℃ - 55℃ (>45℃ De-rating)
Storage Temperature	-30℃ - 60℃
Altitude	≤3000m
Communication Interfaces	Ethernet (Optical Fiber)
Communication Protocols	Modbus TCP
Heat Transfer	Liquid-cooling & Liquid-heating
Fire Suppression System	Temperature sensing, Smoke sensing, Combustible gas detection, Cabin level aerosols (Optional)
	Ventilation system, Water fire protection (Optional), Explosion vent panels (Optional)
Certifications	IEC 60730-1 Annex H, IEC 62933, IEC 62477, EN 61000-6, IEC 62619, UN38.3
Optional	Dehumidifier
Optional	AC/DC dual-path power supply

A single energy storage unit may consist of one SCDC and multiple ESSs. The number of ESS can be flexible configured based on the actual needs. Please refer to the table below for the technical parameters of the energy storage unit.

Table 5-2 System (EU) Product Parameters(H601)

Model	**HSL3G0U-06400EU**	**HSL3G0U-09600EU**	**HSL3G0U-12800EU**	**HSL3G0U-16001EU**	**HSL3G0U-19201EU**
DC Side					
Rated Energy	6400.409 kWh	9600.614 kWh	12800.82 kWh	16001.024 kWh	19201.228 kWh
Battery Configuration	416S4P*2	416S4P*3	416S4P*4	416S4P*5	416S4P*6
Cell Type	3.2V/601Ah				
Nominal Voltage Range	1123.2V - 1497.6V				
Rated Charge/ Discharge Rate	0.25P				
Rated Charge/ Discharge Power	1600kW	2400kW	3200kW	4000kW	4800kW
Number of Parallel Branches	4/2 (Optional)	6/3 (Optional)	8/4 (Optional)	10/5 (Optional)	12/6 (Optional)
Auxiliary Power Supply					
Auxiliary Power	3L/N/PE, 400/230V,50Hz,**30/43/57/70/84**kVA Optional: L/N/PE,230V,50Hz,2.5kVA				

 Copyright © Beijing HyperStrong Technology Co., Ltd.

System Parameter					
Dimensions(L*W*H)	**Shoulder to shoulder** 06400EU/09600EU/12800EU:6532/9573/12614mm (0, -6) *2438mm (0, -5) *2896mm (0, -5) **Shoulder to shoulder & back-to-back** 12800EU: 6532mm (0, -6) * 5026mm (0, -5) * 2896mm (0, -5) 16001EU/19201EU: 9573mm (0, -6) * 5026mm (0, -5) * 2896mm (0, -5)				
Weight	50000kg	74000kg	98000kg	122000kg	146000kg
Applicable Scope	Power grid side, new energy side, industry and commerce				
Degree of protection	IP55				
Operating Humidity	0 - 95%RH (No condensation)				
Operating Temperature	-30℃ - 55℃ (>45℃ De-rating)				
Storage Temperature	-30℃ - 60℃				
Altitude	≤3000m				
Communication Interfaces	Ethernet (Optical Fiber)				
Communication Protocols	Modbus TCP				
Heat Transfer	Liquid-cooling & Liquid-heating				
Fire Suppression System	Temperature sensing, Smoke sensing, Combustible gas detection, Cabin level aerosols (Optional)				
	Ventilation system, Water fire protection (Optional), Explosion vent panels (Optional)				
Certifications	IEC 60730-1 Annex H, IEC 62933, IEC 62477, EN 61000-6, IEC 62619, UN38.3				
Optional	Dehumidifier				
Optional	AC/DC dual-path power supply				

Table 5-3 System (EU) Product Parameters(C565)

Model	**HSL3A0U-06017EU**	**HSL3A0U-09025EU**	**HSL3A0U-12034EU**	**HSL3A0U-15042EU**	**HSL3A0U-18051EU**
DC Side					
Rated Energy	6017.024 kWh	9025.536 kWh	12034.048 kWh	15042.56 kWh	18051.072 kWh
Battery Configuration	416S4P*2	416S4P*3	416S4P*4	416S4P*5	416S4P*6
Cell Type	3.2V/565Ah				
Nominal Voltage range	1123.2V - 1497.6V				
Rated Charge/ Discharge Rate	0.25P				
Rated Charge/	1504kW	2256kW	3008kW	3760kW	4512kW

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Discharge Power					
Number of Parallel Branches	4/2 (Optional)	6/3 (Optional)	8/4 (Optional)	10/5 (Optional)	12/6 (Optional)
Auxiliary Power Supply					
Auxiliary Power	3L/N/PE, 400/230V,50Hz,**30/43/57/70/84**kVA Optional: L/N/PE,230V,50Hz,2.5kVA				
System Parameter					
Dimensions (L*W*H)	**Shoulder to shoulder** 06017EU/09025EU/12034EU:6532/9573/12614mm (0, -6) *2438mm (0, -5) *2896mm (0, -5) **Shoulder to shoulder & back-to-back** 12034EU: 6532mm (0, -6) *5026mm (0, -5) *2896mm (0, -5) 15042EU/18051EU: 9573mm (0, -6) *5026mm (0, -5) *2896mm (0, -5)				
Weight	50000kg	74000kg	98000kg	122000kg	146000kg
Applicable Scope	Power grid side, new energy side, industry and commerce				
Degree of Protection	IP55				
Operating Humidity	0 - 95%RH (No condensation)				
Operating Temperature	-30℃ - 55℃ (>45℃ De-rating)				
Storage Temperature	-30℃ - 60℃				
Altitude	≤3000m				
Communication Interfaces	Ethernet (Optical Fiber)				
Communication Protocols	Modbus TCP				
Heat Transfer	Liquid-cooling & Liquid-heating				
Fire Suppression System	Temperature sensing, Smoke sensing, Combustible gas detection, Cabin level aerosols (Optional)				
	Ventilation system, Water fire protection (Optional), Explosion vent panels (Optional)				
Certifications	IEC 60730-1 Annex H, IEC 62933, IEC 62477, EN 61000-6, IEC 62619, UN38.3				
Optional	Dehumidifier				
Optional	AC/DC dual-path power supply				

 Copyright © Beijing HyperStrong Technology Co., Ltd.



Figure 5-1 Description of Product Appearance

Note: The figure is for reference only, different configurations can be deployed in actual conditions, which is based on the project protocol.

Salt Spray Resistance and Seismic Description

The total thickness of the BESS container paint film shall meet the requirements of C5 level, and the overall neutral salt spray resistance time shall reach 1440h and above after three spraying and three drying processes in accordance with ISO12944-6 standards.

The seismic design code for BESS Container complies with the requirements of ASCE7-22 CARLIFRONIA MERCED COUNTY.

In order to address the diverse seismic design requirements worldwide, the most provision of design codes listed in the following table shall be applied. However, the precise structural and foundation designs will depend on the specific project conditions, including soil condition, rock types, ground motion parameters and relevant local laws and regulations.

Regions	Design Codes
US/Canada/Mexico	IEEE Std 693-2018
Europe	IEC 60068-3-3-2019
China Domestic	GB 50260-2013
Japan	JEAG 5003-2019
Australia/New Zealand	SAA AS 1170.4/SNZ NZS 1170.5
Chile (Reference for South America)	Nch2369-2003

5.2 Structural Appearance and Internal Layout of ESS

HyperBlock M is a modularized energy storage product, with a basic structure of one SCDC and several ESSs which are assembled on site. Each ESS is a standard 10ft container. The appearance and dimension figures of the container are illustrated as follow:

 Copyright © Beijing HyperStrong Technology Co., Ltd.

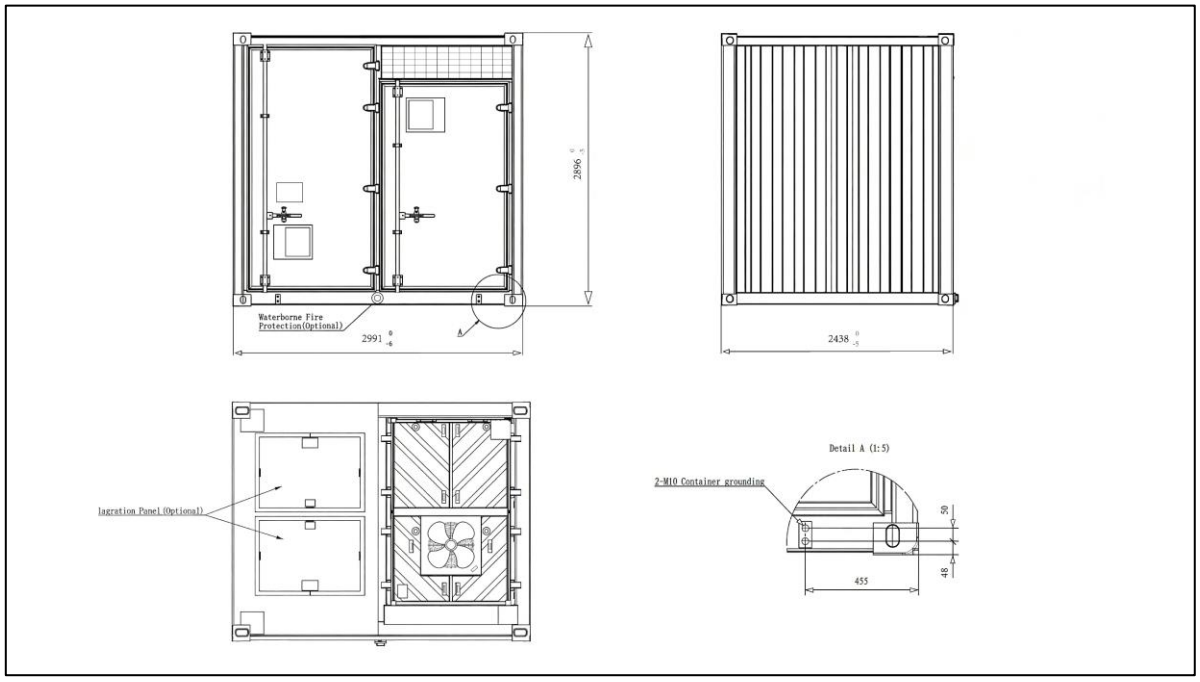


Figure 5-2 1 Appearance and Dimension of a Container

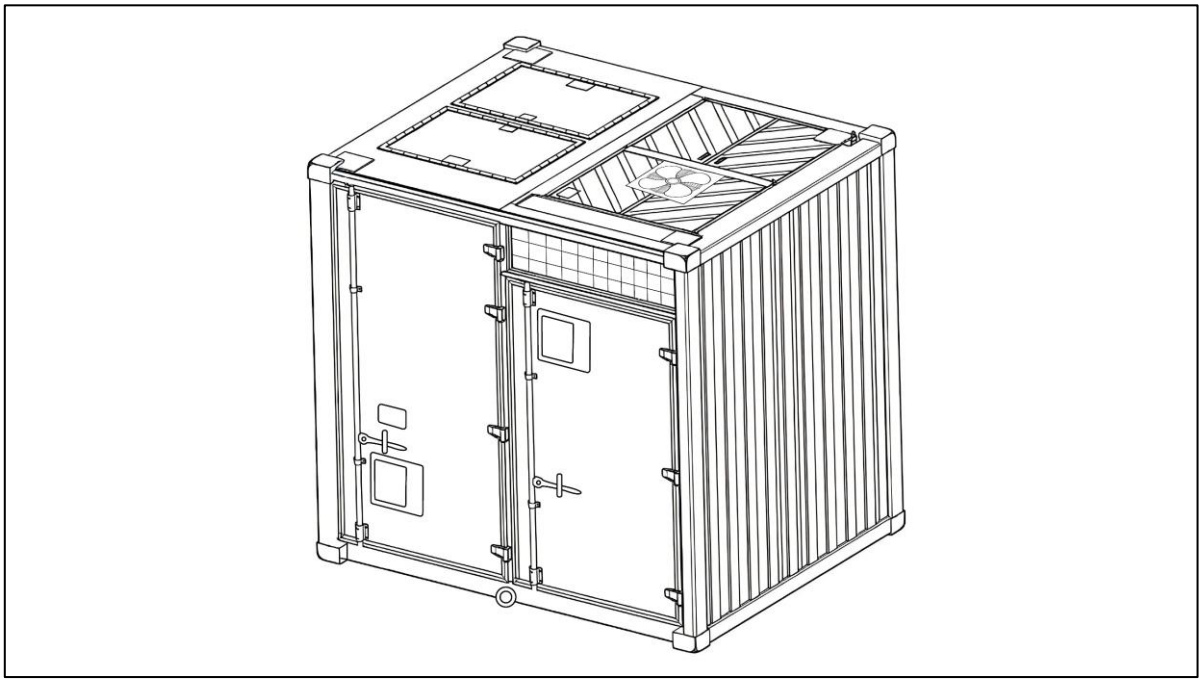


Figure 5-2 2 Overall Appearance

 Copyright © Beijing HyperStrong Technology Co., Ltd.

The SCDC is installed on the side wall of the ESS. The appearance and dimension figures of the container with a control cabinet are illustrated as follow:

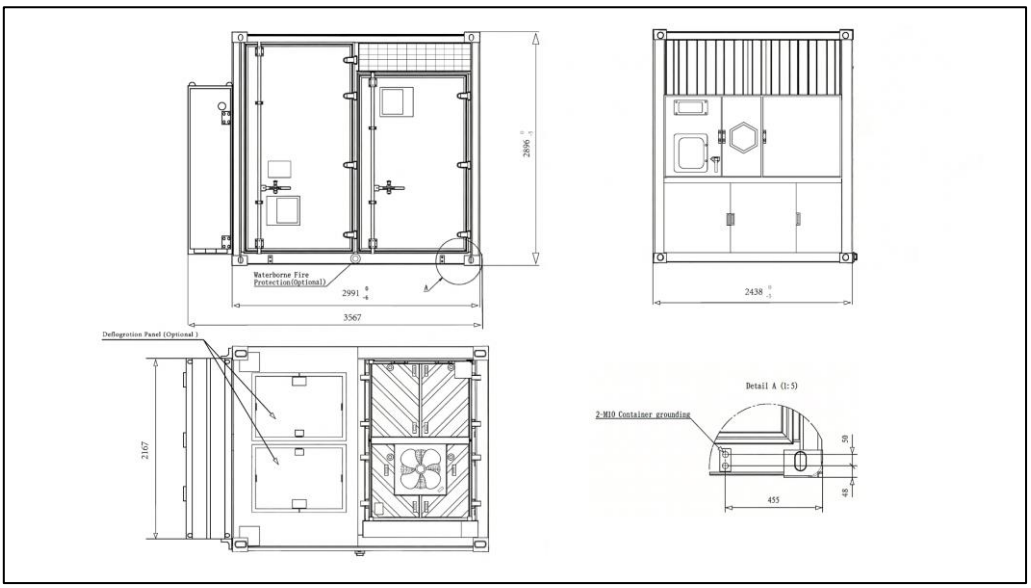


Figure 5-3 1 Appearance and Dimension of a SCDC

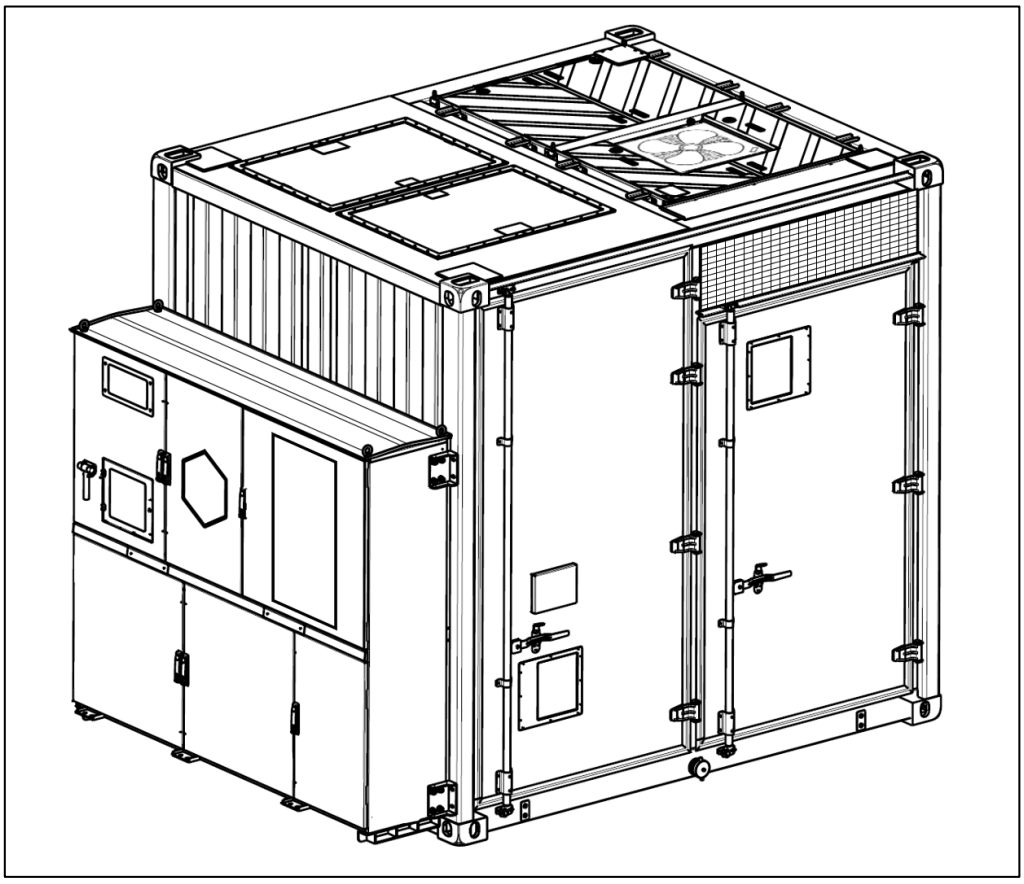


Figure 5-3 2 Location of the SCDC

 Copyright © Beijing HyperStrong Technology Co., Ltd.

⚠ CAUTION

- The nameplate contains important parameter information related to the ESS, which should be protected during transportation, installation, maintenance, overhaul and other operations. Destruction or removal is strictly prohibited.

⚠ CAUTION

- Emergency Stop Button: In the event of an emergency, press this button to disconnect the electrical connections inside the ESS container.

Internal Design of Energy Storage Systems：

The internal layout of the ESS is illustrated as follow.

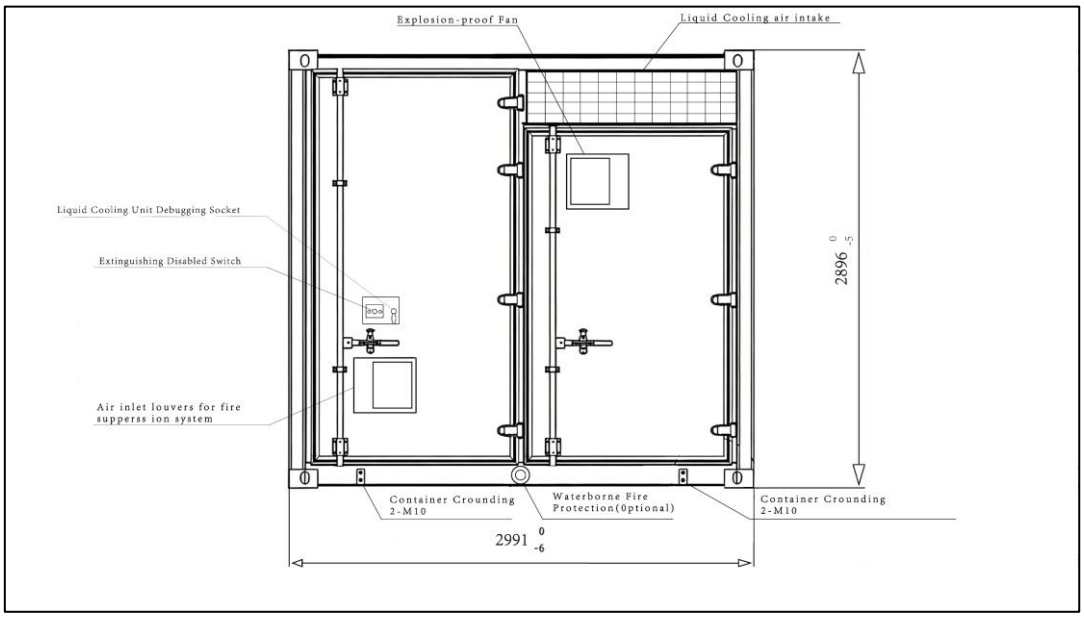


Figure 5-4 Internal Layout 1

 Copyright © Beijing HyperStrong Technology Co., Ltd.

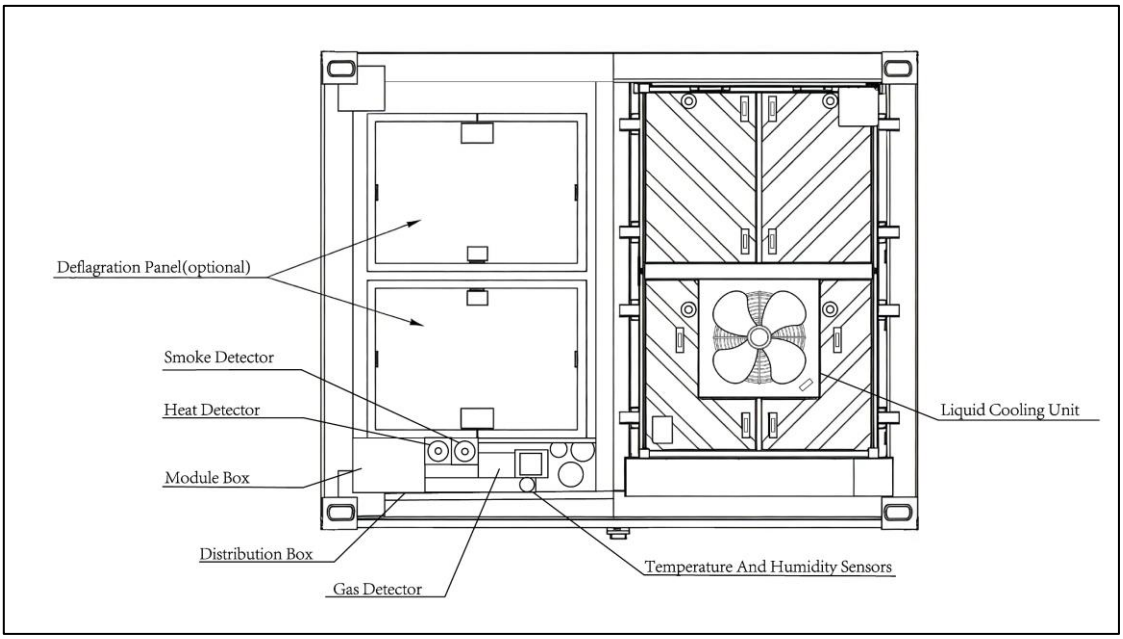


Figure 5-4 Internal Layout 2

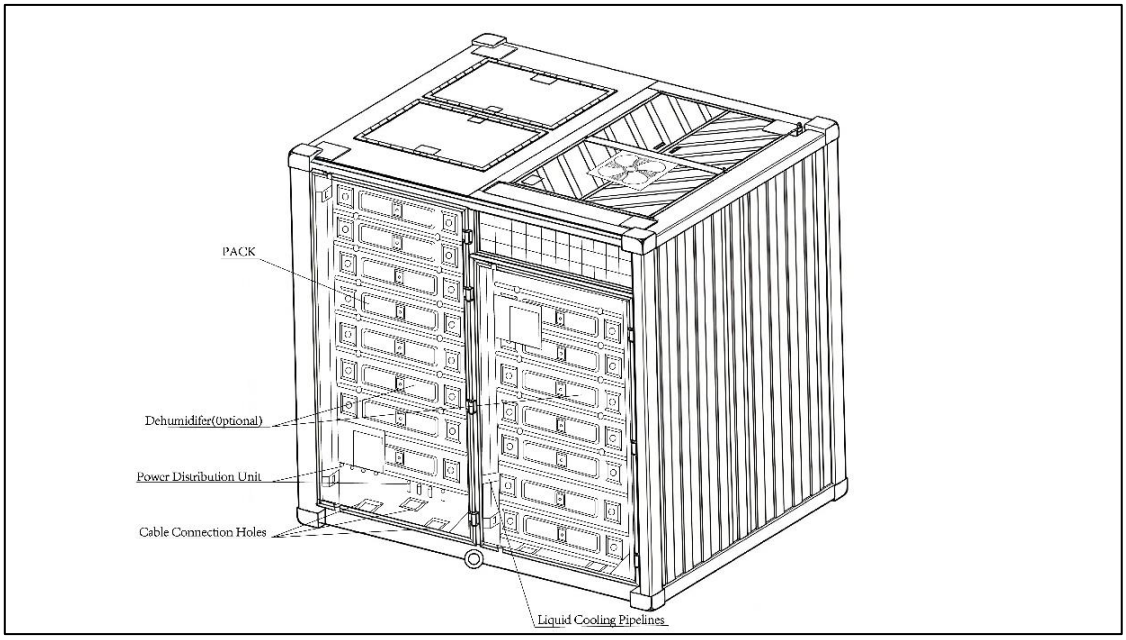


Figure 5-4 Internal Layout 3

*The above figures are for reference only, please refer to the actual product for details.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

5.3 Primary Topology of ESS

The primary circuit topology diagram of the energy storage unit is as follows. It consists of multiple ESSs and one SCDC. Each ESS internally contains four battery racks. Every two battery racks are connected to one PDU, which includes components such as the main circuit contactor, fuse, pre-charge loop, semi-fused switch, and BCMU. In the PDU, the two battery racks are combined into one branch circuit for external output. A single ESS contains two PDUs, which provide a total of two branch circuits for external output. The rated power of each branch circuit is 400 kW.

The output branch circuits from each container are connected by cables to the SCDC. Within the SCDC, various consolidation schemes can be selected based on site-specific requirements, including: isolating switches, fuses with isolating switches, circuit breakers, or direct copper busbar output. Additionally, the SCDC is equipped with SPDs on each branch circuit.

The DC power cables that are output from the SCDC are connected to the DC side of the PCS.

A schematic diagram of the product is shown in the figures below.

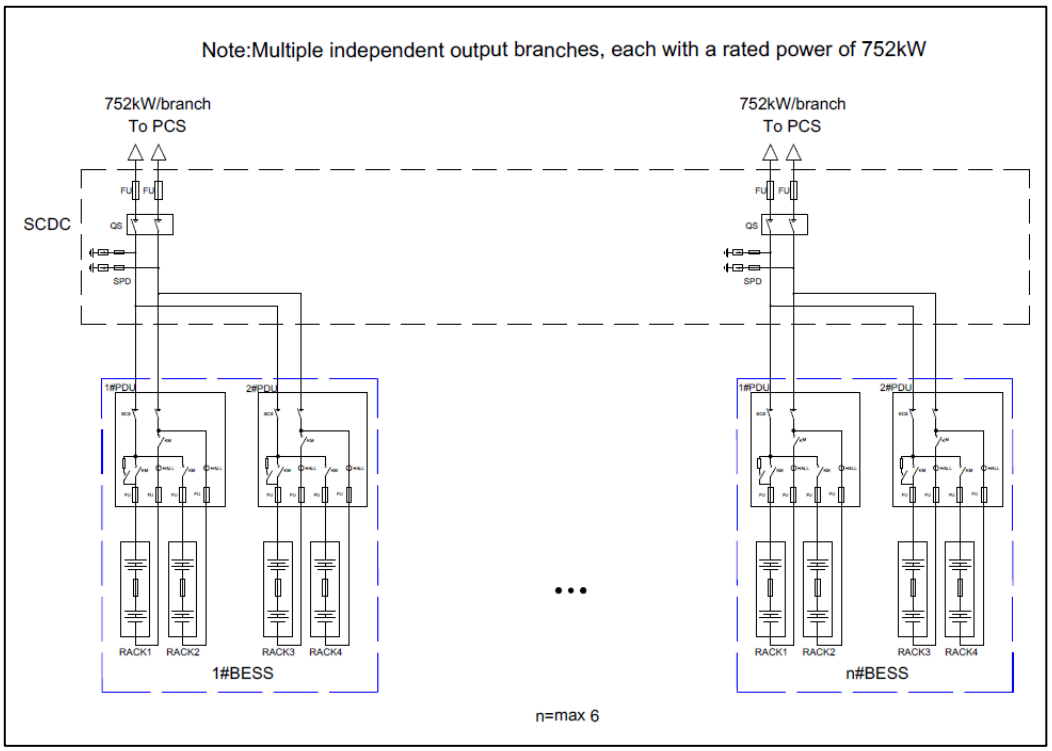


Figure 5-5 Schematic Diagram of Product-12 Racks

 Copyright © Beijing HyperStrong Technology Co., Ltd.

5.4 Power Supply Topology of ESS

The auxiliary power supply topology diagrams for the energy storage unit are shown in Figure (a) and (b) below. As illustrated in Figure (a), it is the auxiliary power supply topology of the SCDC, the external 400VAC/50Hz power supply from the site and station is connected to the lower port of QF1 within the SCDC. In the SCDC, the power distribution branch circuits are as follows:

- QF2-QF7: Provide 3L+N power to the ESS units, primarily provide power for the liquid cooling units.
- The front end of QF22 is UPS AC output side: This branch circuit provides backup power for the ESS units.
- QF14: This is a reserved interface for an external backup power source. If a 230Vac backup power supply is available at the site and station, it should be connected here.
- The remaining branch circuits are internal distribution circuits within the SCDC, mainly providing power to components such as the UPS, air conditioning unit, fire suppression system, fans, etc.

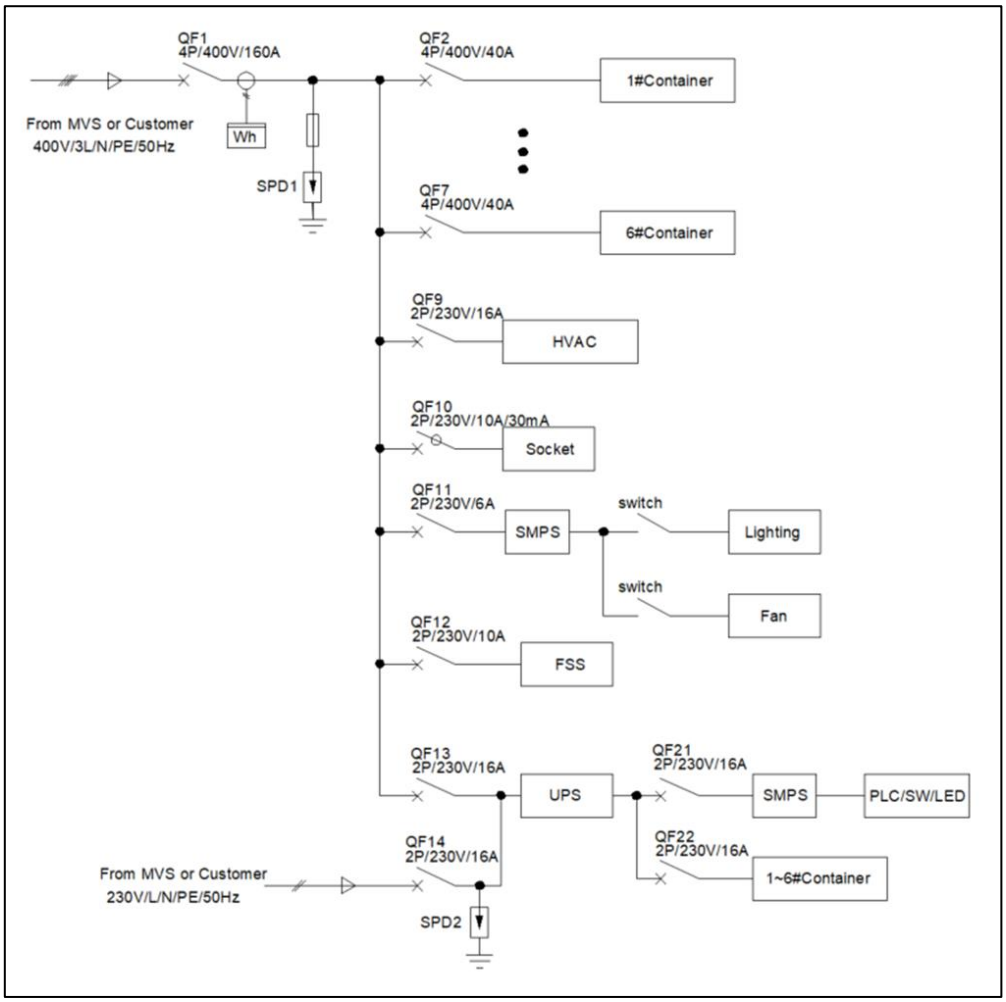
Figure (b) is the power distribution topology within each ESS unit, with the following branch circuits:

- QF1 (Upper port): Receives the 3L+N power supply from the SCDC, mainly used to provide power to the liquid cooling unit.
- QF2: Primarily provides power (L+N) to the dehumidifier.
- QF3 (Upper port): Receives power from the backup supply branch circuits from the SCDC. This is mainly used to provide power to the 24VDC switching power supply within the ESS. The 24VDC mainly provided power to critical components such as the control system and fire suppression fans in the ESS.
- QF4: the circuit breaker for the fire suppression branch circuit.

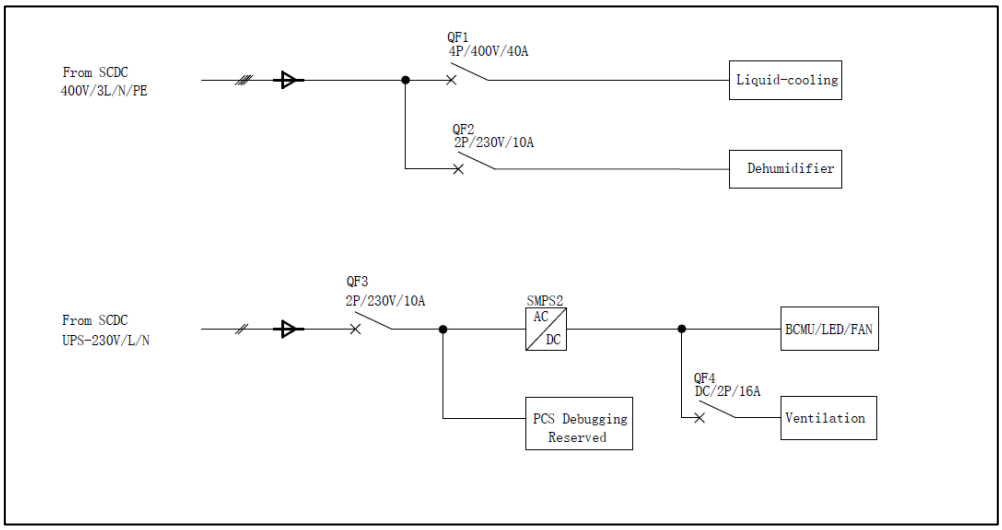
The SCDC and each ESS units are interconnected by cables for establishing the power distribution network. Notably. The power supply of each liquid cooling unit in each ESS is independent, which has individual connections from the SCDC respective ESS. The backup power from the UPS is connected in a configuration of daisy-chain (hand-in-hand) due to the lower capacity. The supply that originates from the SCDC is connected to ESS #1, then from ESS #1 to ESS #2, and so forth.

The power supply circuit is shown in the following figure.

 Copyright © Beijing HyperStrong Technology Co., Ltd.



(a)



(b)

Figure 5-6 Power Supply Topology Diagram.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

5.5 Communication Topology of the ESS

The communication topology diagram of the energy storage unit is illustrated as below:

- Each ESS contains two BC mainboards, each of them manages two battery racks respectively.
- Among them, BC mainboard #2 communicates with the liquid cooling unit, dehumidifier (optional), temperature and humidity sensors, and combustible gas detectors by Modbus RTU protocol to achieve information exchange.
- Similarly, BC #2 reads information from the I/O Module within the ESS via CAN communication. The primary function of this module is to monitor statuses within the ESS, such as door access status, waterlogging sensor status, and fire suppression fan/louver status.
- BC #1 and BC #2 are connected by an Ethernet cable. BC #2 is then connected to the exchanger in the SCDC by another Ethernet cable. Consequently, the external data interface for each ESS is a single Modbus TCP/IP network interface.
- Within the SCDC, the BA is the core component of the entire system. Its main functions are as follows:
  - It exchanges data with each ESS via Modbus TCP/IP.
  - It serves as the data interface for the entire system to external entities (EMS, cloud platform, external PCS, external devices, etc.).
  - It is the control core of the entire SCDC. It reads information from components such as the air conditioning unit, temperature and humidity sensors, and UPS via communication, and monitors real-time changes in status signals within the SCDC (door access, circuit breaker status, etc.).

The secondary topology of this product consists of BMU, BCMU, and BAMU. The communication modes between these components are:

- BMU and BCMU: Daisy chain.
- BCMU and BAMU: Ethernet.

The communication topology diagram is shown in Figure 5-7: System Communication Topology Diagram.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

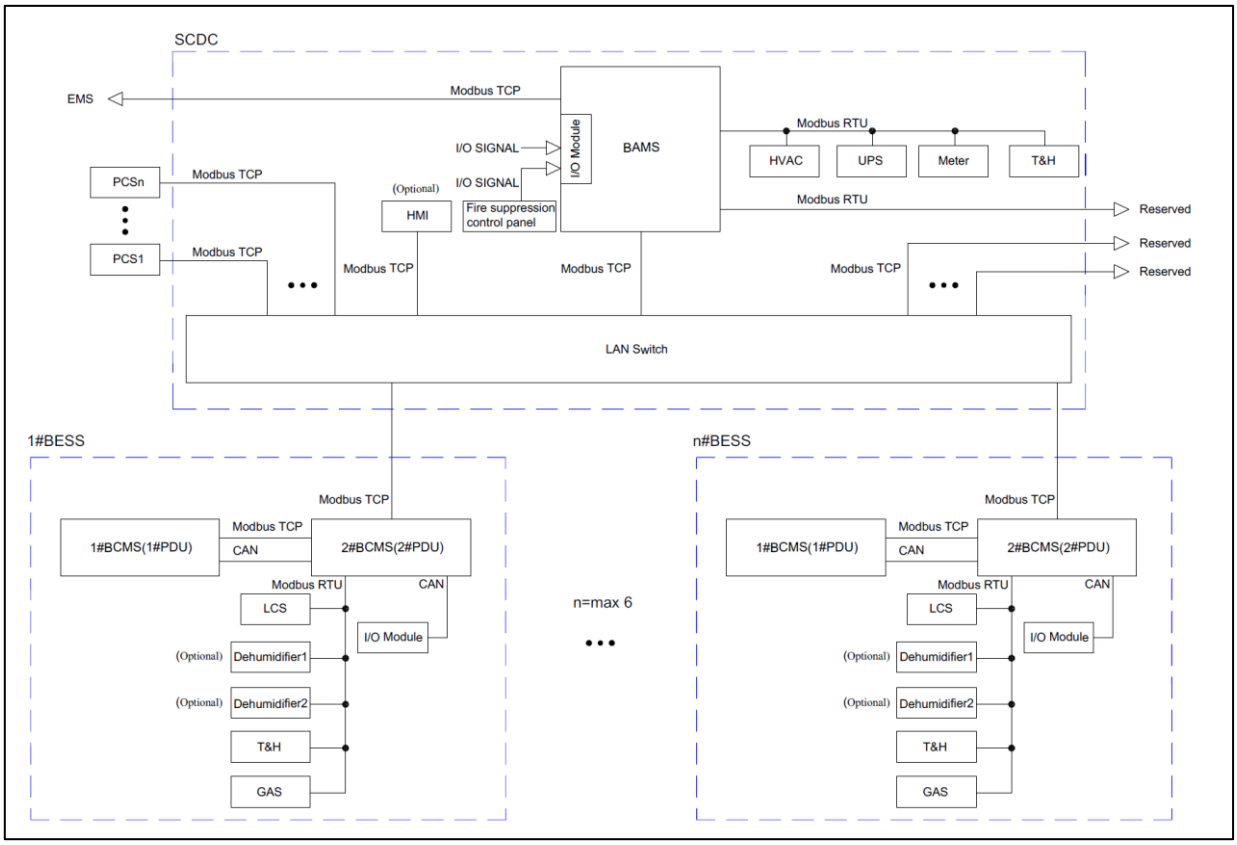


Figure 5-7 System Communication Topology Diagram -12 Racks (For Reference Only)

5.6 Rack

As illustrated below, each ESS contains four battery racks. Each battery rack consists of four battery packs. There are two power distribution units (PDU) in the ESS. Each PDU is connected to the power and communication harnesses of two battery racks, and manages the two racks.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

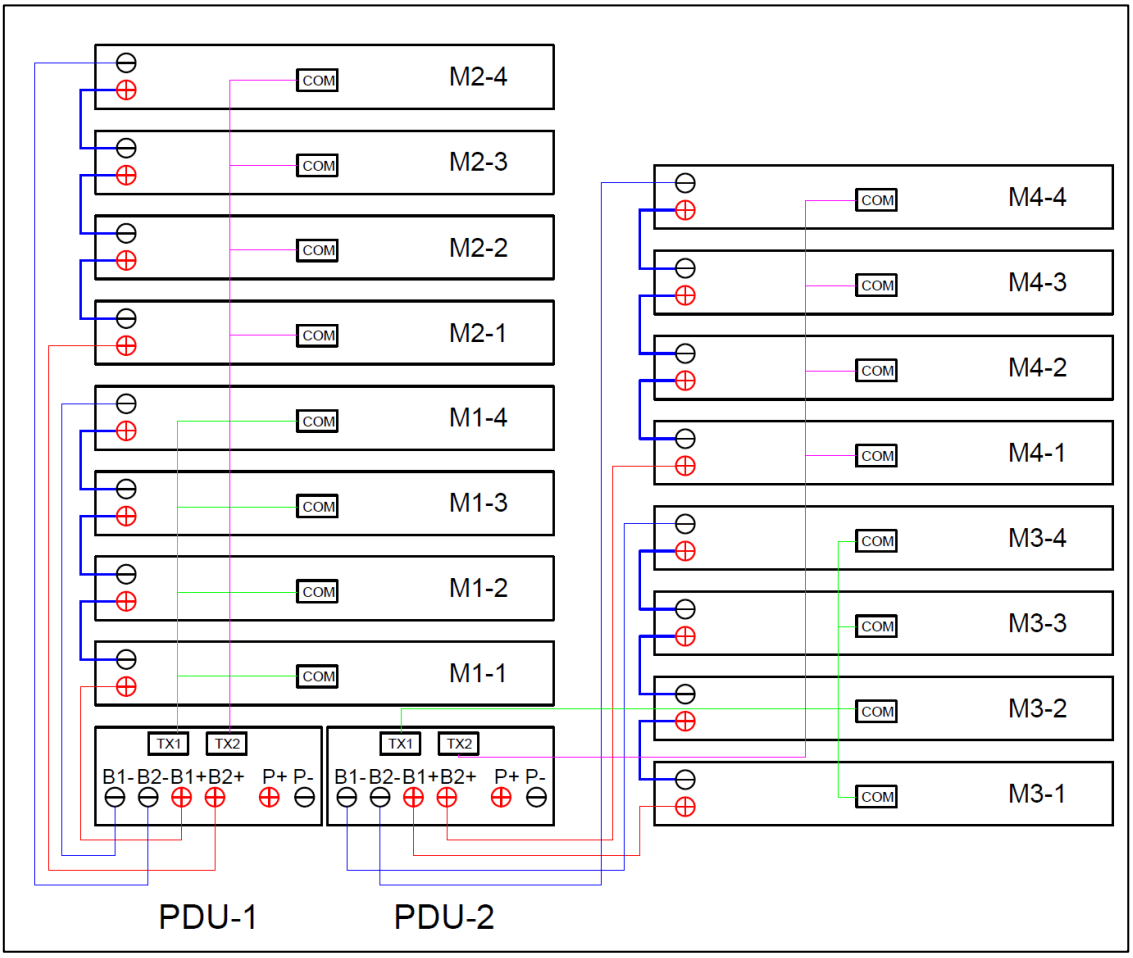


Figure 5-8 Battery Rack Wiring Diagram (For Reference Only)

Table 5-4 Product Parameters

No.	Specifications	Parameters
1	Type/Model	/
2	Cell Type	LFP565Ah
3	Cell Quantity	416
4	Nominal Voltage	1331.2Vd.c.
5	Rated Capacity	565Ah
6	Total Storage Energy	752.128kWh
7	Rated/Max.Charge/Discharge Power	188.032kW
8	Connecting Type	(104S)4S
9	Upper Limit Charging Voltage	1497.6Vd.c.
10	Final Discharge Voltage	1123.2Vd.c.
11	Rack Maximum Short-circuits Current	11kA(@2ms)
12	Nominal Current	150A
13	Recommend Current	150A
14	Operating Temp.	-30℃ to 55℃
15	Storage Temperature [℃]	-20℃-45℃ (one month)

Copyright © Beijing HyperStrong Technology Co., Ltd.

		0℃-35℃ (six months)
16	Dimension	H1358±2, W1274±2, L2236±3 (mm)
17	Weight	≤4900kg
18	Discharge Cut-Off Voltage [V]	1123.2
19	Enclosure Protection (Ip)	IP65 for PDU, IP67 for Battery packs
20	Heat Transfer	Liquid-cooling & Liquid-heating
21	Liquid Coolant Type	Ethylene glycol solution
22	Altitude	≤3000m

No.	Specifications	Parameters
1	Type/Model	/
2	Cell Type	LFP601Ah
3	Cell Quantity	416
4	Nominal Voltage	1331.2Vd.c.
5	Rated Capacity	601Ah
6	Total Storage Energy	800.051kWh
7	Rated/Max.Charge/Discharge Power	200.012kW
8	Connecting Type	(104S)4S
9	Upper Limit Charging Voltage	1497.6Vd.c.
10	Final Discharge Voltage	1123.2Vd.c.
11	Rack Maximum Short-circuits Current	11kA(@1ms)
12	Nominal Current	150A
13	Recommend Current	150A
14	Operating Temp.	-30℃ to 55℃
15	Storage Temperature [℃]	-20℃-45℃ (one month) 0℃-35℃ (six months)
16	Dimension	H1358±2, W1274±2, L2236±3 (mm)
17	Weight	≤4900kg
18	Discharge Cut-off Voltage [V]	1123.2
19	Enclosure Protection (Ip)	IP65 for PDU, IP67 for Battery packs
20	Heat Transfer	Liquid-cooling & Liquid-heating
21	Liquid Coolant Type	Ethylene glycol solution
22	Altitude	≤3000m

5.7 Pack

The pack is designed as a battery box with a well-organized structural layout, attractive appearance, and excellent heat dissipation performance. Each 1P104S battery pack consists of four 1P26S battery modules, collection harness, BMU,

 Copyright © Beijing HyperStrong Technology Co., Ltd.

related electrical components and structural components, etc.

The battery pack is shown in Figure 5-9.

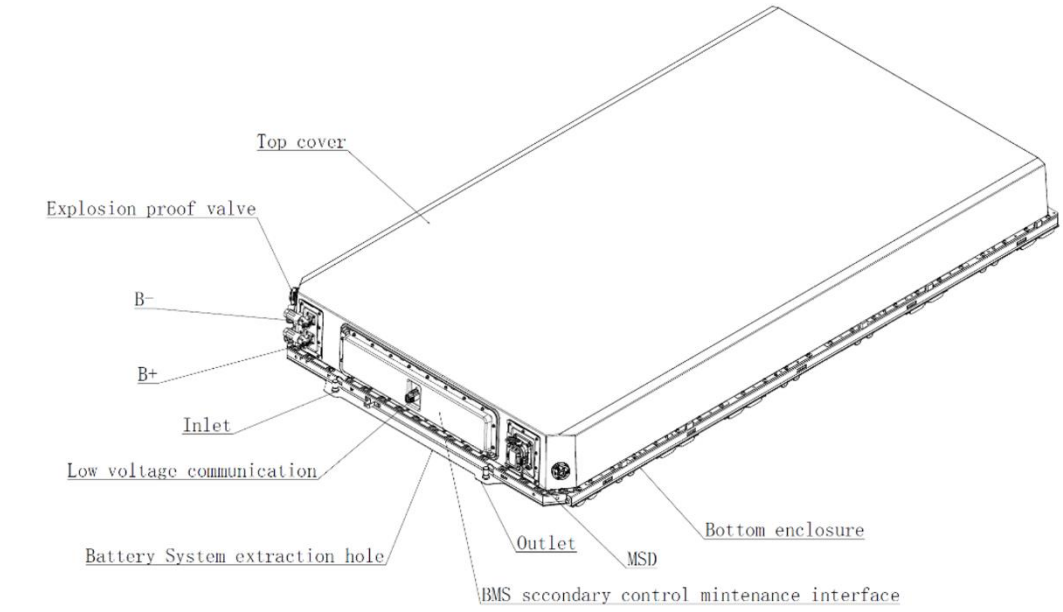


Figure 5-9 Battery Pack Diagram

Table 5-5 Technical specifications of battery module(C565)

No.	Specifications	Parameters	Remarks
1	Cell Type	LFP-565Ah	
2	Configuration	104S1P	
3	Rated Capacity	565Ah	
4	Rated Energy	188.032kWh for 565Ah cells	0.25P/0.25P, 25℃±2℃, BOL
5	Rated Voltage	332.8Vdc	
6	Voltage Range	260~379.6Vdc (>0℃)	
7	Recommended Voltage Range	280.8~374.4Vdc	Cell: 2.7~3.6Vdc
8	Charge and Discharge Rate	≤0.25P	
9	Operating Temperature	Charge: 0~60℃ Discharge: -30~60℃	Cell level, suggest operation with thermal management
10	Storage Temperature	-20-45℃ (one month) 0℃-35℃ (six months)	
11	IP Rating	IP67	
12	Dimensions	2227±15mm(D)*1250±10mm(W)*259±5mm(H)	See Figure 2
13	Weight	1200kg	±1.5%
14	Cooling/Heating	Liquid cooling & liquid	

 Copyright © Beijing HyperStrong Technology Co., Ltd.

	Method	heating	
15	Voltage Acquisition	104 ports	
16	Temperature Acquisition	104 ports	26 points for each submodule.
17	SOC At Shipment	≤30%SOC	

No.	Specifications	Parameters	Remarks
1	Cell Type	LFP-601Ah	
2	Configuration	104S1P	
3	Rated Capacity	601Ah	
4	Rated Energy	200.012kWh for 601Ah cells	0.25P/0.25P, 25℃±2℃, BOL
5	Rated Voltage	332.8Vdc	
6	Voltage Range	260~379.6Vdc (>0℃)	
7	Recommended Voltage Range	280.8~374.4Vdc	Cell: 2.7~3.6Vdc
8	Charge and Discharge Rate	≤0.25P	
9	Operating Temperature	Charge: 0~60℃ Discharge: -30~60℃	Cell level, suggest operation with thermal management
10	Storage Temperature	-20℃-45℃ (one month) 0℃-35℃ (six months)	
11	IP Rating	IP67	
12	Dimensions	2227±15mm(D)*1250±10mm(W)*259±5mm(H)	
13	Weight	1200kg	±1.5%
14	Cooling/Heating Method	Liquid cooling & liquid heating	
15	Voltage Acquisition	104 ports	
16	Temperature Acquisition	104 ports	26 points for each submodule.
17	SOC at Shipment	≤30%SOC	

5.8 Power Distribution Unit (PDU)

The Power Distribution Unit (PDU) is a control unit for distributing battery energy, containing BCMU, positive and negative contactors, fan cooling circuits, and high and low voltage connectors. All contactors are capable of receiving control commands from the BCMU, which can collect individual cell voltages and temperatures reported by BMUs within each battery pack in the rack. After data collection and analysis, the BCMU reports the data to the BAMU. The PDU connects the high and low voltage

 Copyright © Beijing HyperStrong Technology Co., Ltd.

circuits in series with each battery pack, serving as the input and output interface for the power and control circuits of the battery rack.

The appearance and dimensions of PDU is shown in the figures below.

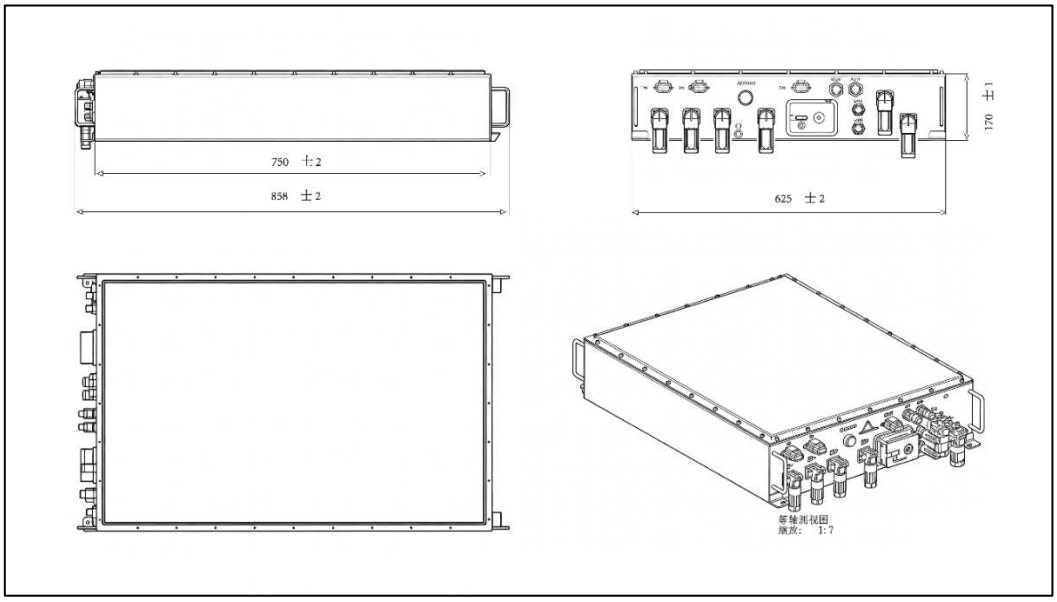


Figure 5-10 Appearance and Dimensions of PDU (For Reference Only)

No.	Specifications	Parameters	Remarks
1	Maximum Operating Voltage	1500VDC	
2	Maximum Continuous Current	170A/branch	
3	Dimensions(W*D*H)	625±2mm(W)*750±2mm(D)*170±1mm(H)	
4	Weight	39±5kg	
5	Power Supply	24VDC	External power supply
6	Communication	Ethernet, Daisy Chain, CAN	Ethernet for communication with the BAMS
7	Display Function	Power and control indicators	

5.9 System Control and Distribution Cabinet (SCDC)

The System Control and Distribution Cabinet (SCDC) in an energy storage system serves as the critical hub that connects battery racks to the Power Conversion System (PCS). It not only ensures the safe collection and unified transmission of DC power from multiple battery racks more importantly, but also integrates core system functions such as underlying protection, precise monitoring, and intelligent control.

The SCDC is primarily divided into five functional areas, as illustrated in Figure 5-11.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

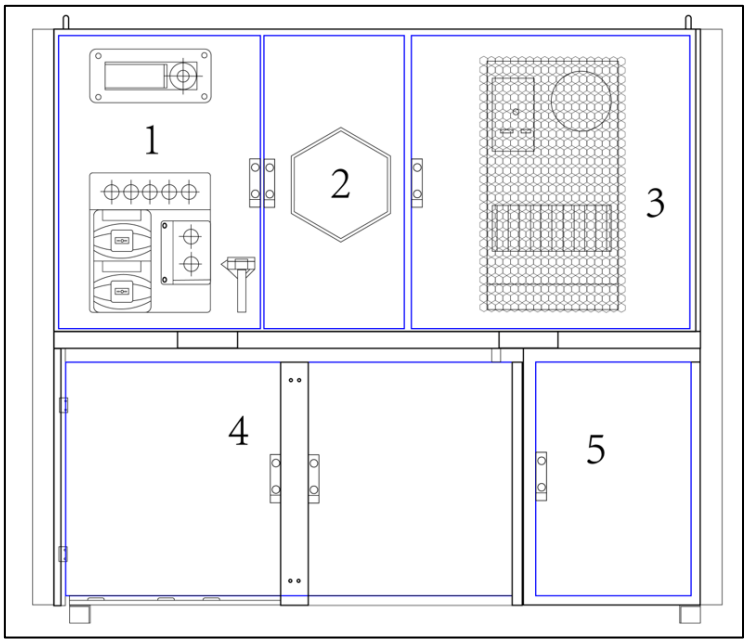


Figure 5-11 Front View of SCDC

As shown in the figure above, the functions of each area are described below:

1. Fire Suppression Control Area

   This area includes the fire suppression control panel, as well as communication and power supply connections for fire detectors. The front panel integrates the system emergency stop button and manual controls for the fire ventilation system.

2. UPS Area

   The Uninterruptible Power Supply (UPS) is located in this area, providing power to critical loads within the control cabinet and each battery container. The front panel integrates equipment status indicators and a human-machine interface (HMI) screen.

3. Power Supply and Control Area

   This area primarily consists of internal power distribution switches and the BMS controllers. The air conditioning unit is located on the front of this section. The cabinet is equipped with circulating turbulator fans, forming the internal temperature control system.

4. Main Circuit Busbar Area

   This area is mainly composed of isolation switches and surge protection devices (SPD). It is used for the DC busbar connection between each battery container and the PCS.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

5. Secondary Wiring Area

   This area internally integrates quick-connect terminals, network cable interfaces, and AC auxiliary power terminals. All external wiring, including AC auxiliary power supply as well as secondary communication and control cables, is uniformly connected to the outside from this area.

The functions of SCDC are illustrated in Figure 5.12.

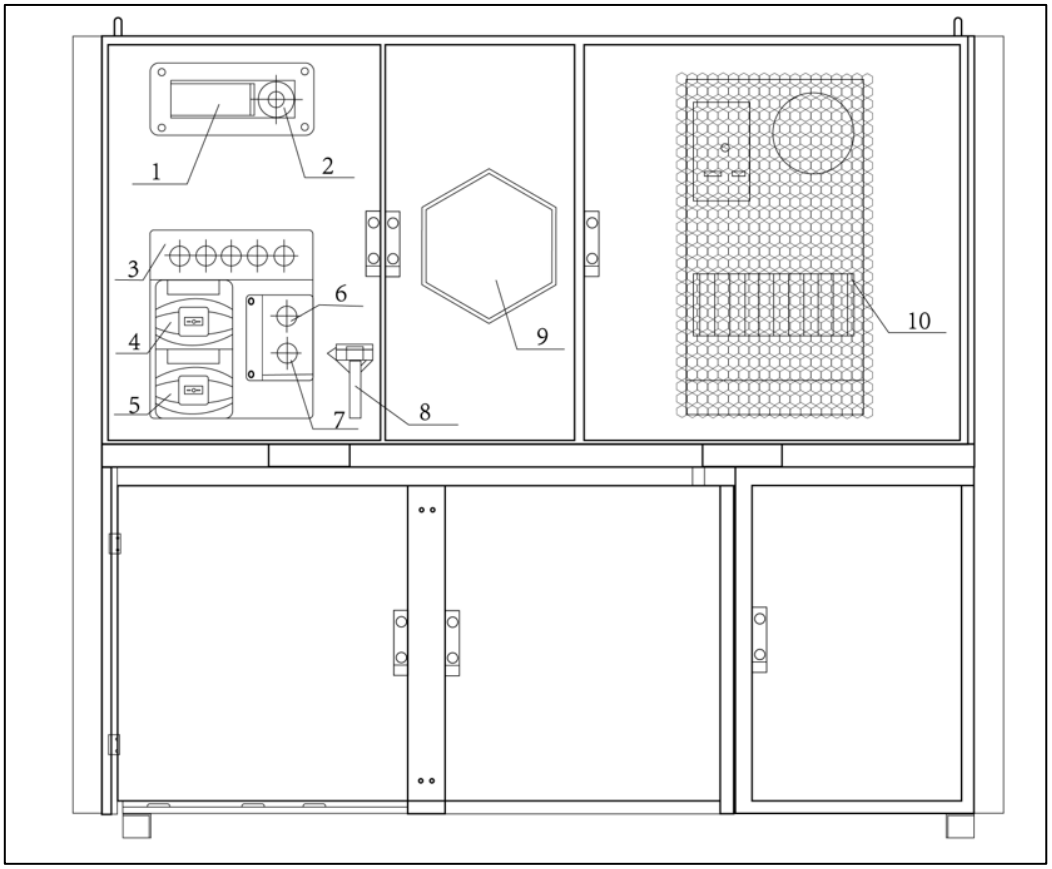


Figure 5-12 Panel Layout of SCDC

 Copyright © Beijing HyperStrong Technology Co., Ltd.

No.	Descriptions	Functions/Definitions
1	Gas Release Indicator	Combustible Gas Release Indicator
2	Combined Aural and Visual Alarm	Audio-Visual Alarm
3	Manual Rotary Switch	Fire Ventilation System Manual Start/Stop Rotary Switch
4	Manual Release Button	Battery Container Aerosol Manual Release Button
5	Manual Abort Button	Battery Container Aerosol Release Abort Button, Long Press to Abort
6	Remote/Local Rotary Knob	Switchover between EMS Remote Control and Local HMI or VIEW control
7	Emergency Stop Button	System in Emergency Shutdown Status
8	Emergency Hammer	For Emergency Use, Break the Fire Button Panel
9	LED Indicator	For More Information, Refer to the Indicator Light Status Definitions
10	Air Conditioner	Temperature Control Device of Control Busbar Cabinet

LED Indicators

Indicator	Status	Status Description
Blue	Solid	The System is Powered On and the equipment is functioning normally
	Blinking	The System or Parts of the System are Currently Starting or Stopping
Red	Solid	System Fault/Emergency Stop Fault
	Blinking	System-Level Alarm
Green	Solid	System Operating
	Blinking	System Running with Alarms

5.10 Fire Suppression System (FSS)

The fire suppression system of the ESS consists of a fire detection and alarm system, a fire extinguishing system, an active ventilation system, and a water fire protection system. Due to the modular design, the main control unit of the Fire Suppression System (FSS) is installed within the SCDC. Each ESS is equipped only with components such as smoke detectors, thermal detectors, and combustible gas detectors. The FSS main unit is connected to the components in each ESS via dedicated fire suppression system cables, forming a complete and integrated fire suppression system.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

The HyperBlock M adopts a modular assembly configuration consisting of one System Control and Distribution Cabinet (SCDC) and multiple energy storage container units. Accordingly, the fire suppression system's master plan follows a design of "1+n": 1 refers to addressable fire control panel and n refers to peripheral devices (smoke detectors, thermal detector, I/O modules, etc.).

The HyperBlock M system's fire suppression components are distributed across the System Control and Distribution Cabinet (SCDC) and individual energy storage container units. Within each SCDC, the system integrates one addressable fire control panel, one smoke detector, three dual-channel input/output modules, an audible and visual alarm unit, a fan start/stop rotary switch, and an emergency Start Button.

For every energy storage container unit, the fire suppression package includes one smoke detector, one thermal detector, one combustible gas detector, six single input/output modules, three aerosol fire suppression system, and a ventilation system including fans and motorized louvers. As an optional safety feature, deflagration panels can be installed on container roofs.

This configuration has a modular design, the SCDC serves as the central control hub while each container maintains autonomous fire detection and suppression capabilities. The single Input/Output Modules enable comprehensive monitoring and control integration, while the optional deflagration panels provide additional protection against potential gas buildup in accordance with NFPA 68 standards for explosion protection. The aerosol suppression system offers targeted fire extinguishing capability specifically designed for lithium-ion battery applications.

The addressable fire control panel connects to all peripheral devices via a two-wire bus system to enable data communication and control.

The addressable fire control panel interfaces with the energy storage system's control system through two communication methods: fault dry contact signals and Modbus 485 protocol, ensuring redundant data exchange.

When multiple addressable fire control panels are deployed, they interconnect with the station-level host controller via RS485 communication, collectively forming a station-level fire suppression network.

The connections between SCDC and ESS are illustrated as Figure 5-13.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

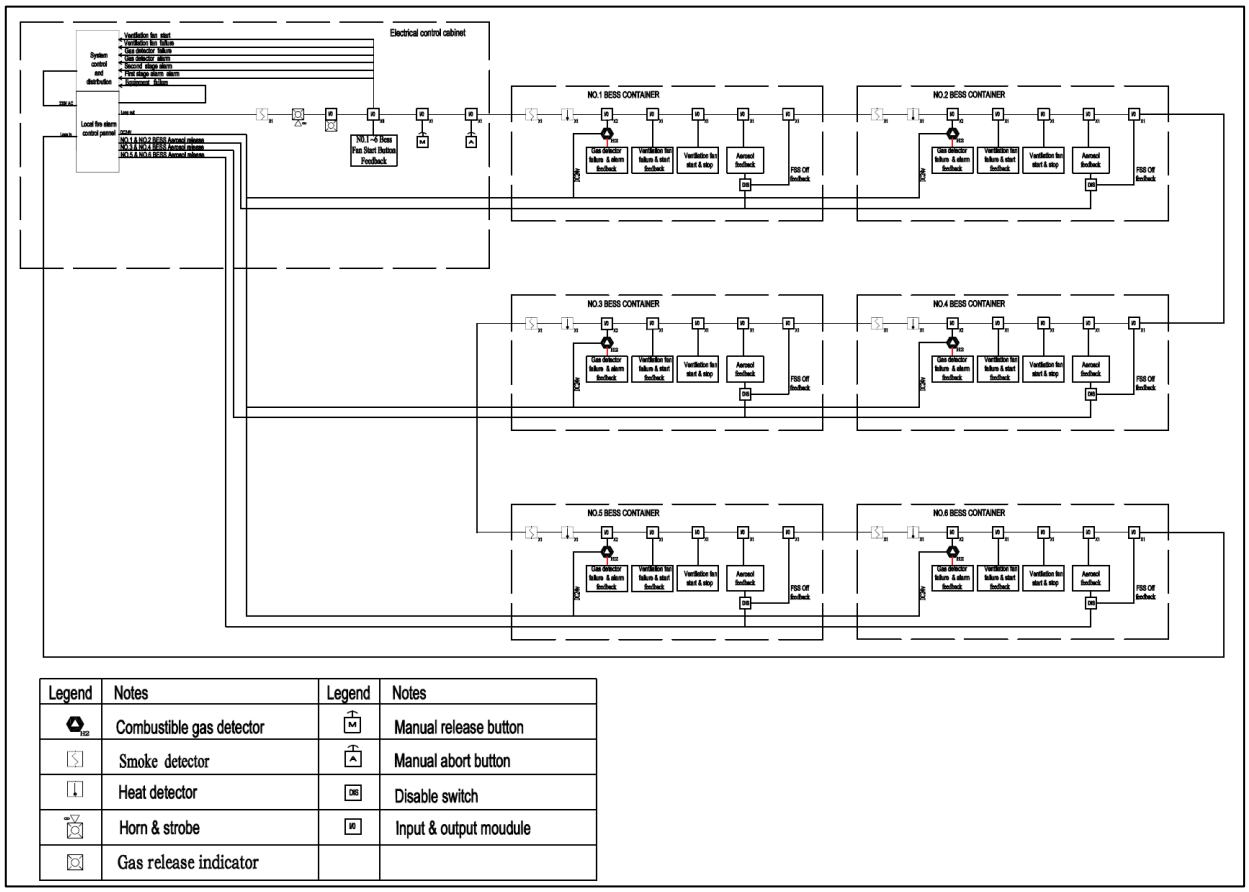


Figure 5-13 Connections between SCDC and ESS

Fire Detection and Alarm System

Each battery prefabricated cabin has a set of fire detection and alarm system, which is integrated with the gas fire extinguishing system. The technical requirements of a fire detection and alarm system are as follows:

- Sensors and Detectors Combustible gas detection, smoke fire detectors, and heat detectors are installed in the battery container.
- Sound and Light Alarms: The system has a prefabricated sound and light alarm outside the cabin. When the gas fire extinguishing apparatus is activated, the alarm issues a warning and transmits the alarm signal to the maintenance personnel at the station or remote operation center according to the degree of fire.
- The fire suppression system is equipped with a deactivation switch, allowing the electrical circuit of the aerosol fire suppression system to be isolated prior to maintenance or servicing of the fire protection system. The electrical cabin is equipped with fire alarm control panel.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

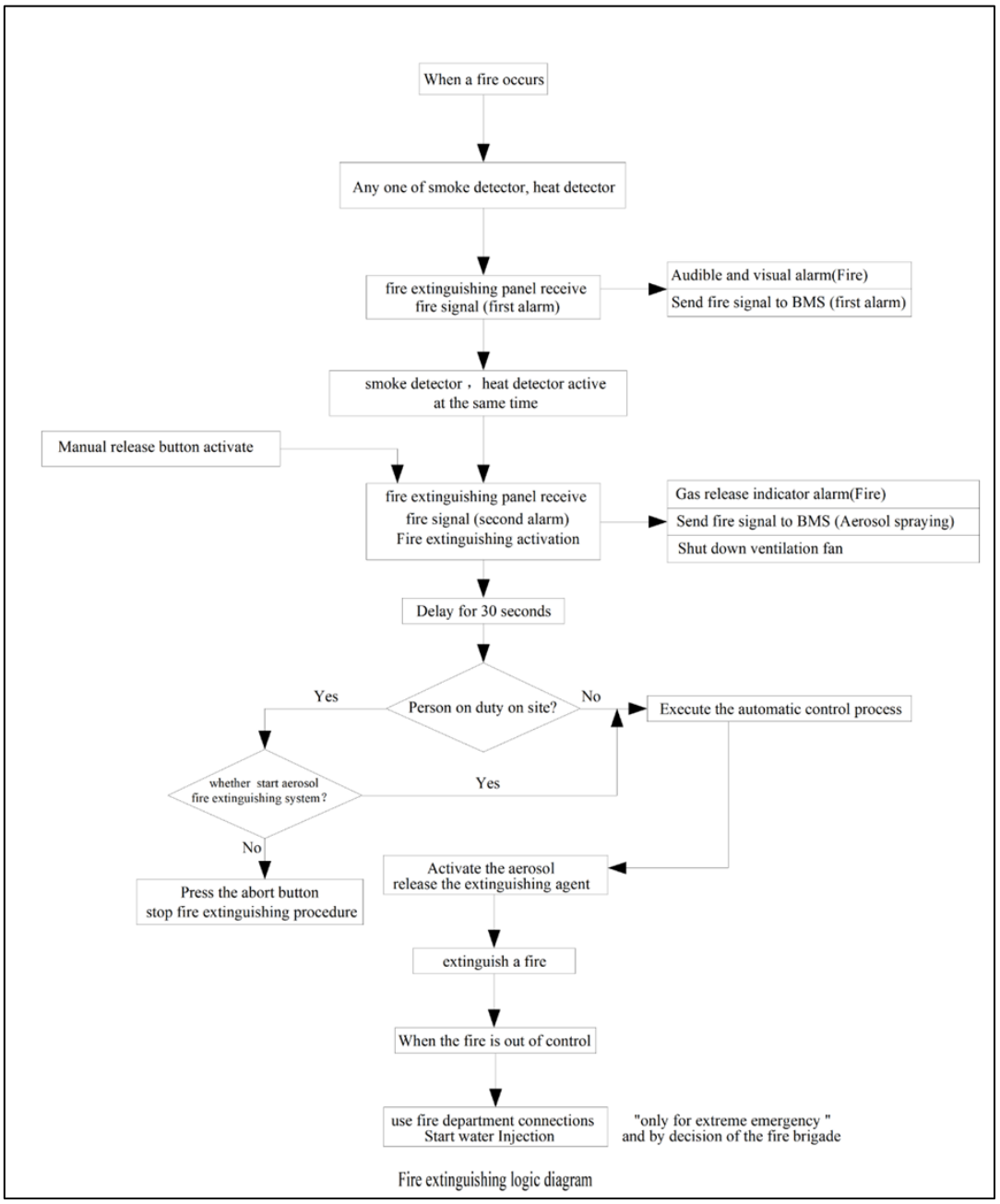


Figure 5-14 The Working Principal Flow Chart of Aerosol Extinguishing System

Aerosol Fire Extinguishing System

An aerosol extinguisher system includes an aerosol extinguishing device, an emergency release button, a disable switch, and other electrical components. The aerosol fire extinguishing system has the following features:

The aerosol fire extinguishing system has the following features:

- A Class 2 fire alarm has been triggered. The fire alarm control panel activated the aerosol fire suppression system, initiating a 30-second

 Copyright © Beijing HyperStrong Technology Co., Ltd.

countdown. Following the countdown, the aerosol fire suppression system completed its discharge. Higher fire extinguishing efficiency and rapid response time.

- The emergency start switch allows manual activation of aerosol discharge to suppress the fire.
- All power supply and control cables are independently arranged and galvanized steel pipes or metal channel boxes for added protection.

The control logic of the aerosol fire extinguishing system is as follows:

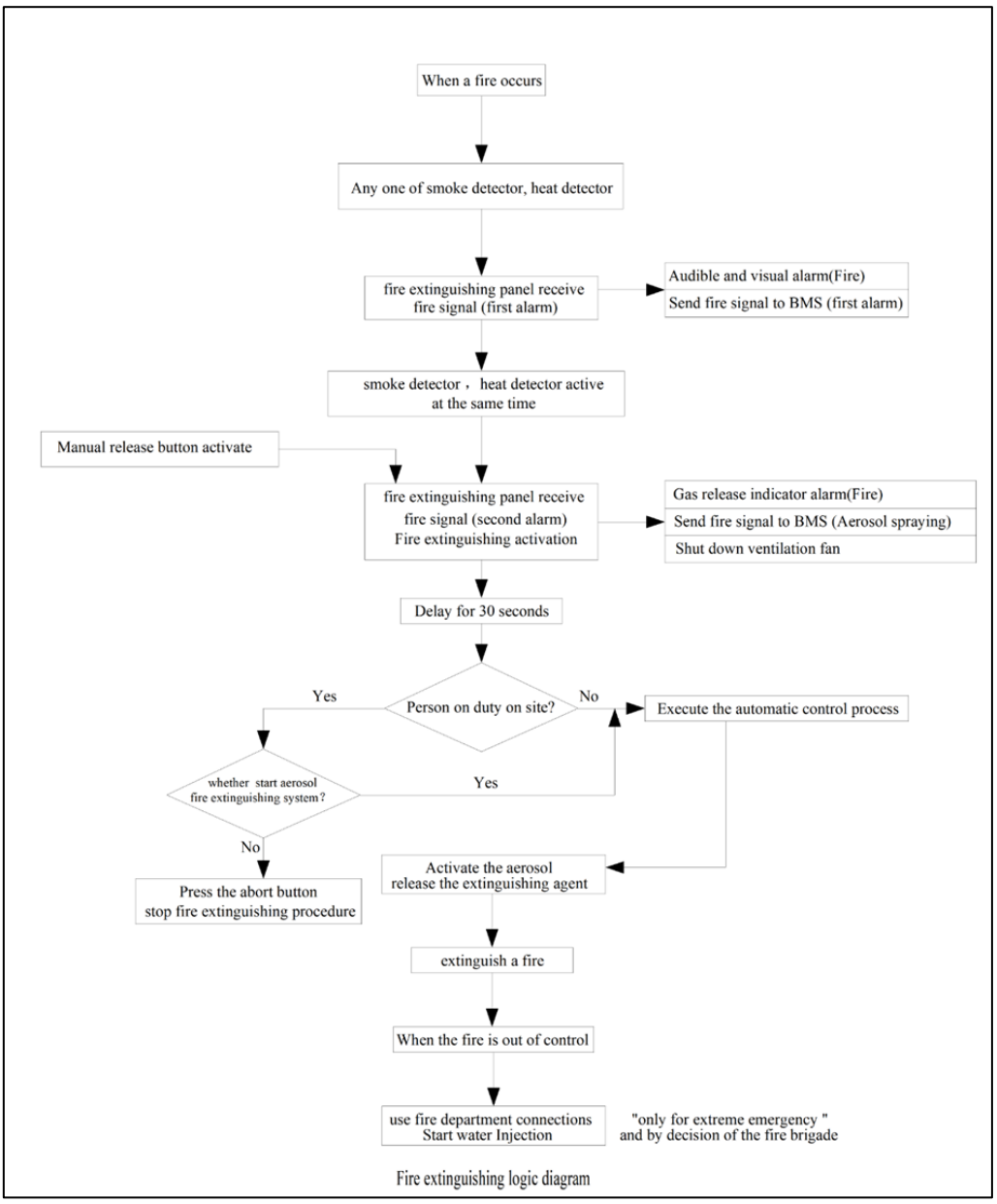


Figure 5-15 The Working Principal Flow Chart of Aerosol Extinguishing System

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Note: The current operational logic of the fire extinguishing system is as follows:

1. When any smoke detector or temperature detector reaches the alarm threshold, the fire alarm control panel triggers a Level 1 fire alarm signal and activates audible and visual alarms. The BMS receives the fire alarm signal.
2. When any smoke detector or temperature detector simultaneously reaches the alarm threshold, the fire alarm control panel triggers a Level 2 fire alarm, activating audible and visual alarms. The BMS receives the Level 2 fire alarm signal. After a 30-second countdown on the fire alarm control panel, the aerosol fire suppression system deploys.

Remark:

1. Inspection, testing, and maintenance of the aerosol fire suppression system should be in accordance with EN15276.

Active Ventilation System

The container is additionally equipped with an active ventilation system that works in conjunction with the combustible gas monitoring module to promptly address safety hazards caused by excessive concentrations of combustible gases. The ventilation system can be manually activated with the fan start-stop button.

The active ventilation system is turned off by default to maintain the container's sealing level of IP55 or above. This prevents the entry of external dust, rainwater, etc., which can guarantee the safe and stable operation of the energy storage system under normal conditions. When the combustible gas detector is triggers, an activated signal is sent, the energy storage system enters the protection shutdown state. At this point, the active ventilation system is activated, which can rapidly reduce the internal pressure and the concentration of combustible gases, thereby ensuring the safety of the entire energy storage system.

The active ventilation system features high-efficiency, ventilation capability, while also possesses both safety and reliability.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

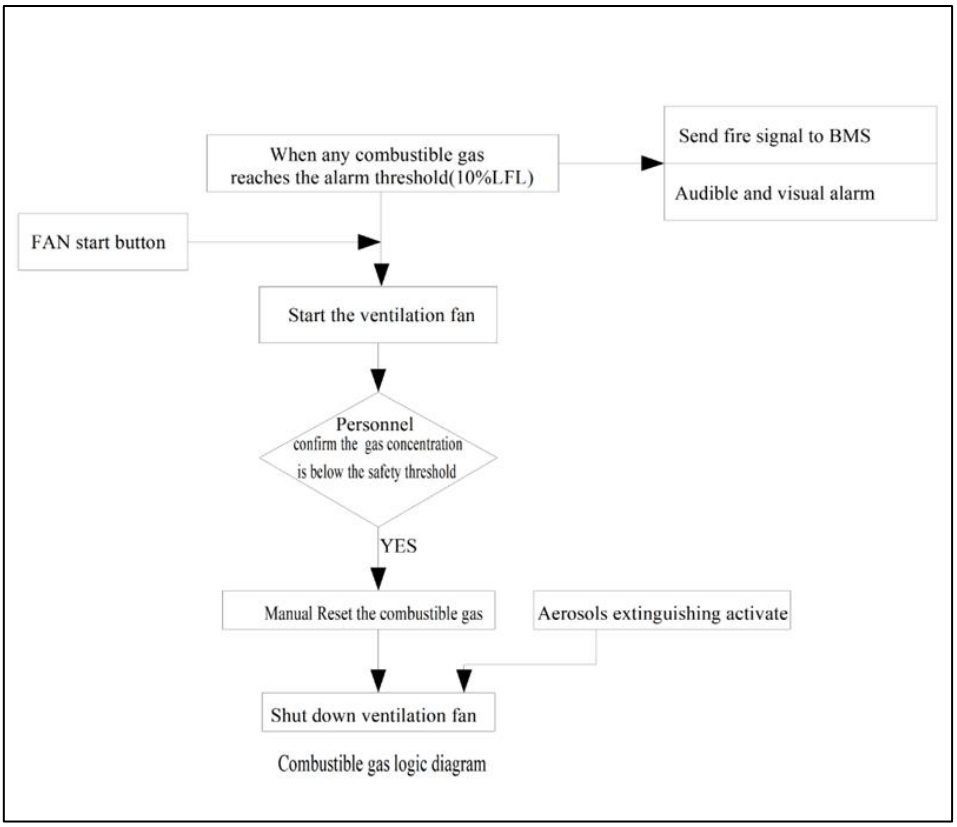


Figure 5-16 Ventilation System Working Principal Flow Chart

Note:

1. When any gas detector detects hydrogen concentration in the environment reaching the alarm threshold, the intake louvers and exhaust fan activate simultaneously to reduce the gas concentration in the environment.
2. During fan operation, if the fire alarm control panel registers a Level 2 fire alarm, the fan shuts off immediately.

Explosion Relief System

The explosion vent is mounted on top of the battery container. In the event of an explosion, when the container pressure reaches 0.1 bar, the explosion vent opens along a predetermined area to release the pressure from the container to the surrounding area.

ESS Water Fire Protection

The rapid liquid injection system is the last line of defense for energy storage fire protection, which is mainly to prevent the fire spread and minimize the fire amplification and reduce the damage. Each battery container is equipped with:

- One standard water fire-fighting interface.
- Container-level piping and nozzles.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

This setup enables rapid liquid injection function to ensure the safety of personnel and equipment. The system uses one 2½" STORZ connector, conforming to EU. standard hose access requirements. If the interfaces do not match during use, you may select the compatible interface yourself.

Access Precautions of Battery Accident-Fire Water

1. After the site risk assessment is completed, firefighters should evaluate whether they can approach the battery container to connect the fire-fighting water interface for safety. Utilize a fire hose with a DN65 interface size for water injection.
2. The water pressure should be within 0.2MPa ~ 1.2MPa, and plenty of water shall be continuously supplied for fire suppression, cooling and flooding.

5.11 Thermal Management System (TMS)

The Thermal Management System (TMS) enables optimal temperature control, enhancing product performance, extending battery life, reducing auxiliary power consumption, and increasing the overall value of the product. This TMS is custom-designed based on the temperature characteristics of each container's components within the energy storage system: the battery container utilizes a liquid cooling system to heat and cool the batteries, while the electrical cabinet employs forced-air cooling via air conditioning for heat dissipation. Both the battery and electrical cabinets are equipped with dehumidification devices to maintain the operating environment within a reasonable humidity range.

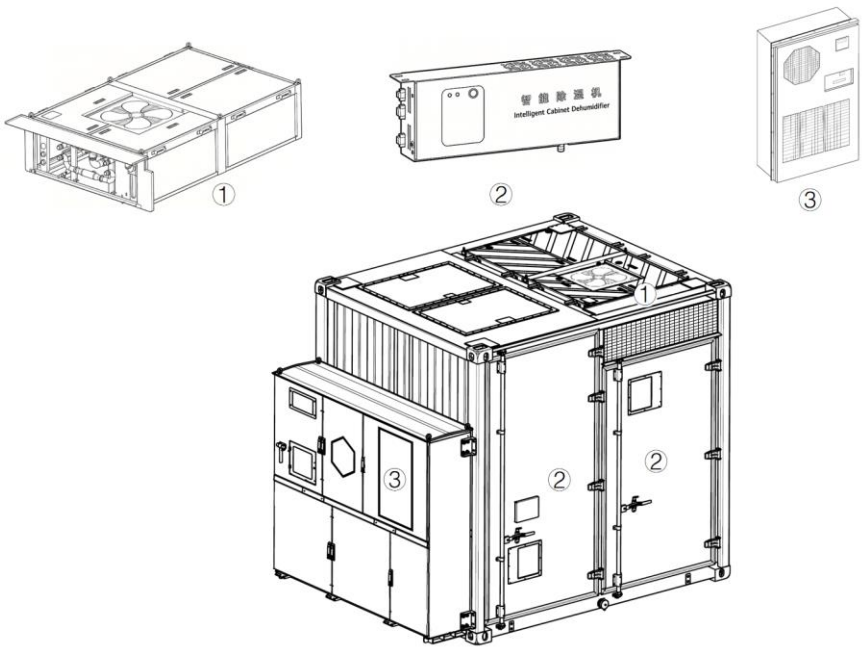


Figure 5-17 Schematic Diagram of Thermal Management System For ESS

 Copyright © Beijing HyperStrong Technology Co., Ltd.

No.	Cabinet	Thermal Management Protocol	Device	Capabilities	Quantity	Standard Configuration or Not
1	Battery Cabinet	Heating	Liquid Cooling Unit	12kW	1	Yes
		Cooling		15kW		
2		Dehumidifying	Dehumidifier	1L/day	2	Optional (based on project configuration)
3	SCDC	Heating	Air-cooled air conditioner	1kW	1	Yes
		Cooling		2kW		
		Dehumidifying		4L/day		

Liquid-Cooling Unit

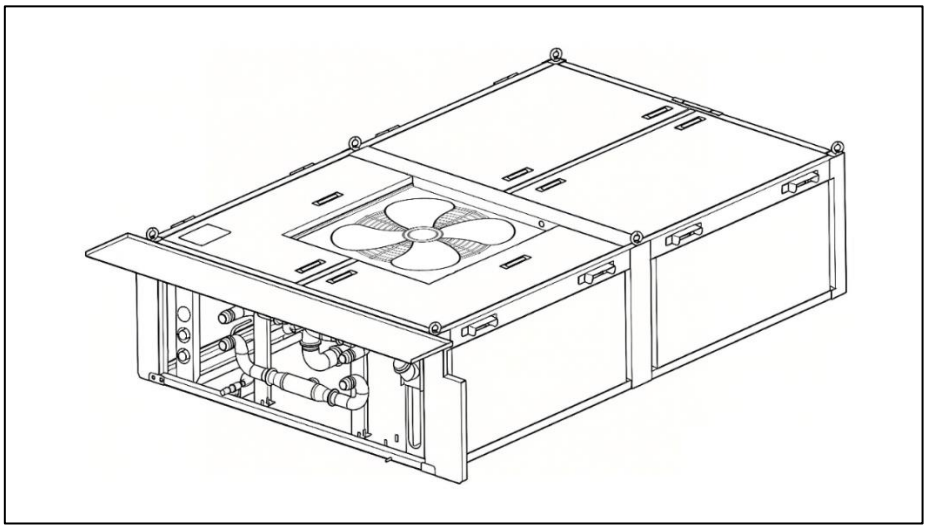


Figure 5-18 Liquid-cooling Unit

The liquid cooling unit features an integrated design, incorporating communication with the upper-level computer, alarm functionality, and other key features. This AC-powered refrigeration unit offers high reliability, simplified installation, and does not require complex debugging.

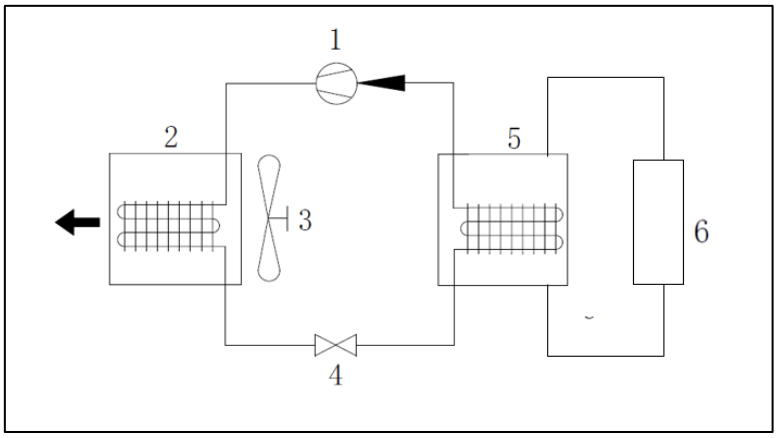
Key Characteristics of the Liquid Cooling Unit:

- The unit achieves efficient refrigeration for low-power systems through an intelligent temperature control strategy, which optimizes compressor start-up frequency and implements a load-based frequency conversion strategy.
- The unit is equipped with a heating device; the unit activates heating up when the system temperature falls below a specified threshold.
- The unit is equipped with an automatic fluid replenishment device, which replenishes fluid automatically when it is short of fluid.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- A safety valve is integrated into the system to release pressure when it exceeds safe limits, ensuring the safety of the system's pipeline.
- The system includes liquid leakage detection functionality, issuing alarms in the event of a leak to enable early identification and resolution by personnel.
- A wind shield is added around the air outlet of the liquid cooling unit, which can prevent the hot air blown out by the liquid cooling unit from re-entering the unit's air inlet through the installation gaps between the unit and the container, because the hot air would otherwise affect the cooling efficiency.

The operating system principle of the liquid-cooling unit is as follows



1. Compressor
2. Condenser
3. External circulation fan
4. Throttle element
5. Chiller
6. Battery coolant pipeline network

Description of Refrigerant Pipeline

The refrigerant (R32) within the liquid cooling unit pipeline of closed system is used for cooling the load and flows sequentially through the plate heat exchanger, compressor, condenser, and expansion valve before returning to the condenser. In the plate heat exchanger, the refrigerant absorbs heat from the load (the glycol mixture in the battery liquid cooling pipeline) and vaporizes, simultaneously generating cooling capacity. The compressor draws in and compresses the gaseous refrigerant from the plate heat exchanger. The compressed high-temperature, high-pressure refrigerant then enters the condenser, where it is cooled by a cooling medium (such as air). The condenser liquefies the refrigerant, releasing heat. The cooled refrigerant then passes through the expansion valve, where its pressure is reduced. The cooling medium returns to the plate heat exchanger, and the gas/liquid mixture vaporizes again, absorbing heat to achieve refrigeration.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Description of Cooling Medium Pipeline

The pump draws the cooling medium (glycol mixture) from the battery cooling pipeline. After being cooled by the refrigeration system, the cooling medium is then delivered to the load. The cooling medium with high-temperature enters the plate heat exchanger for heat exchange. The cooling medium that is cooled by the plate heat exchanger continues to circulate. This heat exchange process cycles continuously.

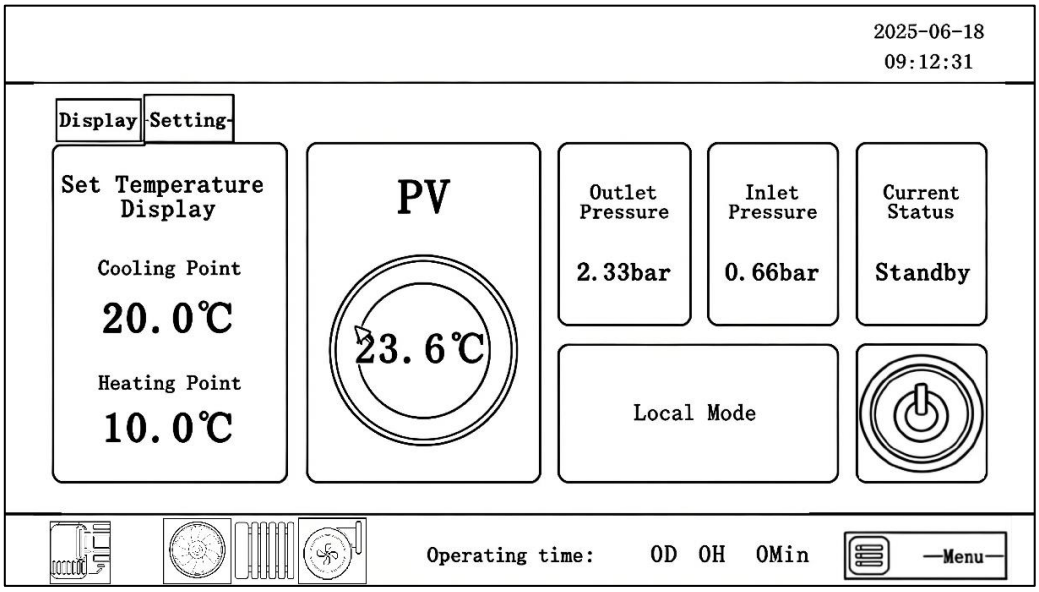
Key Parameters of Liquid Cooling Unit

Items	Technical Parameters	Units
Rated Cooling Capacity@L45 W20	15000	W
Rated Heating Capacity	12000	W
Maximum Electrical Power	12900	W
Refrigerant	R32	
Rated Flow Rate	180	L/min
Operating Voltage & Frequency	380/480 50/60	V, Hz
Communication Protocol	RS485	
Dimensions	W×D×H：1200×1900×450	Mm

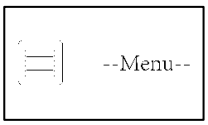
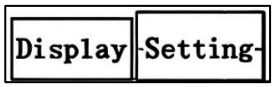
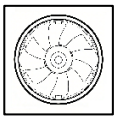
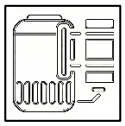
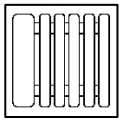
Control Panel of Liquid Cooling Unit

The liquid cooling unit is equipped with a detachable control panel. After connection via an aviation connector, the control panel can be used to query the unit's operating status and detailed parameters. Please the following figures for detailed descriptions of the control panel buttons and icons:

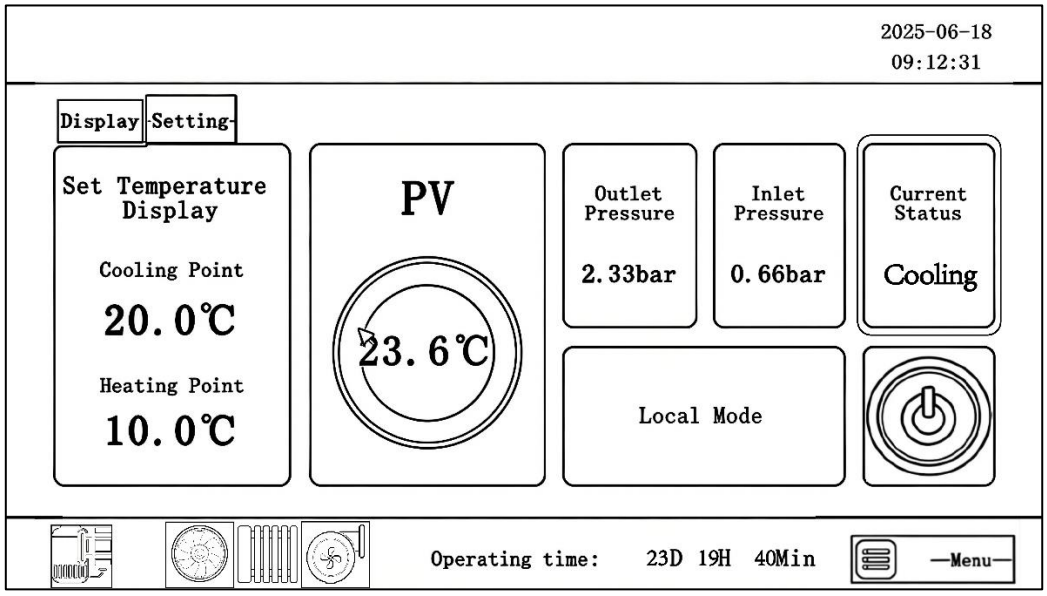
Standby Interface



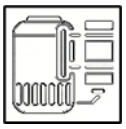
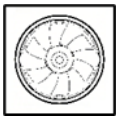
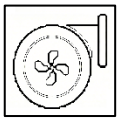
 Copyright © Beijing HyperStrong Technology Co., Ltd.

Icons	Descriptions
	Enter the menu interface.
	Display/Set local cooling and heating set points.
	In case of a system fault, it will flash on the main interface. Tap it to query the current fault information; the icon is hidden when there is no fault.
	For indicating the fan operation status.
	For indicating the compressor operation status.
	For indicating the electric heater operation status.

Work Interface



 Copyright © Beijing HyperStrong Technology Co., Ltd.

Icon	Name	Descriptions
	Compressor	If this icon keeps flashing and turns green, it indicates that the compressor of the unit has been started.
	Fan	If this icon keeps flashing and turns green, it indicates that the fan of the unit has been started.
	Pump	If this icon keeps flashing and turns green, it indicates that the pump of the unit has been started.

Air Conditioning System

The air conditioning equipment provides functions such as temperature control, which can maintain the electrical cabinet temperature within the range of 0-45℃. The key functions include:

- Cooling is activated when the ambient temperature is above 40℃, to ensure continuous and effective operation of the equipment under high-temperature conditions.
- Heating is activated when the ambient temperature is below 5℃, to prevent the startup issues caused by low temperatures.

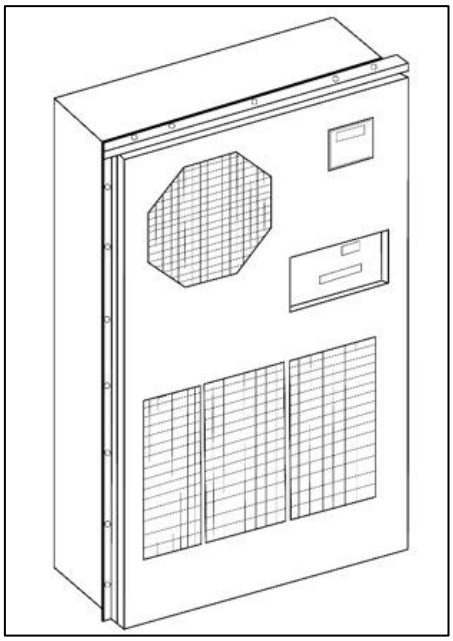
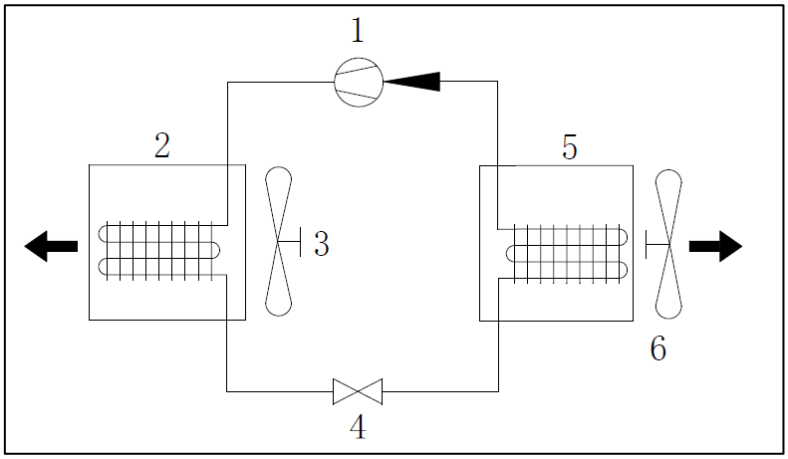


Figure 5-19 Air conditioning System

 Copyright © Beijing HyperStrong Technology Co., Ltd.

The system principle of the air conditioning system unit is as follows.



1. Compressor
2. Condenser
3. External circulation fan
4. Throttle element
5. Evaporator
6. Internal circulation fan

Description of Refrigerant Pipeline

The refrigerant (R513A) within the air-conditioner cooling medium pipeline of closed system is used for cooling the load and flows sequentially through the plate heat exchanger, compressor, condenser, and expansion valve before returning to the condenser. In the evaporator, the refrigerant absorbs heat from the load (devices in electrical cabinet with high temperature) and vaporizes, simultaneously generating cooling capacity. The compressor draws in and compresses the gaseous refrigerant from the evaporator. The compressed high-temperature, high-pressure refrigerant then enters the condenser, where it is cooled by a cooling medium (such as air). The condenser liquefies the refrigerant, releasing heat. The cooled refrigerant then passes through the expansion valve, where its pressure is reduced. The cooling medium returns to the evaporator, and the gas/liquid mixture vaporizes again, absorbing heat to achieve refrigeration.

Description of Cooling Medium Pipeline

The fans draw the gas with high temperature from the electrical cabinet. After being cooled by the refrigeration system, the gas with low temperature is then delivered to electrical cabinet to control the environment temperature and cool the heating device in the electrical cabinet.

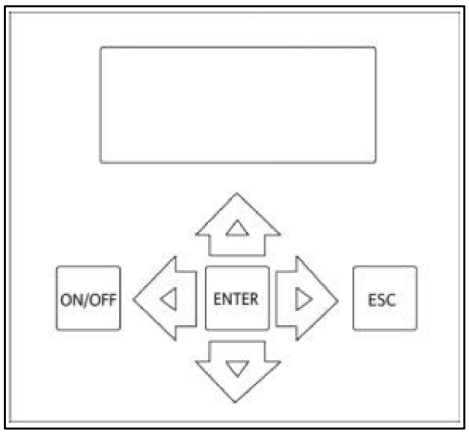
 Copyright © Beijing HyperStrong Technology Co., Ltd.

Main Parameters

Technical Data	Value	Unit
Rated Cooling Capacity @L35 L35	2100	W
Air Volume, Internal Circulation	650	M³/h
Refrigerant Type	R513a	
Rated Voltage	230±10%, 50/60	V, Hz
IP Protection Classification	IP55	
Ambient Air Temperature	-40~55	℃
Installation Attitude	Door Mounted	
Dimensions(H*W*D)	783×483×257	mm

Control Panel of Cooled Air Conditioner

The air-cooled air conditioner is equipped with the control panel. The operating status and detailed parameters of the air conditioner can be queried through the control panel. The button description of control panel is illustrated as below.



Buttons	Descriptions
ON/OFF	Power ON/OFF: Press and hold this button for about 5 seconds to turn on/ off the device.
▲	Up-arrow: To select the previous record/menu or increase the setting value (only for passwords).
▼	Down-arrow: Select the next record/menu or reduce the setting value (only for passwords).
◀	Left-arrow: Increase the setting value or select the previous data when performing password operations.
▶	Right-arrow: Reduce the setting value or select the next data when performing password operations.
ENTER	Enter: Confirm the input.
ESC	Esc: Return to the previous screen menu.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Dehumidifier

The dehumidifier provides functions such as humidity control, which can maintain the humidity in the battery container and electrical room within 80% RH.

The key functions are as follows:

- When the ambient humidity exceeds 80% RH, it starts dehumidification to ensure that the battery system and electrical components in the electrical room can operate effectively under appropriate humidity conditions.

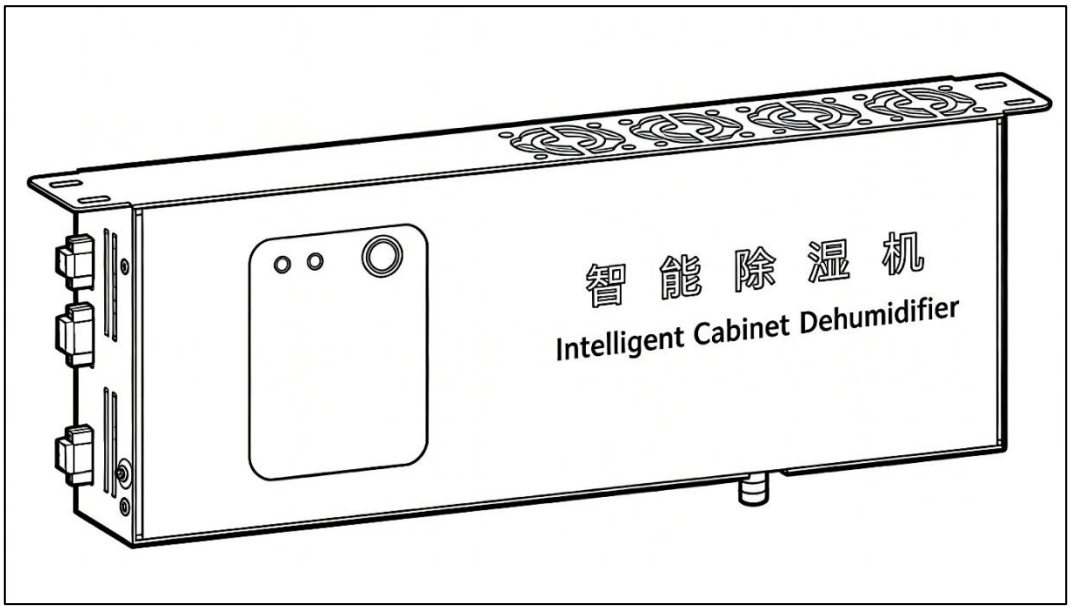


Figure 5-20 Dehumidifier

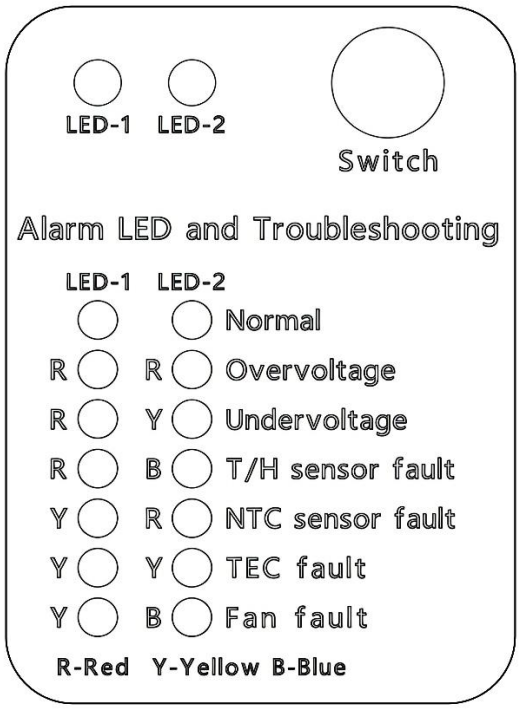
Main Parameters

Technical Data	Value	Unit
Dehumidification capacity	1000±10%, 35℃ 85%RH	mL/D
Rated voltage	85V~265, 50/60	V, Hz
IP Protection classification	IP20	
Ambient air temperature	5~55	℃
Installation attitude	Door Mounted	
Dimensions(H*W*D)	142×368×50	mm

Display of Dehumidifier

When the product is powered on, it enters the self-check mode, and the alarm indicator light and the button indicator light remain constantly on; after the self-check is completed, it enters the working state, and all the indicator lights go out.

 Copyright © Beijing HyperStrong Technology Co., Ltd.



Description of Indicator Lights

The unit is equipped with one operation indicator light (green button light) and two alarm indicator lights (tri-color lights). The operation indicator shows the unit's operating status, while the alarm indicators show abnormal conditions.

- During dehumidification: The operation indicator is illuminated.
- When not dehumidifying: The operation indicator is off.
- In Case of Anomaly: The specific fault condition is indicated by the alarm lights. Detailed fault descriptions are provided in the table below.

No.	Alarm Light 1	Alarm Light 2	Abnormalities
1	OFF	OFF	None
2	Red	Red	Overvoltage
3	Red	Yellow	Undervoltage
4	Red	Blue	Fault of Temperature and Humidity Sensor
5	Yellow	Red	Fault of NTC Sensor
6	Yellow	Yellow	Fault of refrigerator
7	Yellow	Blue	Fault of Fans
Note: If multiple abnormal states exist simultaneously, they will be displayed in turns, with each state being displayed for 1 second.			

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Pipeline Network

The liquid cooling pipeline system has been optimized from the previous Three-level structure (primary, secondary, and tertiary pipe) to a configuration consisting only of secondary and tertiary pipelines. The new design connects the secondary pipelines directly to the liquid cooling unit, which can reduce the number of pipeline connection points, thereby lowering the risk of leaks. This network evenly distributes the coolant to each battery pack, ensuring uniform heat dissipation across all battery cells.

An automatic air vent valve is installed at the top of the secondary pipeline, which can detect the accumulation of gas in the upper section of the pipeline through an internal float mechanism or similar device: when gas accumulates to a certain amount, the buoyancy of the float will be reduced, thereby opening the valve to release the gas; after the gas is exhausted, the liquid will flow into the valve body and the float will rise and automatically close the valve.

The liquid cooling system operates within the following pressure limits:

- Maximum hydrostatic pressure: 1.5bar
- Maximum working pressure: 3.5bar

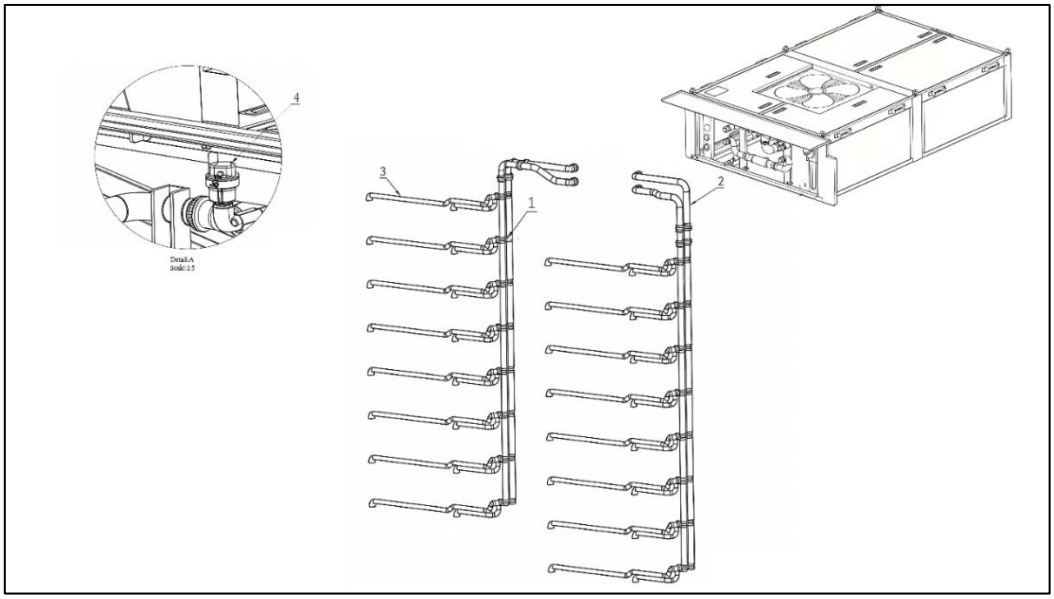


Figure 5-21 Pipeline Network

No.	Name of Components
1	Double Flow Stop Valve
2	Secondary Pipeline
3	Tertiary Pipeline
4	Vent Valve

 Copyright © Beijing HyperStrong Technology Co., Ltd.

6. Equipment Installation

6.1 ESS Lifting

Basic Information of Equipment

- Overall dimensions (L*W*H): 10-feet container (high box) 2,991mm*2,438mm*2,896mm.
- Lifting position of the container: corner pieces, top lifting.
- Weight of container box: approximately 3.8 tons.
- Net internal load capacity: approximately 20.2 tons
- The total weight of the battery prefabricated container (including batteries): approximately 24 tons (4 racks). The lifting operation shall be conducted with a maximum load capacity of 30 tons, while taking into account the variations in the lifting process and the number of modules among different models.

Lifting Precautions

⚠ CAUTION

- During the lifting and transportation of the container, all standard operating procedures (SOPs) and regulations related to safety shall be strictly complied with in the country or region where the project is located.
- The Company shall not be responsible for any personal injury or property damage resulting from the violation of relevant requirements, SOPs or regulations.
- Throughout the entire process of BESS lifting, it is essential to strictly comply with the safe operating practice of the crane.
- A clear safety zone of 5 to 10 meters shall be maintained around the operation area.
- All lifting operations during adverse weather conditions shall be suspended including but not limited to heavy rain, fog, or strong wind.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

The following requirements shall be fulfilled during the lifting of the ESS container:

- All the ESS container doors shall be locked.
- A suitable crane or lifting device shall be selected based on site-specific conditions with adequate load capacity, boom length, and radius of rotation.
- Additional towing or traction equipment may be required when moving across inclined surfaces or slopes.
- The intended route shall be cleared of obstructions such as overhanging trees, power lines, or other impediments.
- The lifting operations of ESS containers should be conducted under favorable weather conditions whenever practicable and shall be suspended in adverse weather, such as heavy rain, fog, or strong wind.
- The operating location shall be secured prior to initiating the lifting process.
- During the lifting and installation operations, professional personnel shall be present to monitor the entire process.
- Warning signs or cordons shall be set up and formed to restrict access of unauthorized personnel.
- All personnel involved in loading, unloading and bolting shall receive appropriate training, particularly about safety
- A suitable crane or lifting device shall be selected based on site-specific conditions to ensure a safety operation. The crane load-bearing capacity ≥100,000kg is preferred.
- The strength of the sling used shall support the weight of the ESS container.
- The length of the sling shall be adjusted according to specific requirements.
- all sling connections shall be secured and safe, please ensure that all sections of the sling which are connected to the corner pieces are equilong.
- The length of the sling should be adjusted appropriately based on the site-specific requirements
- The battery cabin shall not be leaning and the operation shall be smooth during the entire lifting process.
- The four telescopic pillars of the ESS shall be utilized for lifting.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- All necessary auxiliary measures shall be taken to ensure a safe and smooth lifting of the battery prefabrication module.
- The crane shall be operated strictly in accordance with the safe operating practice of the crane during the lifting process of the battery cabin.
- **DO NOT** stand within 5~10m of the operation area. Standing beneath the lifting arm or moving machine shall be strictly prohibited to avoid accidents.
- Lifting operations shall be suspended in adverse weather conditions, such as heavy rain, fog, or strong wind.
- The following figure illustrates the crane operation while lifting the ESS container. The inner dotted circle in the figure indicates the working area of the crane. Standing inside of the red circle shall be strictly prohibited during the operation of the crane.

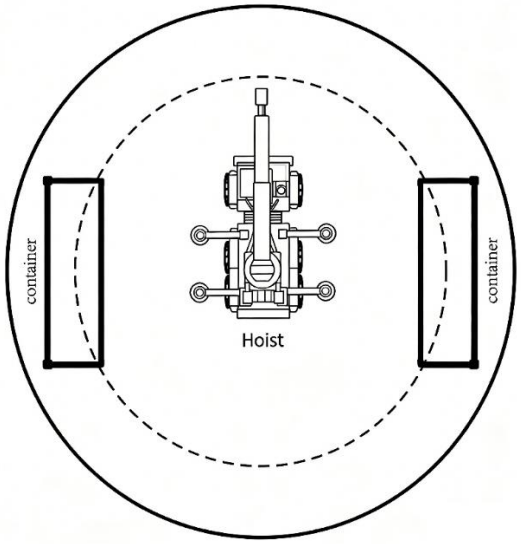


Figure 6-1 Schematic Diagram of the Crane Operation

Container Lifting Process

All operation items shall be carried out in accordance with the following requirements during the lifting of the ESS:

- The ESS shall be lifted vertically and shall not be dragged on the ground, the roof of the lower box, or other surface. The ESS shall not be dragged or pushed on any surface.
- After the ESS is lifted to 300mm high from the support surface, the lifting shall be suspended, and the connection between the spreader and the ESS shall be inspected, and then the lifting can be only continued after secured connections are confirmed.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- During the placement of the ESS, the lowering and landing operations shall be gentle and smooth. It is strictly prohibited to place the ESS out of vertical alignment by flinging the spreader.
- The installation site of the ESS shall be stable, flat, well-drained, and free of obstacles or protrusions.

Description of Specific Lifting Program for Containers

Special care should be taken to protect each side of the container to be scratched. Please refer to the following figure for lifting guidance.

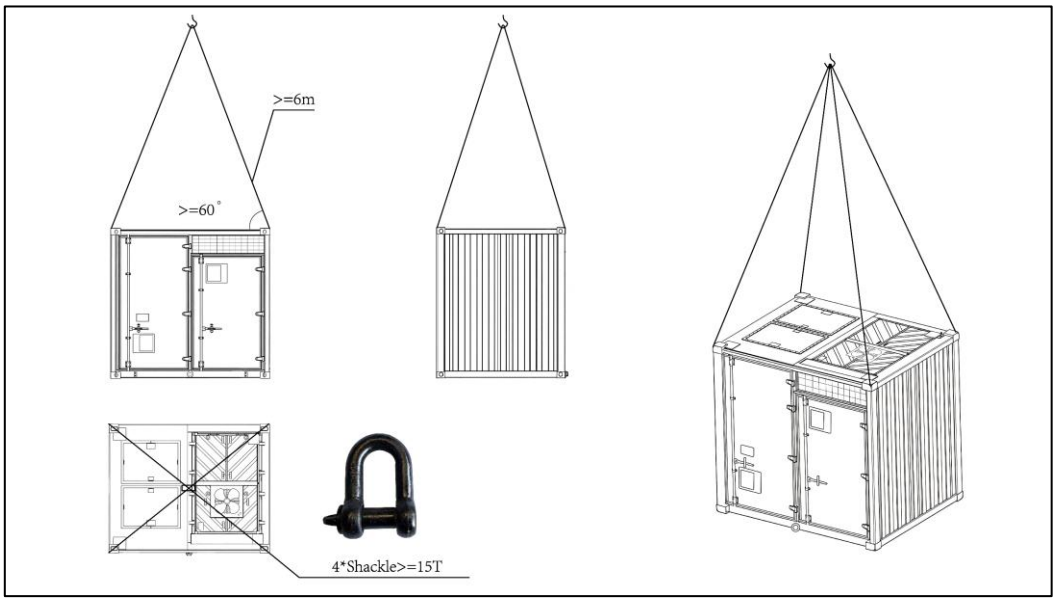


Figure 6-2 Container Support and Locations of Fixing Points

*The figure above is for reference only. Please refer to the actual product for details.

Part Name	Parameter
Sling	Sling ≥ 6 meters, load > 15T;
Shackle	Shackle ≥ 15T;

Bottom Fixing

Welding fixation method

The container bottom shall be welded and secured to the foundation. After the container is welded, anti-corrosion treatment shall be applied and the bottom edges shall be sealed.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

6.2 Mechanical Installation

⚠ CAUTION

- During the entire process of mechanical installation, relevant standards and requirements of the project site shall be strictly followed.

Pre-Installation Inspections

- Inspection of deliverables
  - Check the completeness of the deliverables According to the attached packing list.
- Inspection equipment
  - Verify that the received container is consistent with the model that as ordered.

Ensure that there is no damage on the ESS container and internal equipment.

⚠ CAUTION

Only an intact ESS container shall be installed!

Please confirm the following items prior to installation:

- The unit itself is intact.
- The equipment in the ESS is intact.

If any problem is recognized, contact the transporter or the Company immediately.

Environmental Requirements of Installation

Site Selection Requirements

When selecting the installation site, the following factors such as climate, and geological conditions (e.g. stress wave emission, underground water level, etc.) shall be taken into account with due consideration.

- The surrounding environment shall be dry and well-ventilated.
- There should be no tree around the installation site prevent obstruction from the doors of the system, and the air inlet shall not be covered by branches or leaves carried by high wind.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- The site shall be far from areas where toxic and hazardous gases are concentrated, as well as flammable, explosive and corrosive substances.
- The system shall be installed far from residential areas to avoid noise.

Foundation Requirements

⚠ CAUTION

- The overall weight of the ESS is significant, and the conditions of the installation site (primarily geological limitations, environmental factors, and climatic conditions, etc.) shall be thoroughly determined prior to the construction of the foundation. The design and construction of the foundation shall be started only when the evaluation is completed.

An improper foundation construction plan will result in difficulties or complications in the placement of the ESS container, the opening and closing of the doors, and the subsequent operations. Therefore, the foundation of the ESS container shall be designed and constructed in accordance with specific standards in advance to meet the requirements of mechanical support, cable routing, and later maintenance and overhaul.

The following requirements shall be fulfilled during the construction of the foundation:

- The soil at the installation site shall be solid. It is recommended that the relative soil compactness at the installation site shall be ≥98%. If the soil is soft, appropriate measures should be taken to ensure the stability of the foundation.
- The bottom of ditch where the foundation will be constructed shall be compacted and filled to provide adequate and effective support for the cabinet.
- The foundation shall be constructed according to a provided plan or a protocol validated by HyperStrong. The flatness tolerance of the foundation shall be within ±3mm. Specifically, the height difference between the lowest point and the highest point of the upper surface of the foundation shall not exceed **6mm,** as shown in the following figure.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

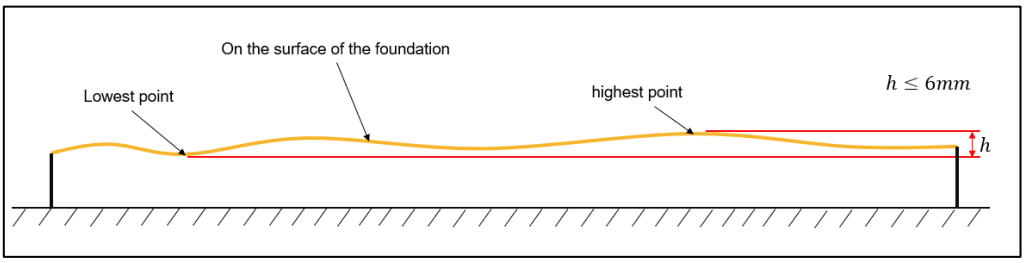


Figure 6-3 Foundation Flatness Requirement

- The energy storage container shall be installed at heightened platform to prevent the foundation and interior equipment from plurosion.
- The cross-sectional area and height of the foundation shall meet the specified requirements.
- Drainage system shall be equipped with in accordance with the local geological conditions.
- A concrete foundation with sufficient cross-sectional area and height shall be constructed. The exploration unit will measure the size of the foundation based on the geological features of the site.
- Cable routing should be considered in advance during foundation construction

---

⚠ CAUTION

- The spoil (earth and rock excavated or dredged from foundation ditch) excavated during foundation construction shall be removed immediately to avoid affecting the subsequent lifting operations of ESS containers.

- The maintenance platform should be constructed around the foundation to facilitate later maintenance.
- According to the location and the size of upper cable entrances and exits of the energy storage containers, sufficient space shall be reserved for the cable troughs on the AC/DC side, which the cable conduits can be pre-embedded during the construction of the foundation.
- The specifications and number of tubes that will be inserted shall be determined according to the cable types and the number of inlet and outlet lines.
- A drainage system should be constructed to prevent waterlogging at the bottom of the energy storage container or interior equipment during the rainy season or heavy precipitation.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- The both ends of all pre-embedded pipes should be temporarily sealed to prevent from contaminants entering which will complicate wiring later.
- After all the cables are connected, the cable entrances, exits, and joints should be sealed with fireproofing mud or other suitable material to prevent from rodent entry.
- Grounding units shall be pre-embedded according to the relevant standards in the country or region where the project is located.

Description of Foundation Form

HyperBlock M Energy Storage System consists of one SCDC (System Control and Distribution Cabinet) and multiple BESS (Battery Energy Storage System) units.

On-site Installation:

- The BESS units must be hoisted and positioned onto the pre-constructed foundations.
- For detailed foundation specifications, refer to the attached drawings.
- To further clarify the foundation requirements, this document provides the following supplementary explanations:

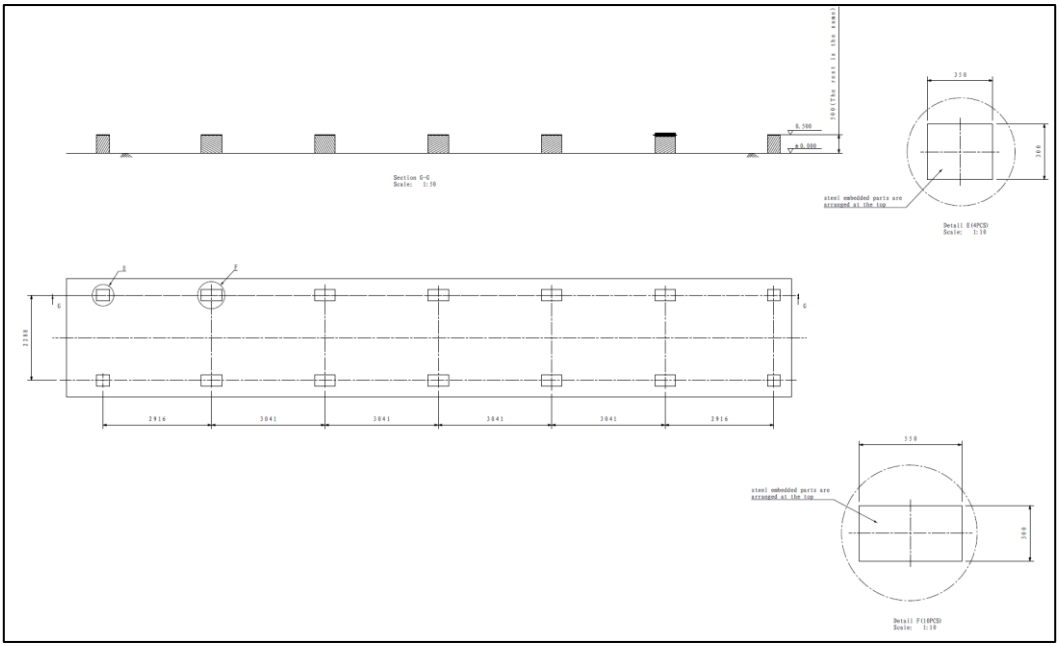
As shown in Figures 6-4(a) below: (a) shows a top view of the foundations for an energy storage system consisting of 6 BESS units (Placed side by side), requiring a total of 14 foundations. The recommended height is at least 500mm above ground level. Foundations are divided into two types: F. The difference is: Type F foundations are larger in size than type E because it needs to support 2 BESS units. The embedded steel plates are flush with the top of the foundations.

As shown in Figures 6-4(b) below, (b) shows a top view of the foundations for an energy storage system consisting of 6 BESS units (Placed side-by-side and back-to-back), requiring a total of 12 foundations. The recommended height is at least 500mm above ground level. Foundations are divided into four types: J, K, M and N. This is because, compared to side-by-side arrangement, as the Type J and Type K foundations need to support two BESS units placed back-to-back.

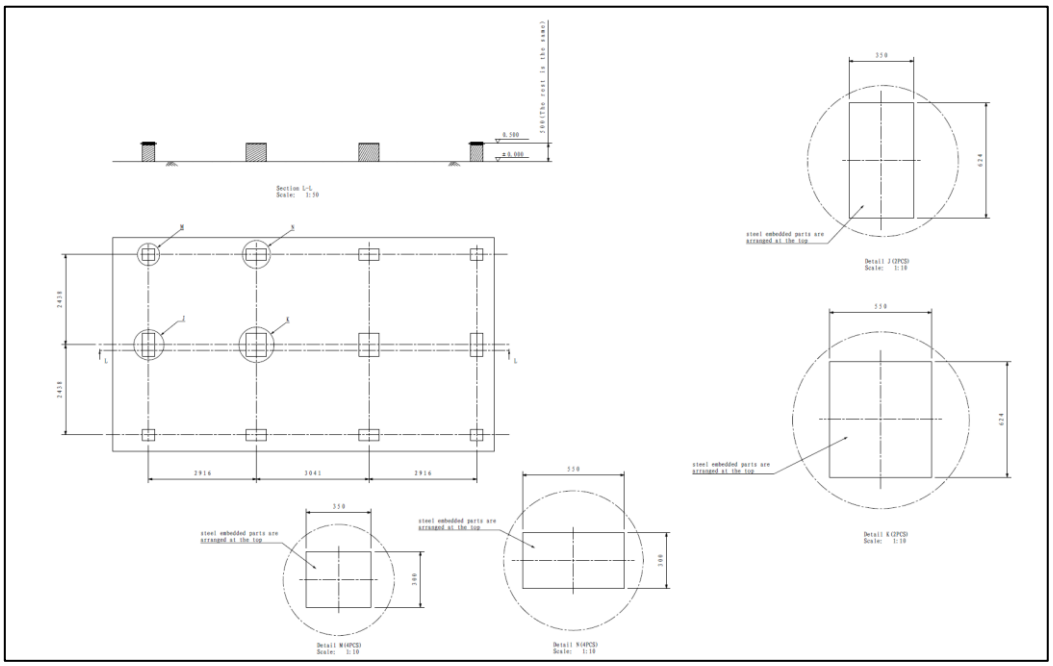
When the energy storage system consists of other quantities of BESS units, the number of foundations can be adjusted accordingly by referring to the following diagram.

It is important to note that the foundations described in this chapter apply to BESS products with bottom pallets installed, as well as to BESS products without bottom pallets. If the BESS products are with bottom pallets, the foundation may take other forms, such as a strip foundation or an even simpler slab foundation.

 Copyright © Beijing HyperStrong Technology Co., Ltd.



(a)



(b)

Figure 6-4 Top View of Foundations

(Energy storage system composed of 6 BESS units)

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Restricted Locations

A pre-assembled battery system shall not be installed:

a) in restricted locations as defined for switchboards in AS/NZS 3000;

b) within 600mm of any exit;

c) within 600mm of any vertical side of a window or building ventilation that ventilates a habitable room;

d) within 600mm of any hot water unit, air conditioning unit or any other appliance not associated with the pre-assembled battery system;

e) within 900mm below any of the items included (b), (c) and (d);

f) in ceiling spaces;

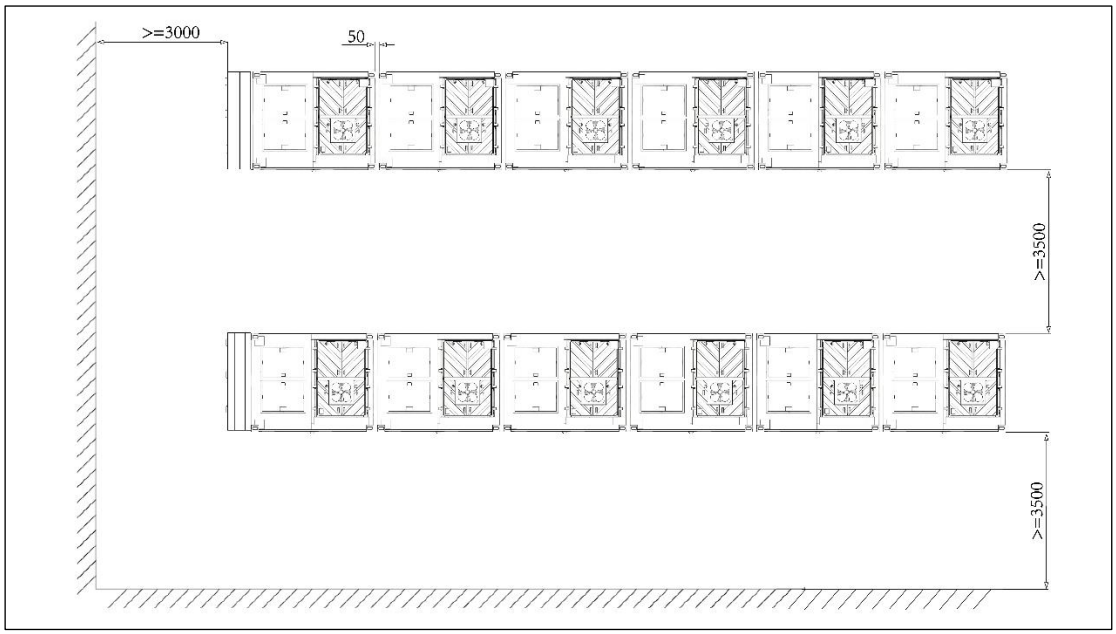
g) in wall cavities;

h) on roofs, except where specifically determined as suitable;

i) under stairways;

j) under access walkways;

k) in any evacuation route or escape route.



Installation Space Requirements

For ease of operation and maintenance, sufficient space should be allocated around the ESS container installation location. The minimum space requirement is illustrated in the figures.

- A maintenance clearance of 3500mm shall be reserved on the container door side to facilitate future operation and maintenance (the foundation shall not extend beyond the container corner fittings).

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- A maintenance clearance of 3000mm shall be reserved on the control cabinet door side to facilitate future operation and maintenance (the foundation shall not extend beyond the container corner fittings).
- The spacing between containers placed side-by-side (left and right) shall be 50mm.
- The spacing between containers placed end-to-end (front and rear) shall be 150mm.



Figure 6-5 Multi Container Installation

*The above figures are for reference only, please refer to the actual product for details.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

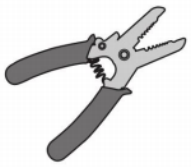
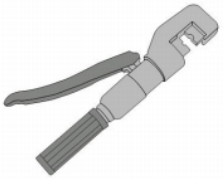
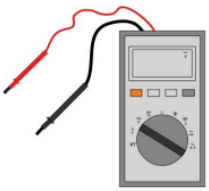
7. Product Electrical Installation

7.1 Setup Tools Preparation

Personal Protective Equipment:

		
Safety gloves	Safety shoes	Safety helmet

Tool Preparation:

		
Screwdriver	Wire stripper	Hydraulic tong
		
Heat gun	Multimeter	M10 socket wrench

 Copyright © Beijing HyperStrong Technology Co., Ltd.

7.2 Door Requirements of Boxes and Equipment Cabinets

All the keys of equipment cabinet door are located in the lower right corner area of the SCDC, placed within a built-in key box as shown in Figure 7-1.

- **Step 1** Retrieve the key of the door for SCDC from the equipment's accompanying accessories.
- **Step 2** Open the cabinet door at the lower right corner of the SCDC. The key box is located inside this cabinet.
- **Step 3** Open the key box. Keys for all fire suppression and battery container doors, as well as for the distribution boxes inside the battery container, are stored within the key box.

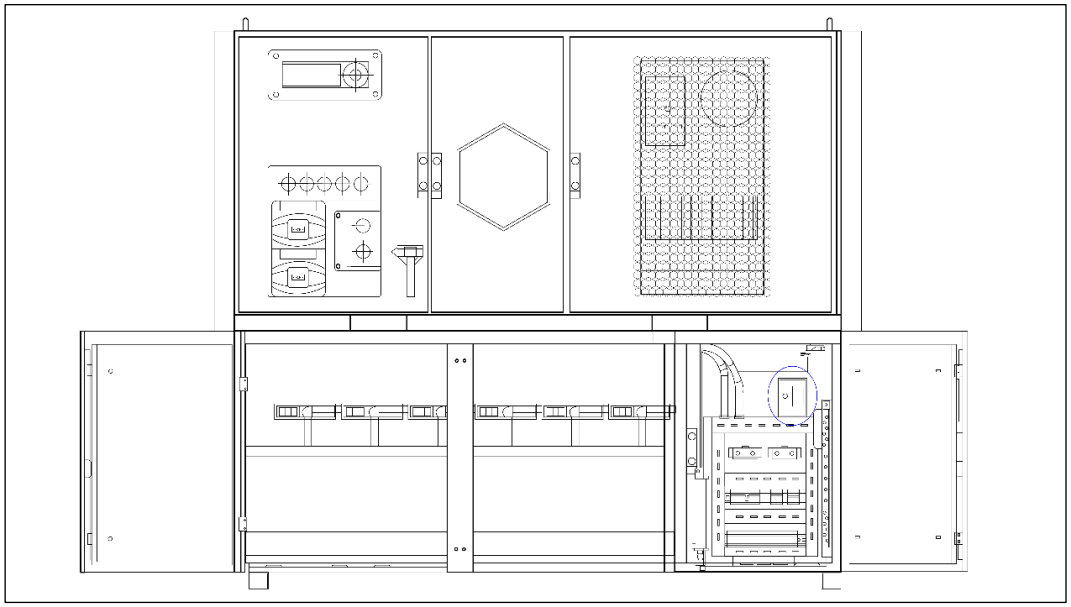


Figure 7-1 Schematic Diagrams of Key Box Location

 Copyright © Beijing HyperStrong Technology Co., Ltd.

7.3 Cable Harness Preparation

The selected cables shall meet the following conditions:

1. Adequate current-carrying capacity. The factors that affect the current-carrying capacity of the conductor include but not limited to environmental conditions, material type of conductor insulation, cable routing, cable material, and cross-sectional area.
2. The cable diameter shall be selected in accordance with the maximum current-carrying capacity, and there should be a margin for the length.
3. All DC input cables shall have the same specification and material.
4. The specifications and materials of three-phase AC output cables shall be the same.
5. Flame-retardant cables shall be used.

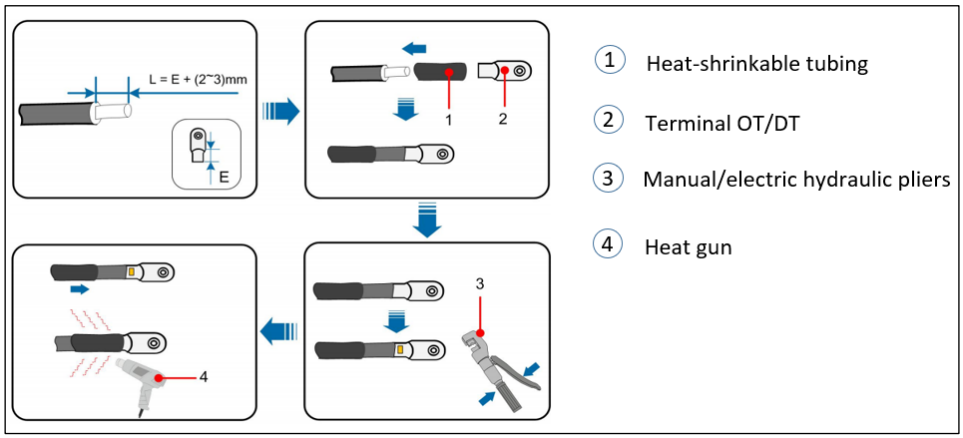


Figure 7-2 Crimping of OT/DT Terminals

Note: The DC power cables are sheathed in metal corrugated conduits. The metal corrugated conduits shall be grounded at a single end, with the grounding point located on the side of SCDC.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

7.4 DC Cable Connections

Each rack converges at the DC busbar with 4/0 AWG DC power cable, the DC lug is shown in the figure below.

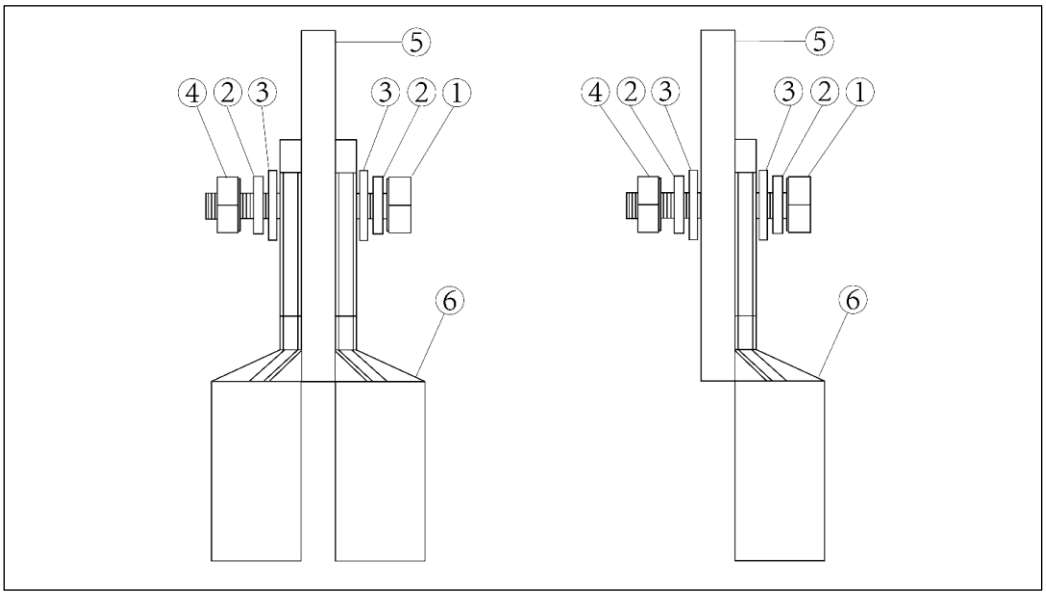


Figure 7-3 DC Connection Terminal

No.	Name	No.	Name
1	Bolt	4	Nut
2	Spring Washer	5	Copper Row
3	Flat Washer	6	Copper Tip

7.5 Cable Access Port

The control cabinet is located at the end of the container. Cables from the container are routed to the bottom of the control cabinet via cable trays or other supporting structures. The cable entry hole at the bottom of the control cabinet is protected by using a tower-type sealing ring. All cables are fed the control cabinet uniformly through this opening. After cable installation is completed, the sealing ring area shall be sealed with fireproofing mud or other suitable material.

The cable inlet of the ESS container is located at the bottom of the SCDC, and the external access cables are uniformly fed from the bottom. The inlet is composed of two pieces of detachable metal plates, which can be drilled according to the actual cable requirements. After wiring, the inlet should be tightly sealed with fireproof mud or other suitable materials.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

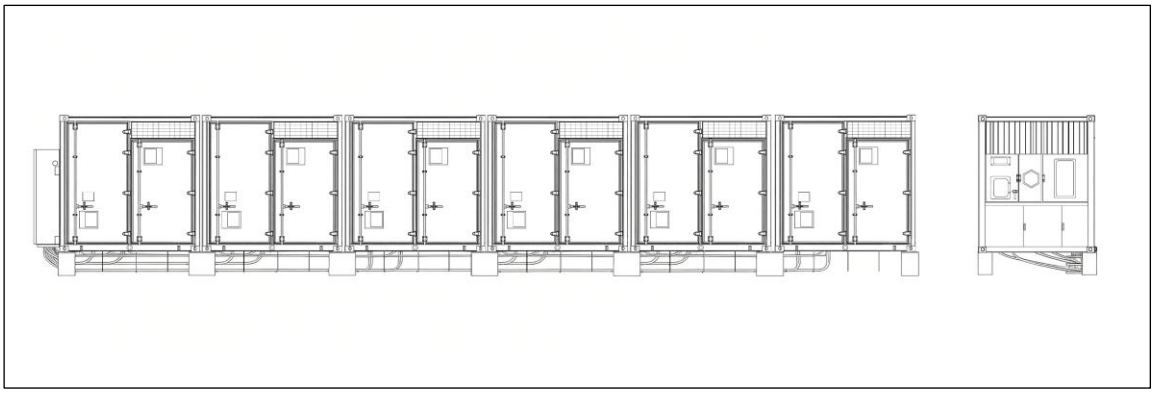


Figure 7-4 Cable Access Port

*The above figures are for reference only, please refer to the actual product for details.

7.6 Container Grounding

There are two grounding methods: welding with grounding flat steel or fixing with a grounding cable.

- Grounding flat steel: Remove the protective tape from the earthing point and weld a piece of hot-dip galvanized flat steel with a size of 50mm×100mm to the earthing point. When the welding is completed, the entire fixing surface shall be sprayed.

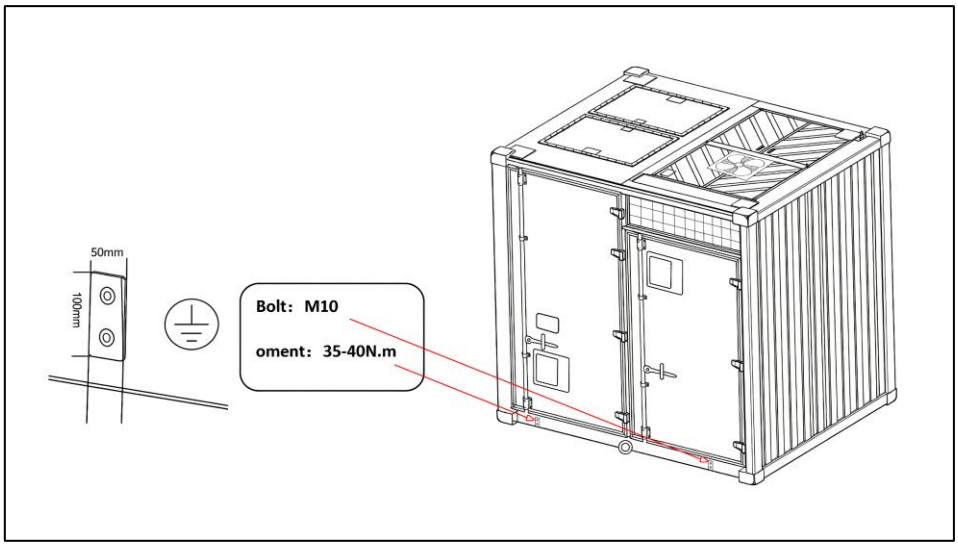


Figure 7-5 Busbar Grounding

- Grounding cable: Reliably connect the two grounding points to the ESS grounding point with a 70mm^2~95mm^2 grounding cable. (Insulating tape is attached to the factory grounding point. The type needs to be removed before wiring).

Crimp the terminals and secure the cables by using M10 bolts.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

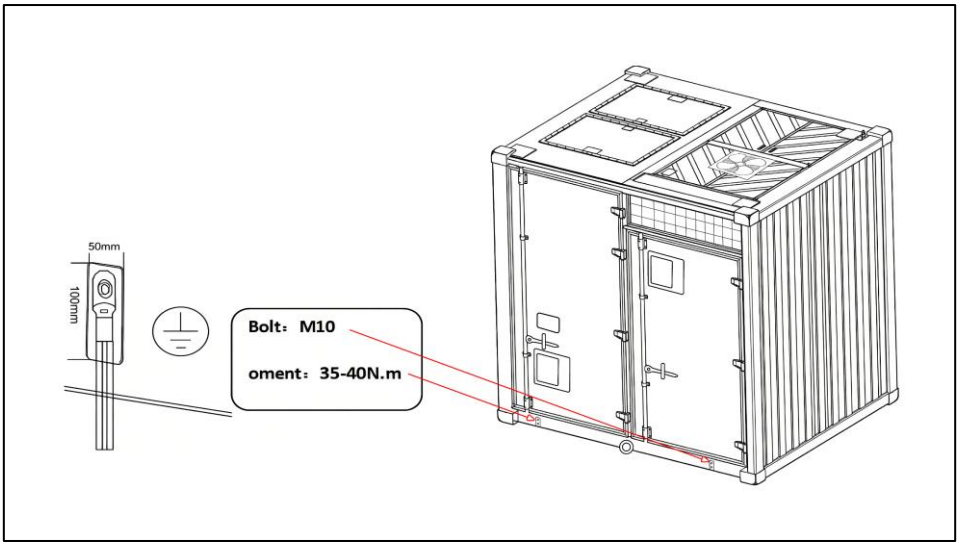


Figure 7-6 Cable Grounding

7.7 SCDC External Interfaces

DC Power Cable Interface

The locations and types of the DC power cable interfaces are as shown in Figure 7-7. The wiring position for the DC power cable is located below each branch circuit switch.

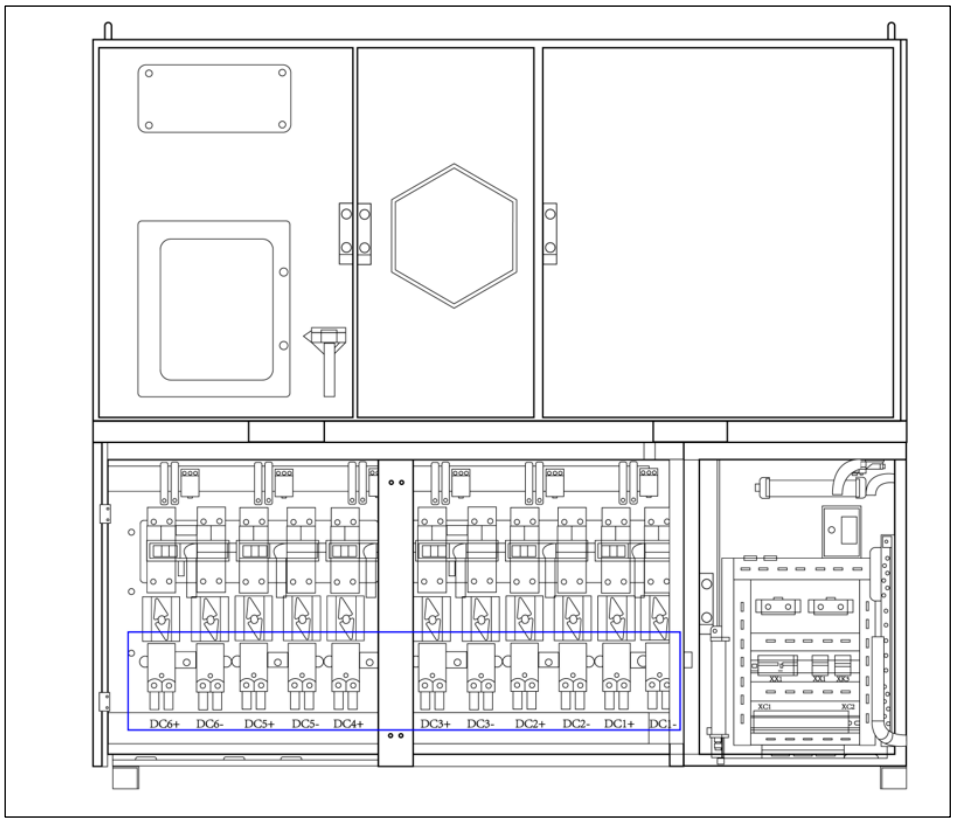


Figure 7-7 Diagram of DC Power Cable Interface

 Copyright © Beijing HyperStrong Technology Co., Ltd.

DC Power Cable Interface(The wire diameter is referenced based on copper conductors.)

Recommended Cable for Single Isolator Switch to PCS and Battery Container					
Interface Code	Rated Voltage	Connecting Interfaces	Recommended Wire Diameter	Termination Point	Terminal Specifications
QS1~QS8	1500 VDC	PCS+	2*185mm^2	DC Side of the PCS	M12 Threaded Hole in Copper Busbar
		PCS-	2*185mm^2		
QS1~QS8		P+	2*120mm^2	PDU (Power Distribution Unit) of the Battery Container	/
		P-	2*120mm^2		

Note: If the customer uses aluminum conductors, it is necessary to further verify the current-carrying capacity of the conductor, the terminal's wiring capacity, and the size of the bulkhead sealing rings.

Operation Steps:

- **Step 1** Open the safety protection panel of the busbar area.
- **Step 2** Guide the cable through the entry port into the busbar connection area and verify the cable polarity.
- **Step 3** Use wire strippers to remove the cable insulation, expose the copper core, and crimp an appropriately sized OT terminal.
- **Step 4** Secure the OT terminal to the connection point using an M16 bolt with a tightening torque of 119–140 Nm.

Caution: Before performing any equipment maintenance or construction work, it is mandatory to confirm that the equipment has been powered off and locked out.

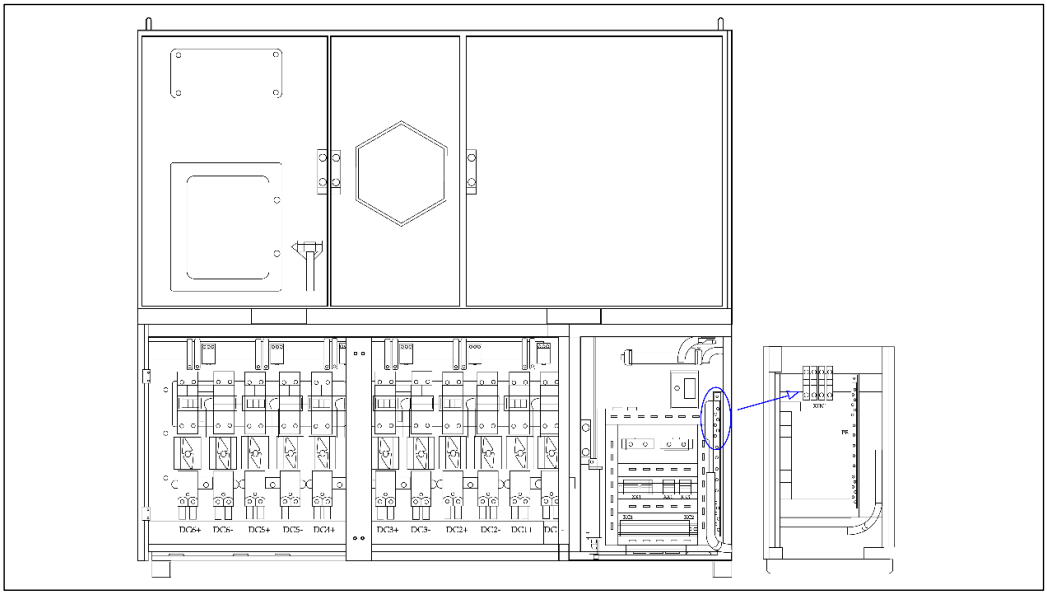


Figure 7-8 Diagram of AC Auxiliary Power Supply Interface

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Main Interface for AC Auxiliary Power Supply

As shown in the Figure 7-8, it is located on the side of the lower right of the SCDC, with the terminal block labeled as XIN1.

The terminal sequence numbers and recommended wire diameter(copper) are listed in the table below:

Connecting Terminal	Rated Voltage /Frequency	Connecting Interfaces	Recommended Wire Diameter	Terminal Specifications
XIN1	400Vac/50Hz	2	50mm^2	M8 Stud, Max. Wire Size: 70mm^2
		4	50mm^2	M8 Stud, Max. Wire Size: 70mm^2
		6	50mm^2	M8 Stud, Max. Wire Size: 70mm^2
		N	50mm^2	M8 Stud, Max. Wire Size: 70mm^2
PE	/	NE	50mm^2	M8 Threaded Hole in Copper Busbar

Note: If the customer uses aluminum conductors, it is necessary to further verify the current-carrying capacity of the conductor, the terminal's wiring capacity, and the size of the bulkhead sealing rings.

Other Power Supply and Communication Control Interfaces

As shown in Figure 7-9, in order to facilitate on-site installation and wiring, all secondary wiring points in the SCDC are located within the electrical cabinet at the lower right corner of the container. These primarily involve power supply and communication interfaces at the container and station levels. For detailed interface definitions, please refer to the following descriptions.

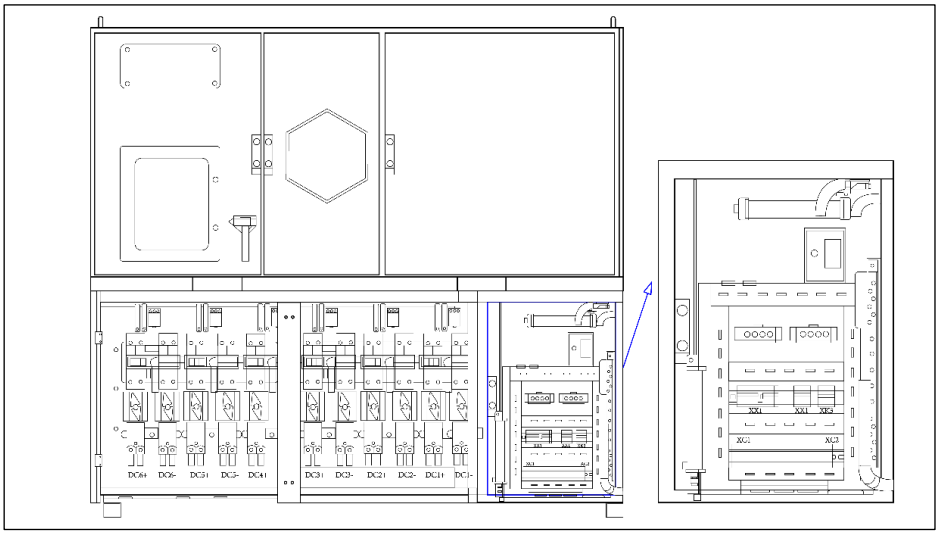


Figure 7-9 Diagram of External Interfaces for SCDC

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Battery Container Power Supply Interface and Station-Level Backup Power Interface (The wire diameter is referenced based on copper conductors.)

Connecting Terminal	Connecting Interfaces	Interface Definition	Definition Name	Definition Description	Recommended Wire Diameter
XC1	1	L1	400Vac Auxiliary Power Supply	Power Supply For #1 Container Liquid Cooling Unit	6mm^2
	2	L2			6mm^2
	3	L3			6mm^2
	4	N			6mm^2
	5	PE			6mm^2
	6	L1	400Vac Auxiliary Power Supply	Power Supply For #2 Container Liquid Cooling Unit	6mm^2
	7	L2			6mm^2
	8	L3			6mm^2
	9	N			6mm^2
	10	PE			6mm^2
	11	L1	400Vac Auxiliary Power Supply	Power Supply For #3 Container Liquid Cooling Unit	6mm^2
	12	L2			6mm^2
	13	L3			6mm^2
	14	N			6mm^2
	15	PE			6mm^2
	16	L1	400Vac Auxiliary Power Supply	Power Supply For #4 Container Liquid Cooling Unit	6mm^2
	17	L2			6mm^2
	18	L3			6mm^2
	19	N			6mm^2
	20	PE			6mm^2
	21	L1	400Vac Auxiliary Power Supply	Power Supply For #5 Container Liquid Cooling Unit	6mm^2
	22	L2			6mm^2
	23	L3			6mm^2
	24	N			6mm^2
	25	PE			6mm^2
	26	L1	400Vac Auxiliary Power Supply	Power Supply For #6 Container Liquid Cooling Unit	6mm^2
	27	L2			6mm^2
	28	L3			6mm^2
	29	N			6mm^2
	30	PE			6mm^2
	31	L	230Vac Uninterruptible Power Supply	230Vac Uninterruptible Power Supply For #1~#6 Container	2.5mm^2
	32	N			2.5mm^2
	33	PE			2.5mm^2
XC2	1	L	230Vac Uninterruptible Power Supply	277Vac Uninterruptible Power Supply from Customer or MVS	2.5mm^2
	2	N			2.5mm^2
	3	PE			2.5mm^2

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Inter-container and Station-Level Fire Suppression Communication Interfaces

Connecting Terminal	Connecting Interfaces	Interface Definition	Definition Name	Definition Description	Recommended Wire Diameter	Terminal Specifications
XX1	1	Aerosol+	Aerosol Release	#1 Container Aerosol Release Signal	1mm²	/
	2	Aerosol-			1mm²	/
	3	Aerosol+	Aerosol Release	#2 Container Aerosol Release Signal	1mm²	/
	4	Aerosol-			1mm²	/
	5	Aerosol+	Aerosol Release	#3 Container Aerosol Release Signal	1mm²	/
	6	Aerosol-			1mm²	/
	7	Aerosol+	Aerosol Release	#4 Container Aerosol Release Signal	1mm²	/
	8	Aerosol-			1mm²	/
	9	Aerosol+	Aerosol Release	#5 Container Aerosol Release Signal	1mm²	/
	10	Aerosol-			1mm²	/
	11	Aerosol+	Aerosol Release	#6 Container Aerosol Release Signal	1mm²	/
	12	Aerosol-			1mm²	/
	13	24V+	24Vdc Auxiliary Power	#1~#6 Container Fire Combustible Gas Detector Power Supply	1mm²	/
	14	24V-			1mm²	/
	15	SLC_L+_A	Fire Communication Bus	Fire Input/Output Module /Detector Communication Bus	1mm²	/
	16	SLC_L+_A			1mm²	/
	17	SLC_ Shield			1mm²	/
	18	SLC_L+_B	Fire Communication Bus	Fire Input/Output Module /Detector Communication Bus	1mm²	/
	19	SLC_L+_B			1mm²	/
	20	SLC_ Shield			1mm²	/
XX2	1	COM	Fire Fault Signal	Output Normally Open Passive Dry Contact Signal	1mm²	Push-in Terminal Block, Max. Wire Size: 4mm²
	2	NO			1mm²	
	3	COM	Gas Detector Concentration Exceeded	Output Normally Open Passive Dry Contact Signal	1mm²	
	4	NO			1mm²	
	5	COM	Gas Sensor Malfunction	Output Normally Open Passive Dry Contact Signal	1mm²	
	6	NO			1mm²	
	7	COM	Fire Level 1 Alarm	Output Normally Open Passive Dry Contact Signal	1mm²	
	8	NO			1mm²	
	9	COM	Fire Level 2 Alarm	Output Normally Open Passive Dry Contact Signal	1mm²	
	10	NO			1mm²	
	11	COM	Fan Status Signal	Output Normally Open Passive Dry Contact Signal	1mm²	
	12	NO			1mm²	
	13	COM	Fan Malfunction	Output Normally Open Passive Dry Contact Signal	1mm²	
	14	NO			1mm²	
	15	RS485A	Fire Communication Bus	Station Level Communication Bus	1mm²	
	16	RS485B			1mm²	
	17	RS485G			1mm²	
	18	RS485A	Fire Communication Bus	Station Level Communication Bus	1mm²	
	19	RS485B			1mm²	
	20	RS485G			1mm²	

 Copyright © Beijing HyperStrong Technology Co., Ltd.

PCS Fault Signal Output and Station-Level Fire Suppression Fault Dry Contact

Connecting Terminal	Connecting Interfaces	Interface Definition	Definition Name	Definition Description	Recommended Wire Diameter	Terminal Specifications
XX3	1	COM	Fault Output To PCS	Output Normally Close Passive Dry Contact Signal	1mm²	Push-in Terminal Block, Max. Wire Size: 4mm²
	2	NC			1mm²	
	3	COM	Fault Output To PCS	Output Normally Close Passive Dry Contact Signal	1mm²	
	4	NC			1mm²	
	5	COM	Fault Output To PCS	Output Normally Close Passive Dry Contact Signal	1mm²	
	6	NC			1mm²	
	7	COM	Fault Output To PCS	Output Normally Close Passive Dry Contact Signal	1mm²	
	8	NC			1mm²	
	9	COM	Fault Output To PCS	Output Normally Close Passive Dry Contact Signal	1mm²	
	10	NC			1mm²	
	11	COM	Fault Output To PCS	Output Normally Close Passive Dry Contact Signal	1mm²	
	12	NC			1mm²	
	14	24V+	24Vdc Auxiliary Power	Reserved Power Supply for HMI Debugging	1mm²	
	15	24V-			1mm²	
	18	RS485A	Serial Communication	Reserved Communication Interface for Equipment Integration	1mm²	
	19	RS485B			1mm²	
	20	RS485G			1mm²	

Inter-container and Station-level Ethernet Data Communication Interface

Connecting Terminal	Interface Definition	Definition Description
NET1	Data Network A	EMS&MVS
NET2	Data Network B	EMS&MVS
NET3	Battery Contain Network Data	#1 Battery Container
NET4		#2 Battery Container
NET5		#3 Battery Container
NET6		#4 Battery Container
NET7		#5 Battery Container
NET8		#6 Battery Container

Note: The definition of Ethernet communication interfaces is subject to adjustment based on the requirements of specific order or projects. Please refer to the relevant technical specification documents for details.

Operating Procedure (3.1-3.4):

- **Step 1** Open the door at the lower right corner of the SCDC
- **Step 2** Route the corresponding power or communication cables from the bottom of the cabinet and connect them to their designated functional terminals
- **Step 3** After completing the connections, ensure proper sealing at all cable entry points.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Note: All inter-container cables are pre-fabricated by the Company, which utilize quick-connect plugs for installation.

7.8 Container External Interfaces

The DC power cables and communication interfaces from the battery container to the SCDC are shown in Figure 7-10 below.

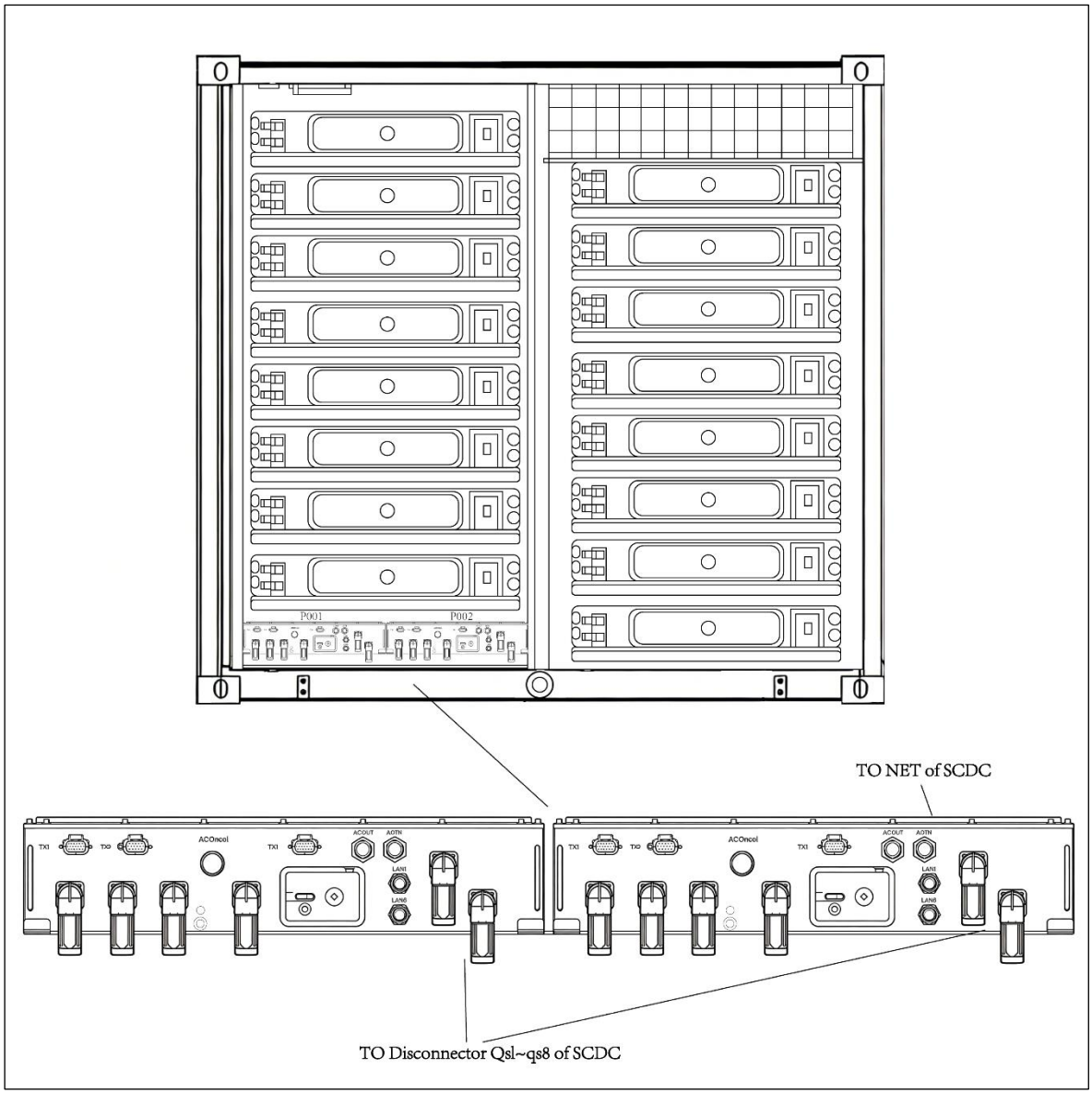


Figure 7-10 DC Power Cables and Ethernet Data Interface

The detailed interface information is provided in the table below. (The wire diameter is referenced based on copper conductors.)

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Recommended Cable Battery Container to SCDC			
Interface Code	Rated Voltage	Connecting Interfaces	Recommended Wire Diameter
PDU1~PDU2	1500VDC	P+	120mm^2
		P-	120mm^2
PDU2	/	LAN2	CAT6e

The inter-container power supply and communication connections are as shown in the figure below.

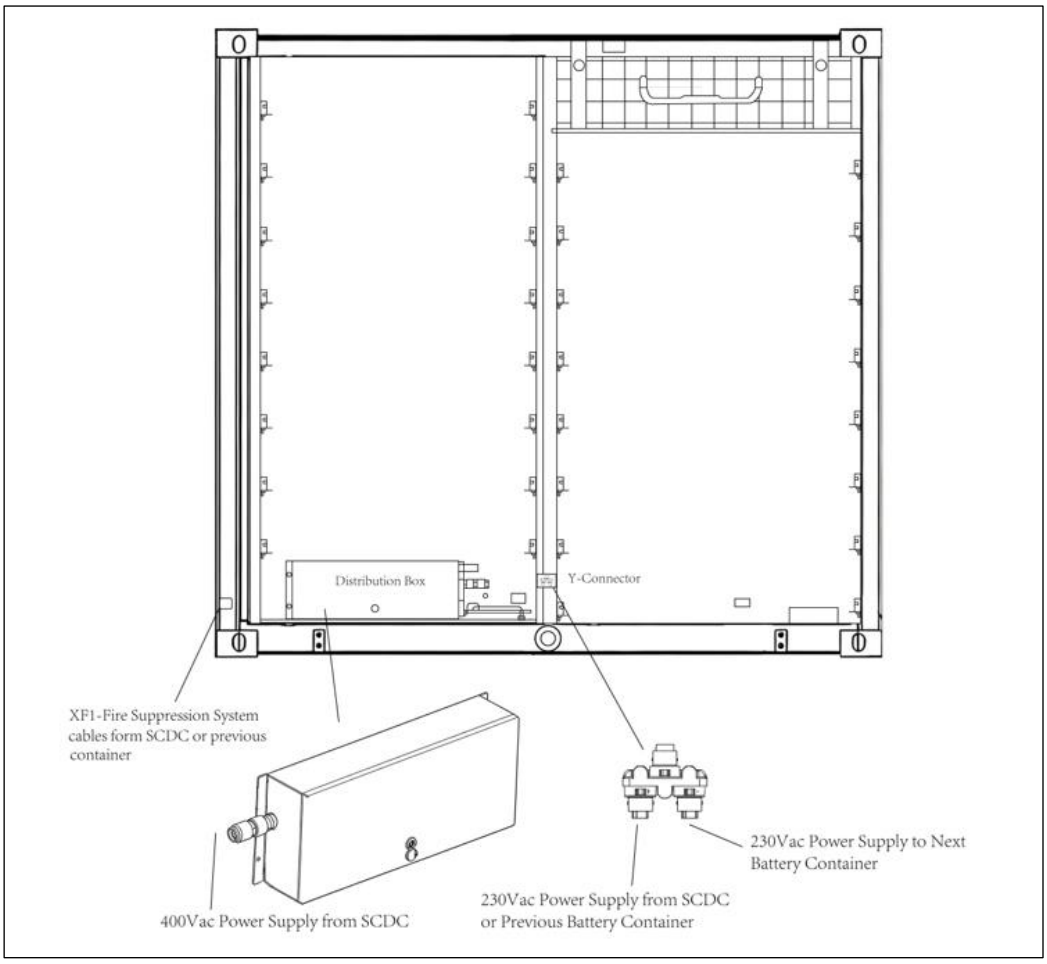


Figure 7-11 Battery Container Power Supply and Fire Suppression Communication Interface

The power supply interface specifications are provided below. (The wire diameter is referenced based on copper conductors.)

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Recommended Cable for Battery Container			
Interface Code	Rated Voltage	Connecting Interfaces	Recommended Wire Diameter
Distribution Box	400Vac	AC AUX	6mm^2
Y-Connector	230Vac	/	2.5mm^2

The fire protection communication interface specifications are provided below.

Connecting Terminal	Connecting Interfaces	Interface Definition	Definition Name	Definition Description	Recommended Wire Diameter
1	1	24V+	Fire Protection Power Supply	Fire Gas Detector Power Supply	2.5mm^2
2	2	24V-			2.5mm^2
3	3	SLC_L+_OUT	Fire Communication Bus	Fire Input/Output Module/Detector Communication Bus	1mm^2
4	4	SLC_L-_OUT			1mm^2
5	5	SLC_Shield			1mm^2
6	6	24V+	Fire Protection Power Supply	Fire Gas Detector Power Supply	2.5mm^2
7	7	24V-			2.5mm^2
8	8	SLC_L+_OUT	Fire Communication Bus	Fire Input/Output Module/Detector Communication Bus	1mm^2
9	9	SLC_L-_OUT			1mm^2
10	10	SLC_Shield			1mm^2

Operating Procedure:

- **Step 1** Open the door of battery container;
- **Step 2** Route the quick-connect plug of the pre-fabricated wiring harness from the bottom of the cabinet and connect it to the corresponding functional location;
- **Step 3** After completing the connections, ensure proper sealing at all cable entry points.

Note: All inter-container cables are pre-fabricated by the Company, which utilize quick-connect plugs for installation.

7.9 Inter-Container Cable Connection Instructions

The energy storage system consists of one SCDC and multiple BESS units. Power circuit cables and communication/signal cables are connected between the SCDC and BESS. The interconnection cables between SCDC and BESS are:

- 1500VDC Power Cable
- 400Vac auxiliary power supply (used to power the liquid cooling unit)
- 220Vac auxiliary power supply (from UPS)
- Ethernet cable
- Fire suppression system cable

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- Schematic diagrams of the control cabinet interface and container interface are detailed in the attached drawings

The customer (or EPC) is required to provide the following: 400Vac (3L+N/PE) power supply cables, station-level 230Vac power supply cables (from UPS, optional), Ethernet cables for SCDC and the station EMS, Ethernet cables between SCDC and PCS, emergency stop signal cables between SCDC and PCS, and fire alarm network cables (optional). Specific requirements are detailed in the interface diagram documentation.

The cables between the SCDC and the BESS can be supplied either by HyperStrong or by the construction contractor. Based on our estimates, the required cable list is provided in the table below. (Note**: This table provides the cable list for an energy storage system consisting of one SCDC and six BESS units. If additional BESS units are included, a separate list must be provided.**)

Table 7-1: Cable List within the Battery Energy Storage System

Main Power Cable				
No	**Name**	**Specification**	**Qty.**	**Remarks**
1	DC1500V/AC1000V 120mm^2 power cable	O-FNOG1-120-2.4m-M12-G-01	1	1#-P+
2	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-2.3m-M12-G-01	1	1#-P-
3	DC1500V/AC1000V 120mm^2 power cable	O-FNOG1-120-3.3m-M12-G-01	1	2#-P+
4	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-3.3m-M12-G-01	1	2#-P-
5	DC1500V/AC1000V 120mm^2 power cable	O-FNOG1-120-5.7m-M12-G-01	1	3#-P+
6	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-5.7m-M12-G-01	1	3#-P-
7	DC1500V/AC1000V 120mm^2 power cable	O-FNOG1-120-6.6m-M12-G-01	1	4#-P+
8	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-6.6m-M12-G-01	1	4#-P-
9	DC1500V/AC1000V 120mm^2 power cable	O-FNOG1-120-9.3m-M12-G-01	1	5#-P+
10	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-9.3m-M12-G-01	1	5#-P-
11	DC1500V/AC1000V 120mm^2 power cable	O-FNOG1-120-10.1m-M12-G-01	1	6#-P+
12	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-10.1m-M12-G-01	1	6#-P-
13	DC1500V/AC1000V 120mm^2 power cable	O-FNOG1-120-12.6m-M12-G-01	1	7#-P+
14	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-12.6m-M12-G-01	1	7#-P-

 Copyright © Beijing HyperStrong Technology Co., Ltd.

15	DC1500V/AC1000V 120mm^2 power cable	O-FNOG1-120-13.4m-M12-G-01	1	8#-P+
16	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-13.4m-M12-G-01	1	8#-P-
17	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-15.5m-M12-G-01	1	9#-P+
18	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-15.5m-M12-G-01	1	9#-P-
19	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-16.1m-M12-G-01	1	10#-P+
20	DC1500V/AC1000V 120mm2 power cable	O-FNBK2-120-16.1m-M12-G-01	1	10#-P-
21	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-18.7m-M12-G-01	1	11#-P+
22	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-18.7m-M12-G-01	1	11#-P-
23	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-19.3m-M12-G-01	1	12#-P+
24	DC1500V/AC1000V 120mm^2 power cable	O-FNBK2-120-19.3m-M12-G-01	1	12#-P-
Ethernet Cable				
No	**Name**	**Specification**	**Qty.**	**Remarks**
1	LAN CAT6a Cable	O-LAN-CAT6a-2.9m-G-01	1	SCDC-1#BESS
2	LAN CAT6a Cable	O-LAN-CAT6a-5.8m-G-01	1	SCDC-2#BESS
3	LAN CAT6a Cable	O-LAN-CAT6a-8.7m-G-01	1	SCDC-3#BESS
4	LAN CAT6a Cable	O-LAN-CAT6a-11.7m-G-01	1	SCDC-4#BESS
5	LAN CAT6a Cable	O-LAN-CAT6a-14.8m-G-01	1	SCDC-5#BESS
6	LAN CAT6a Cable	O-LAN-CAT6a-17.9m-G-01	1	SCDC-5#BESS
220V-Power Cable				
No	**Name**	**Specification**	**Qty.**	**Remarks**
1	Between AP 220V power cable	O-PDX-220V-02-4m	5	220Vac between BESS
2	AP 220V power cable 1	O-PDX-220V-1-02-2.6m	1	SCDC-1#BESS 220V-power cable
480V-Power Cable				
No	**Name**	**Specification**	**Qty.**	**Remarks**
1	AP 400V power cable 1	O-PDX-400V-1-02-	1	SCDC-

 Copyright © Beijing HyperStrong Technology Co., Ltd.

		3.3m		1#BESS 400V-power cable
2	AP 400V power cable 2	O-PDX-400V-2-02-6.1m	1	SCDC-2#BESS 400V -power cable
3	AP 400V power cable 3	O-PDX-400V-3-02-9.1m	1	SCDC-3#BESS 400V -power cable
4	AP 400V power cable 4	O-PDX-400V-4-02-12.1m	1	SCDC-4#BESS 400V -power cable
5	AP 400V power cable 5	O-PDX-400V-5-02-15.2m	1	SCDC-5#BESS 400V -power cable
6	AP 400V power cable 6	O-PDX-400V-6-02-18.3m	1	SCDC-6#BESS 400V -power cable
Fire Suppression System Harness				
No	**Name**	**Specification**	**Qty.**	**Remarks**
1	Fire Protection Cable 1	O-XF1-02-1.5m	1	SCDC-1#BESS FSS cable
2	Between Fire Protection cable	O-XF-01-4m	5	FSS cable n#BESS-n+1#BESS
3	Fire Protection Cable 4	O-XF4-02-16.8m	1	6#BESS-SCDC FSS cable

A suggested cable laying scheme is illustrated in the figure below. As shown in the figure, the ESS is equipped with cable fixing brackets at the base. These brackets can be easily installed underneath the ESS. Cables between the ESS and the SCDC can be routed along these cable brackets and secured. In actual on-site operations, the construction team should retrieve the pre-supplied cables according to the labelling codes and connect them following the designated codes and routing paths. After the connections are completed, the cables should be placed on the cable brackets and secured using cable ties.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

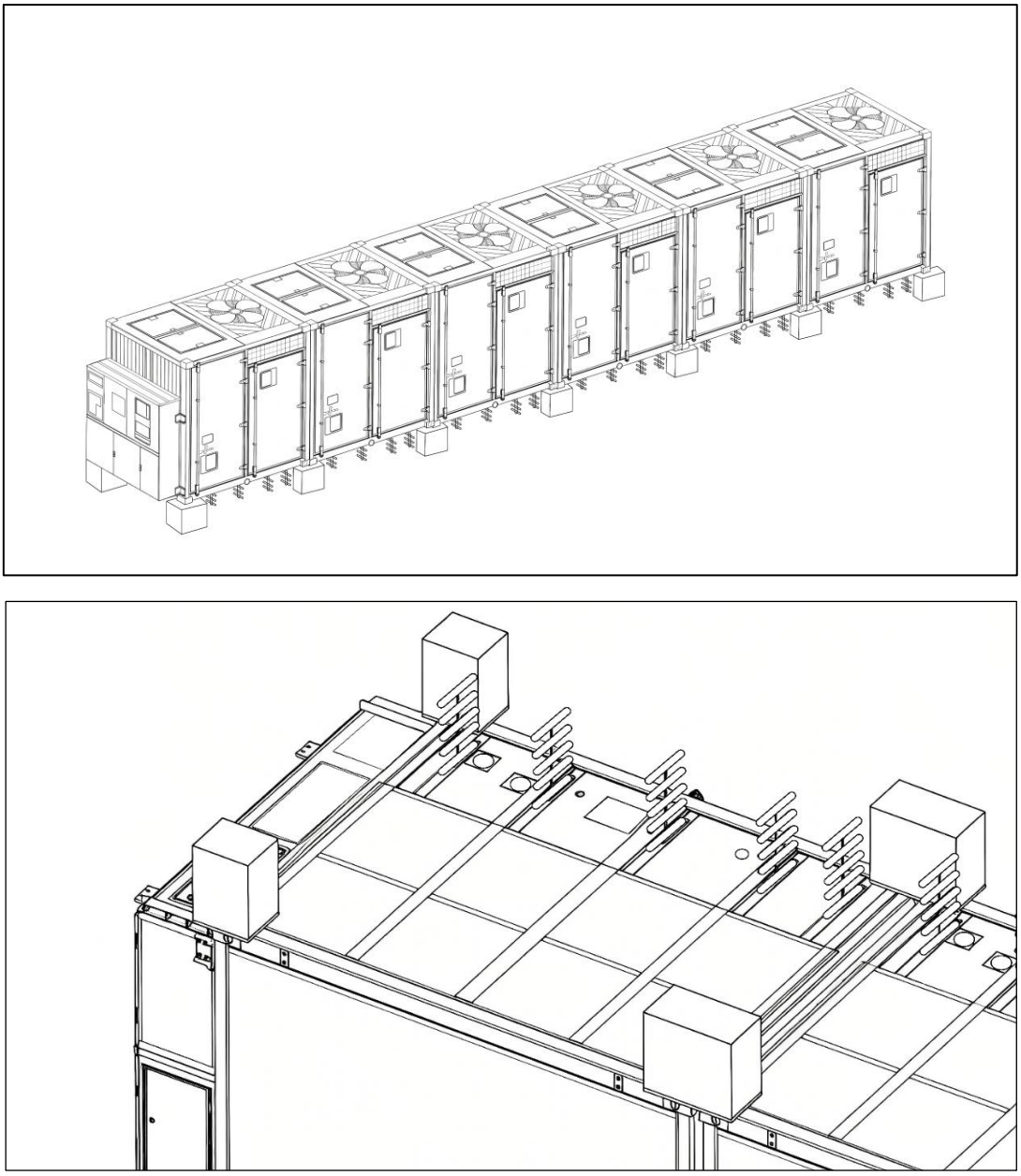


Figure 4 Schematic Diagram of Another Cable Laying Solution

 Copyright © Beijing HyperStrong Technology Co., Ltd.

7.10 UPS Cautions

⚠ CAUTION

Beware of Electric Shock, Explosion or Arcing

- The installation, operation, repair, and maintenance of electrical equipment shall be carried out by qualified personnel.
- Appropriate personal protective equipment shall be worn and the electrical safe operating practice shall be complied with.
- All power supplies of the UPS system shall be disconnected when operating or maintaining equipment.
- Dangerous voltages shall be checked across all terminals, including protective grounding before operating the UPS system.
- The UPS is equipped with an internal power supply. Hazardous voltages may still exist even when the utility power is disconnected. Before installing or servicing the UPS system, please ensure that the unit is in a state of “OFF” and the utility power and battery are disconnected. Please wait five minutes for the capacitors discharge before turning on the UPS.
- The UPS system shall be isolated from the upstream power source by a disconnecting device (disconnector or switch) that complies with local regulations. The disconnecting device shall be installed at a conspicuous and easily accessible location.
- The UPS shall be properly grounded. Due to the possibility of a heavy leakage current, the ground wires shall be connected first.

Failure to follow the instructions may result in severe consequences such as personal injury or death.

⚠ CAUTION

Beware of Equipment Damage

- **Do not** turn on the battery switch until the system is ready to be energized. Avoid over-discharging the battery.
- Adhere to the charging requirements, the storage time of the battery shall not exceed 6 months. If the UPS system remains powered off for an extended period, it is recommended to

 Copyright © Beijing HyperStrong Technology Co., Ltd.

charge the battery of UPS system at least once per month for 24 hours per charge. This periodic charging helps prevent irreversible damage.

Failure to follow the instructions may result in irreversible damage to the UPS.

7.11 Power on/off Process

⚠ CAUTION

Electrical Injuries and Arc flash

- Operators shall hold the necessary legal qualifications.
- Personnel should wear personal protective equipment as required during operation.
- If any problems that may affect the battery or ESS are recognized, please contact after-sales support immediately. Unauthorized modifications to the internal equipment of the battery system shall be strictly prohibited.
- If any abnormalities are observed in the operations of storage battery system, please stop the operation immediately and shut the system down.
- For BESS with long downtime, please perform a **comprehensive and meticulous inspection** to ensure all system indicators meet operation requirements before powering up.

Checks before Powering On:

Before powering up, please double-check the following items

- Verify that the wiring is correct.
- Confirm that the main circuit wiring is securely tightened.
- Ensure that the protective cover in the equipment is firmly installed.
- Confirm that the emergency stop button is released.
- Check and ensure that there is no ground fault.
- Check with a multimeter and ensure that the AC and DC side voltages meet the startup conditions and confirm that there is no overvoltage risk.
- Check and ensure that there are no tools or parts left in the unit.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- Verify that all air inlets and outlets are free of foreign objects or blockages.

Power on Process

System Power-On Procedures

Complete Power-On of Control Busbar Cabinet

1. Close the busbar isolation switches QS1 through QS6 in sequence.
2. Close the main AC power supply switch QF1 of the Control Busbar Cabinet.
3. Close the battery container power supply switches QF2 through QF7 in sequence.
4. Close the control cabinet air conditioning power supply switch QF9.
5. Close the socket outlet power supply switch QF10.
6. Close the cabinet interior lighting and fan switch QF11.
7. Close the power supply switch of fire protection control panel QF12.
8. Close the UPS input and output side switches QF13/QF21/QF22 in sequence, then turn on the UPS battery switch and press the UPS output start button; Where switch QF14 is the externally reserved UPS input power, which is mechanically interlocked with switch QF13. Select the power supply for the UPS input side according to the actual situation of the site.

Battery Container Power-On

1. Open the distribution box of the battery container.
2. Close the miniature circuit breakers QF1 through QF4 in sequence.
3. Press the "DC Control" power button on the high-voltage box panel.
4. Repeat the above three steps sequentially to complete the power-on for each battery container.

System power-on is now completed.

Power off Process:

System Power-Off Procedures

Battery Container Power-Off

1. Open the distribution box of the battery container.
2. Open the miniature circuit breakers QF1 through QF4 in sequence.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

3. Press the "DC Control" power button on the high-voltage box panel.
4. Repeat the above three steps sequentially to complete the power-off for each battery container.

Control Busbar Cabinet Power-Off

1. Open the UPS input and output side switches QF13(QF14)/QF21/QF22 in sequence, press the UPS stop button, and then open the UPS battery switch.
2. Open the fire protection control panel power supply switch QF12.
3. Open the cabinet interior lighting and fan switch QF11.
4. Open the socket outlet power supply switch QF10.
5. Open the control cabinet air conditioning power supply switch QF9.
6. Open the battery container power supply switches QF2 through QF7 in sequence.
7. Open the main AC power supply switch QF1 of the Control Busbar Cabinet.
8. Open the busbar isolation switches QS1 through QS6 in sequence.

System power-off is now completed.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

8. Human Machine Interface (HMI)

This section organizes and elaborates the touchscreen program of the ESS control system, which includes primary data display and essential function operation to assist developers and O&M personnel in understanding the system. The target is to facilitate more effective system maintenance and operation.

8.1 Touchscreen Content and Structure

The ESS touchscreen program consists of six key sections:

- System operation information
- Alarm information
- Event recording
- System diagram
- User management
- System settings

8.2 Touchscreen Program

The touchscreen provides information about the operation of the main equipment of the energy storage control system and facilitates the issuance of control instructions to the system. The connection between the ESS controller and the touchscreen is established via a network cable. The ESS controller transmits the real-time information of each equipment to the touchscreen and the touchscreen sends control instructions to the ESS controller, which then communicates them to the appropriate equipment.

8.3 Basic Environment Requirements for Touchscreen

Equipment Environment Requirements

Requirement Items		**Parameters**
Altitude (m)		≤ 3000m
Installation Location		In an outdoor container
Operating ambient temperature		-20 ~ 45℃
Seismic resistance	Horizontal acceleration	0.2
	Vertical acceleration	0.1
Relative air humidity	The average daily relative humidity shall be ≤ 95%. The average monthly relative humidity shall be≤ 90%. No condensation shall exist in the equipment.	

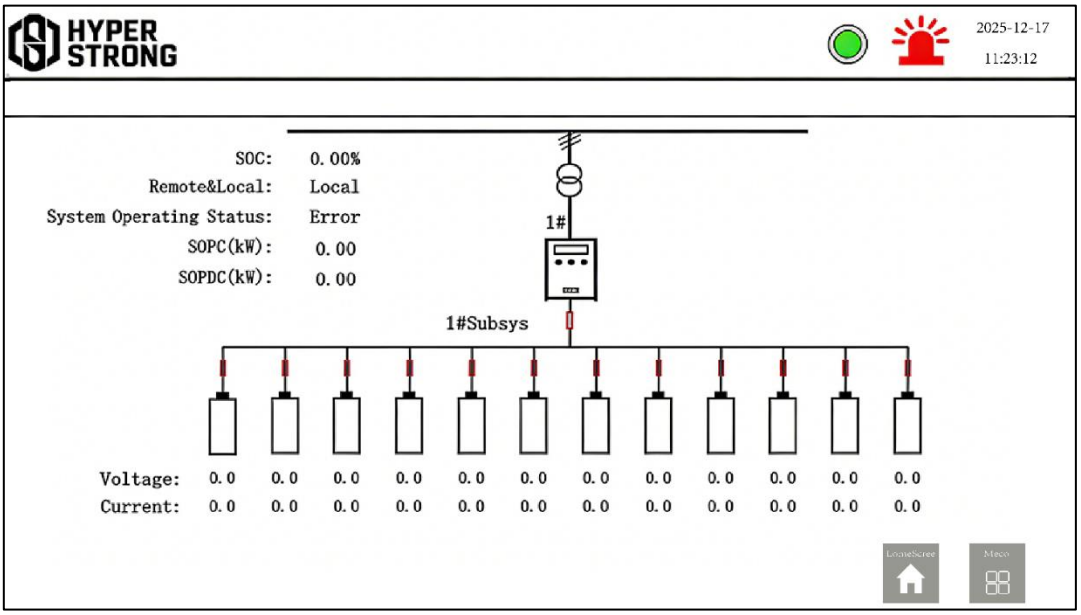
 Copyright © Beijing HyperStrong Technology Co., Ltd.

8.4 Touchscreen Basic Functions

Touchscreen Main Page

The main page of the touchscreen serves as the central hub for system operation and monitoring. It provides the following items:

- A real-time overview of the system performance.
- Quick access to critical features such as alarms, events, and system configurations.
- Easy navigation to other sections for detailed information or adjustments



Once the program is in operation, the first screen displayed is the main page, which provides key information about the ESS control system. This information includes the following items:

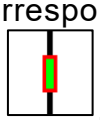
- Subsystem SOC;
- Remote and local states;
- Working states;
- Maximum charging power and discharging power
- Voltage and current of the rack

The topology diagram that centrally locates on the main interface, can visually represent the control structure of the entire ESS. Key elements of the topology diagram include: SCDC and relay switches state:

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- When the interface icon shows  , it indicates that at least one relay of the main positive and main negative relays of the battery rack is disconnected.

- When the icon changes to  , it indicates that both the main positive and main negative relays are closed.

- When all the relay switches of the subsystem are closed. The corresponding busbar will be closed, and the icon  will change to  .

- The icon  indicates that the basic state of the SOC of the rack.

At the top right of the main interface, the following additional information is displayed:

- Communication state of the PLC.
- Alarm display of the ESS.
- Current date and time.

A real-time alarm scrolling display of the ESS is located in the centre of the main interface to ensure that critical alerts are prominently visible.

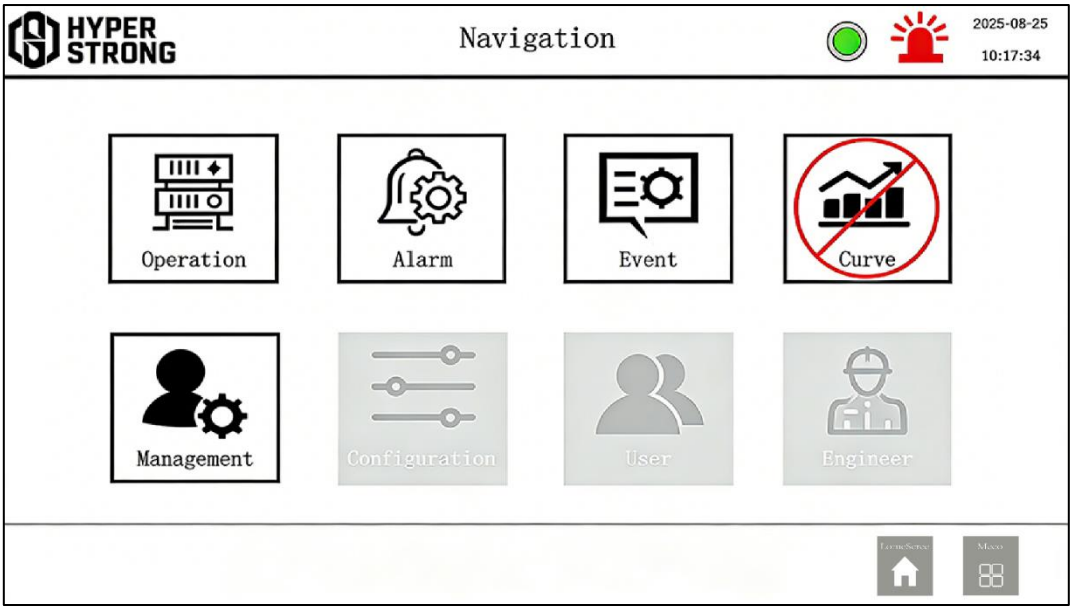
Navigation:

At the bottom right corner of the main interface, users can switch between the Home Screen and Navigation Screen by using 

or 

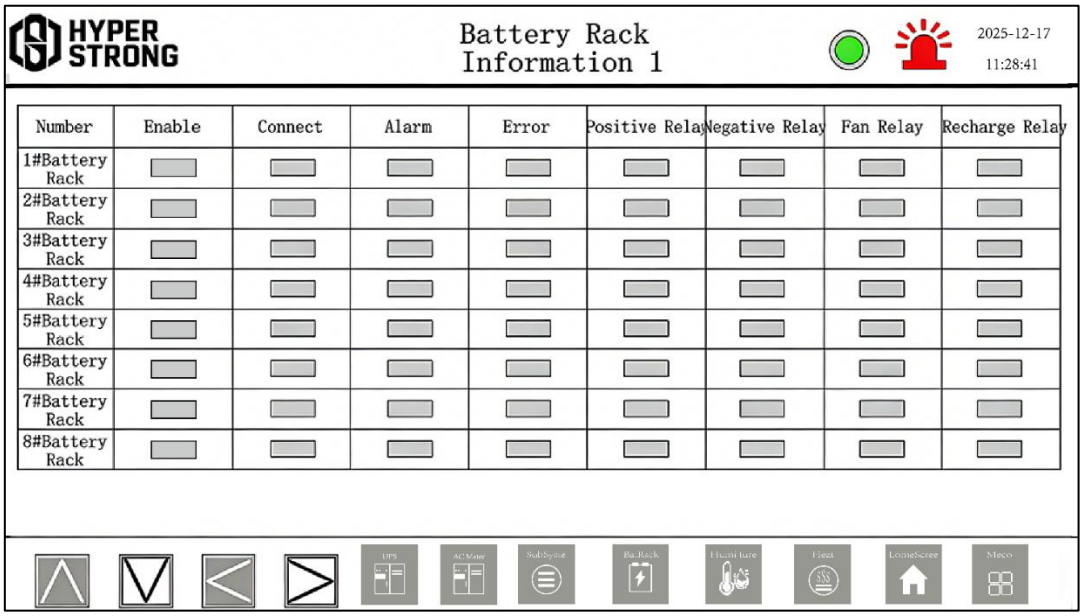
buttons.

 Copyright © Beijing HyperStrong Technology Co., Ltd.



Rack Information Interface

When the user clicks on "Operation" from the main page, the system goes to the Rack Information Interface, which displays detailed operational data for each rack in the BCMU.

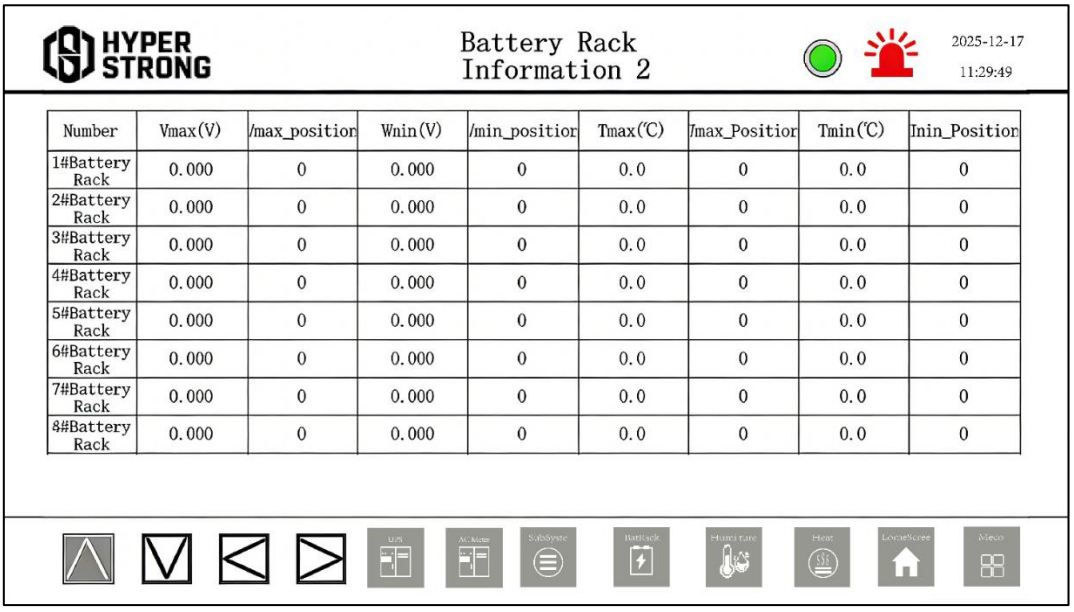


When entering the rack information interface, operators will see the current status of the racks, including “Enable”, “Connect”, “Alarm”, “Error”, “Positive Relay”, “Negative Relay”, “Fan Relay”, and “Pre-charge Relay”.

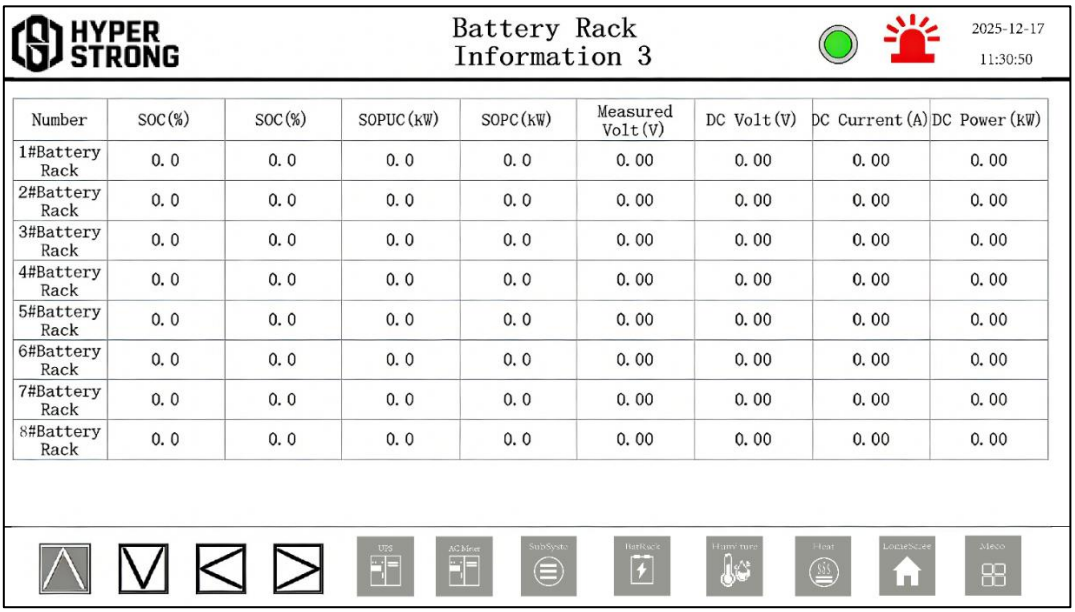
Click on  , it will enter Battery Rack Information 2, Battery Rack Information 3, etc.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Click on  to return to the previous page. Operators can click on  and  to enter any interface to check the operation information of each rack of batteries.



Number	Vmax(V)	/max_position	Vmin(V)	/min_position	Tmax(℃)	/max_Position	Tmin(℃)	Inin_Position
1#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
2#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
3#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
4#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
5#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
6#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
7#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
8#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0



Number	SOC(%)	SOC(%)	SOPUC(kW)	SOPC(kW)	Measured Volt(V)	DC Volt(V)	DC Current(A)	DC Power(kW)
1#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
2#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
3#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
4#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
5#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
6#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
7#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
8#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00

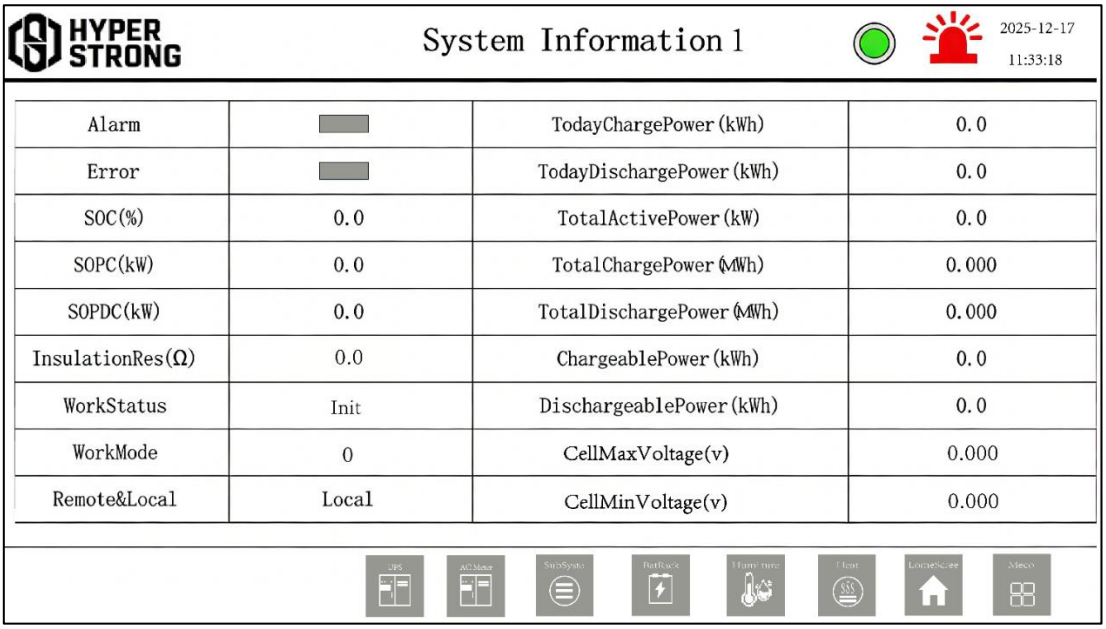
Click on 

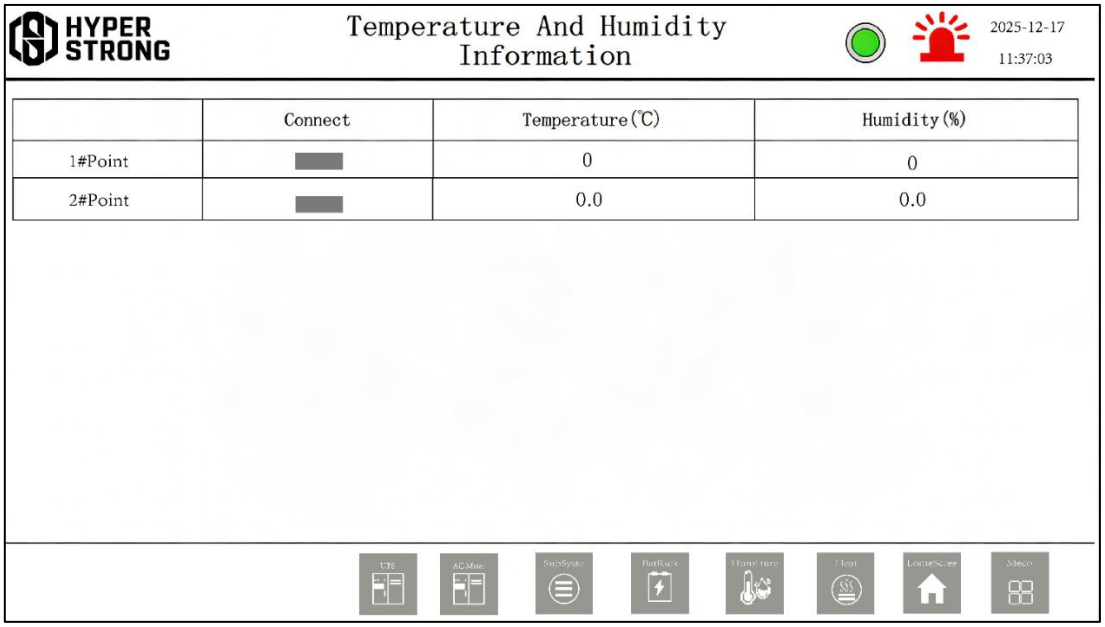
under the battery rack information, it will navigate to the System Information Interface; while clicking on 

, it will go to the Temperature &

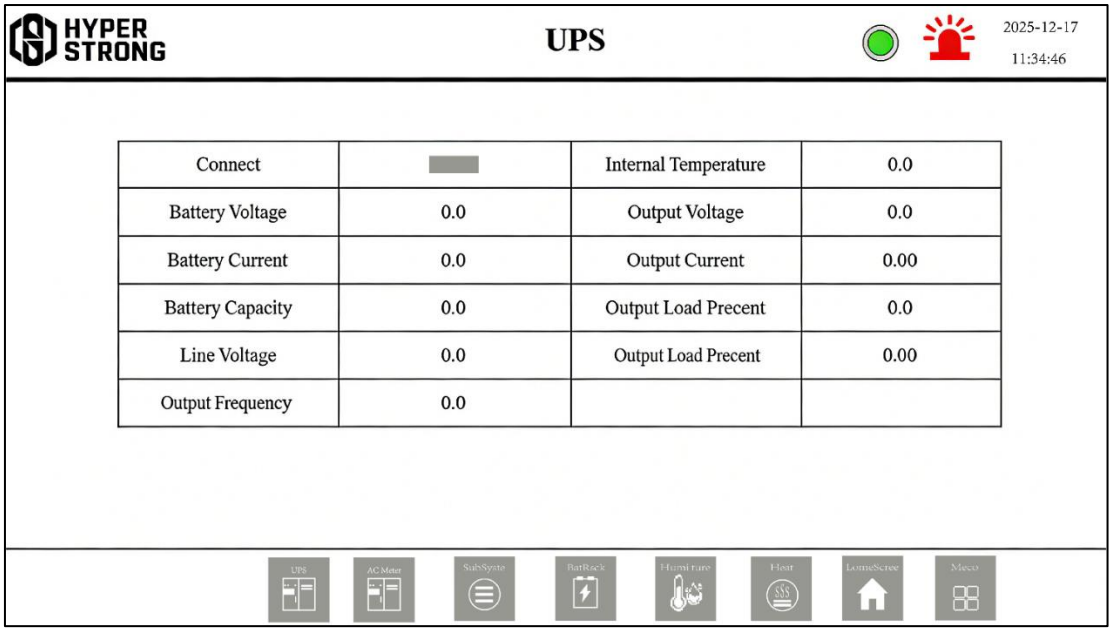
 Copyright © Beijing HyperStrong Technology Co., Ltd.

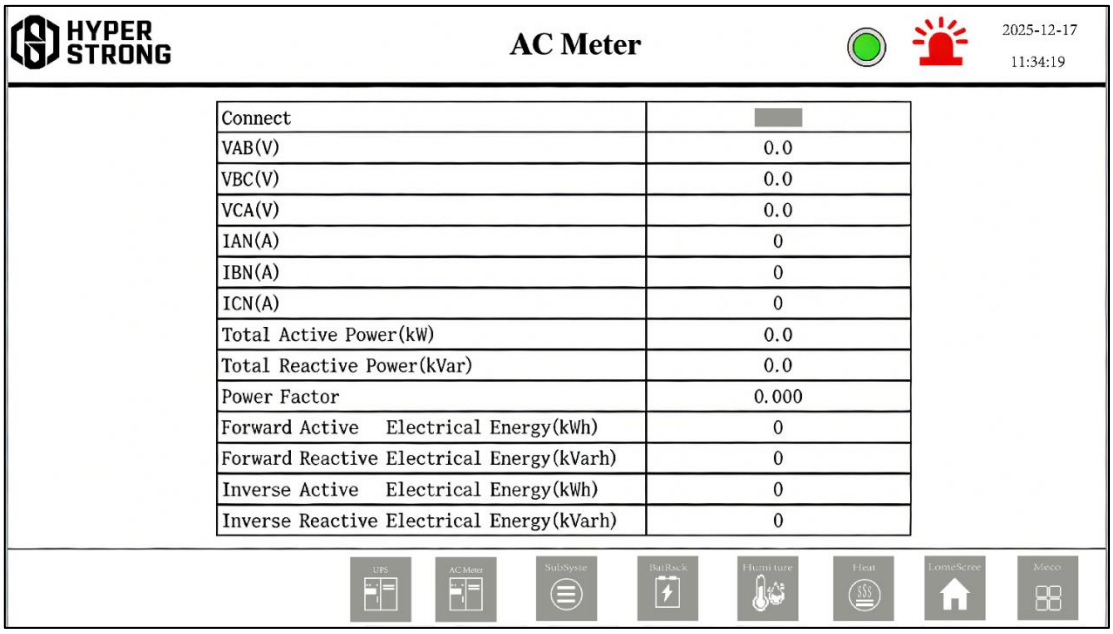
Humidity Information Interface. Operators can also click the icon to review the operation information of each part of the BMS.



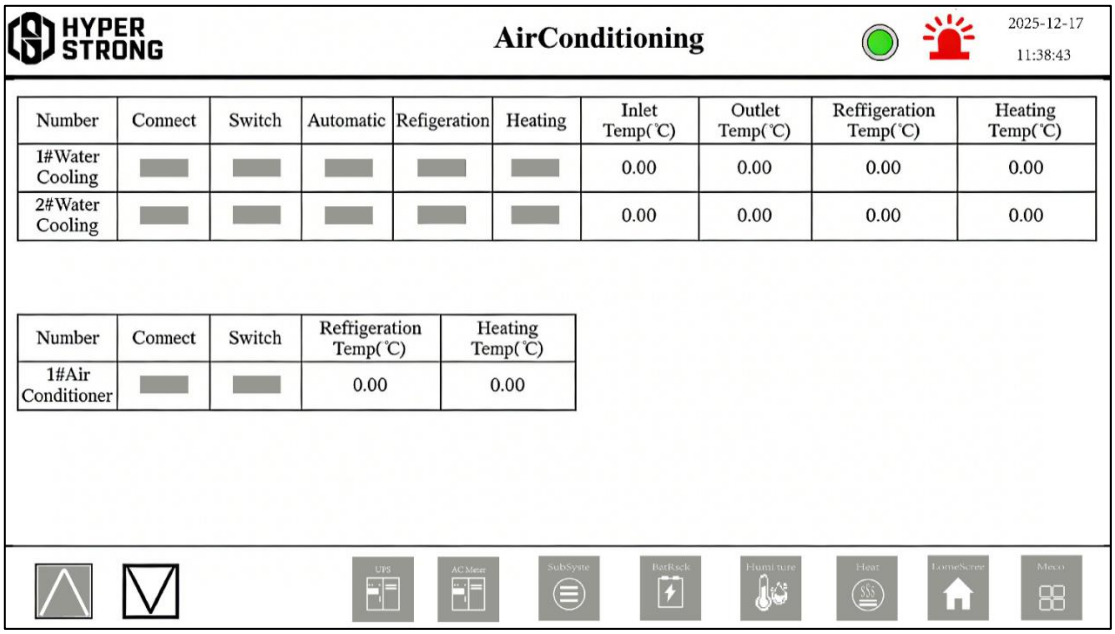


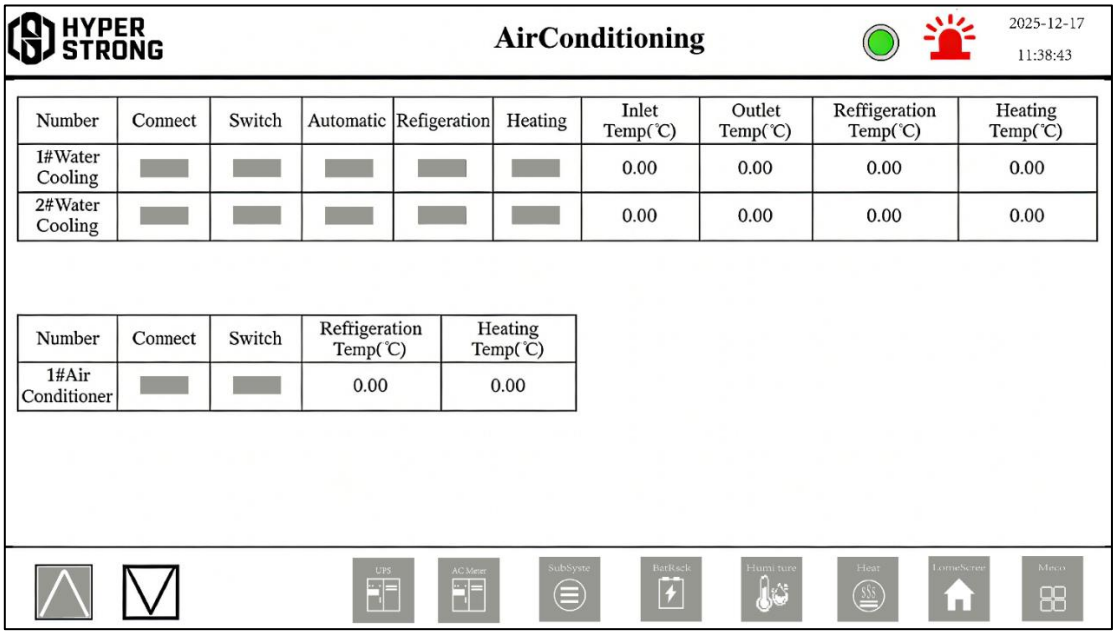
 Copyright © Beijing HyperStrong Technology Co., Ltd.





 Copyright © Beijing HyperStrong Technology Co., Ltd.





Alarm Interface

In the main interface or other interfaces, click on 

to enter the menu. Click “Alarm” to enter the real-time alarm interface and check real-time alarm.

Click on 

to enter the historical alarm information interface and check historical data.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Click on 

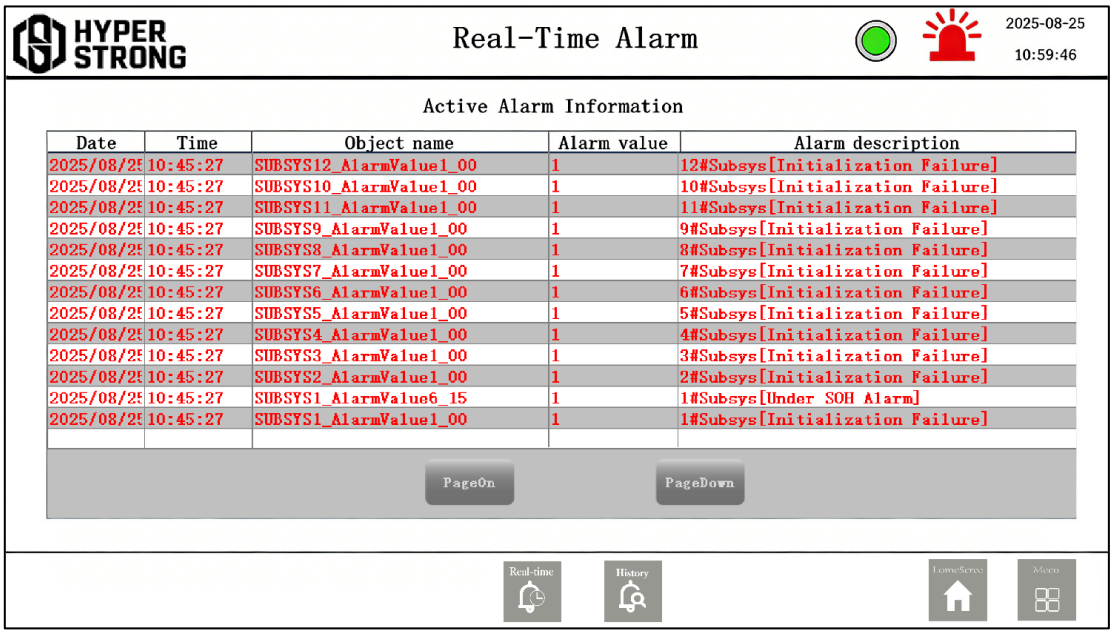
to enter the real-time alarm information interface and check real-time alarms.

The main alarm interface information consists of two sections:

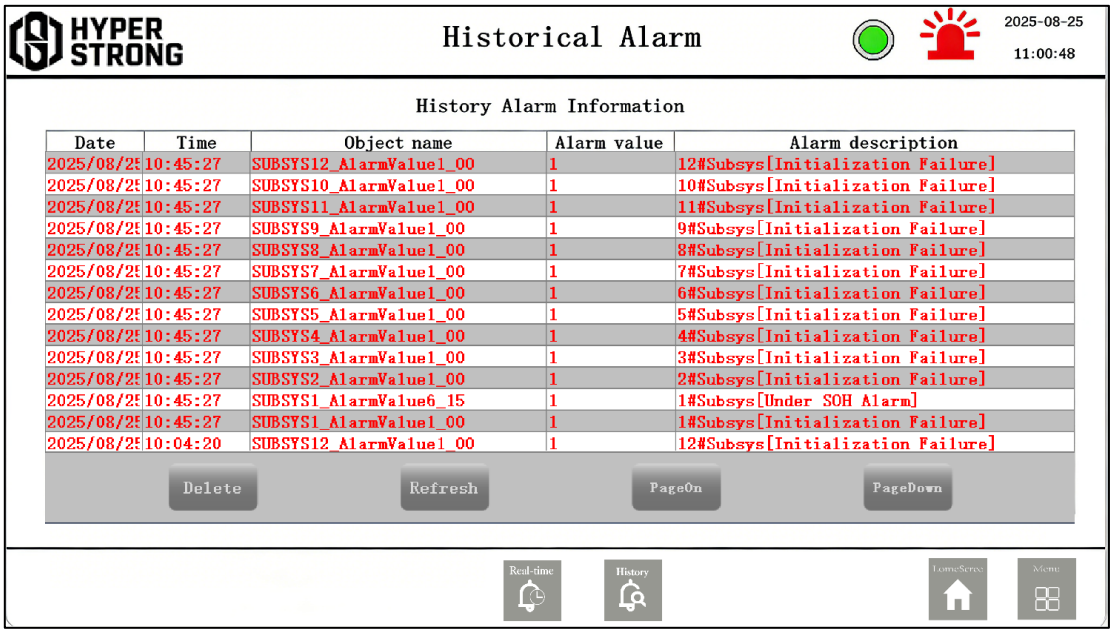
1. Real-time and a historical alarm information display (the alarm information in the picture is only for reference).
2. Historical Alarm information Display

The real-time alarm interface shows up to 14 alarm messages at a time. Users can navigate through additional alarm messages by turning page up or down.

The historical alarm interface also displays up to 14 alarm messages at a time. Users can execute operations such as deleting historical alarm messages, refreshing historical alarm messages, and navigating through messages by turning page up or down.



 Copyright © Beijing HyperStrong Technology Co., Ltd.

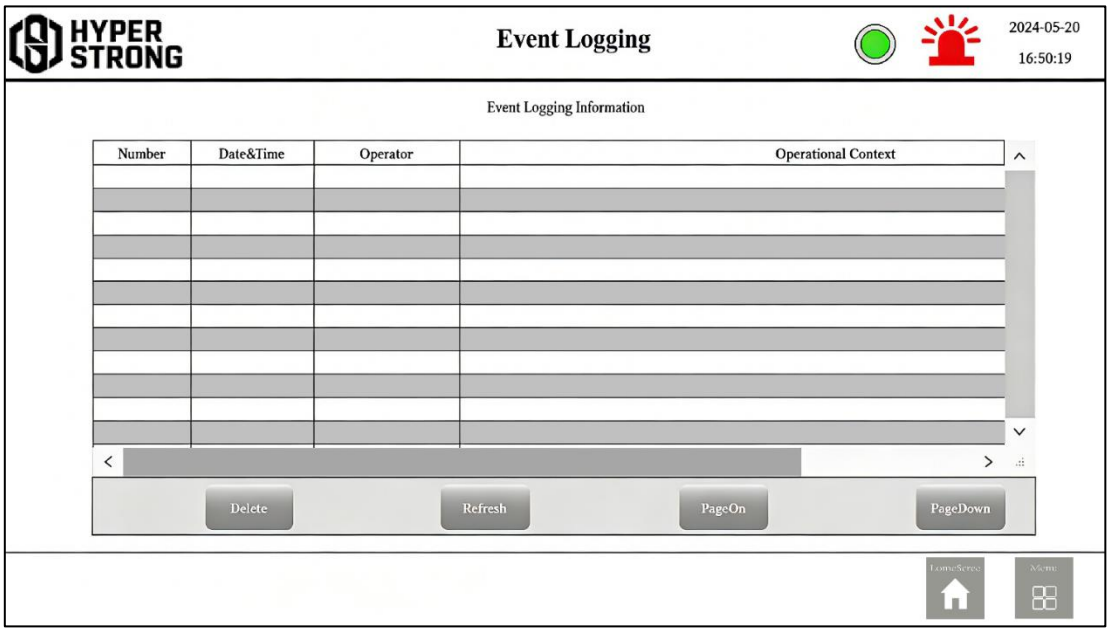


Event Logging Interface

Click on 

of the main page or other pages to access the Menu. Click “Event Logging” to enter the corresponding section and review operation records.

The event logging interface includes the Date & Time, Operator, and Operational Context. Additionally, operators can delete records, refresh the page, and navigate up and down the page.

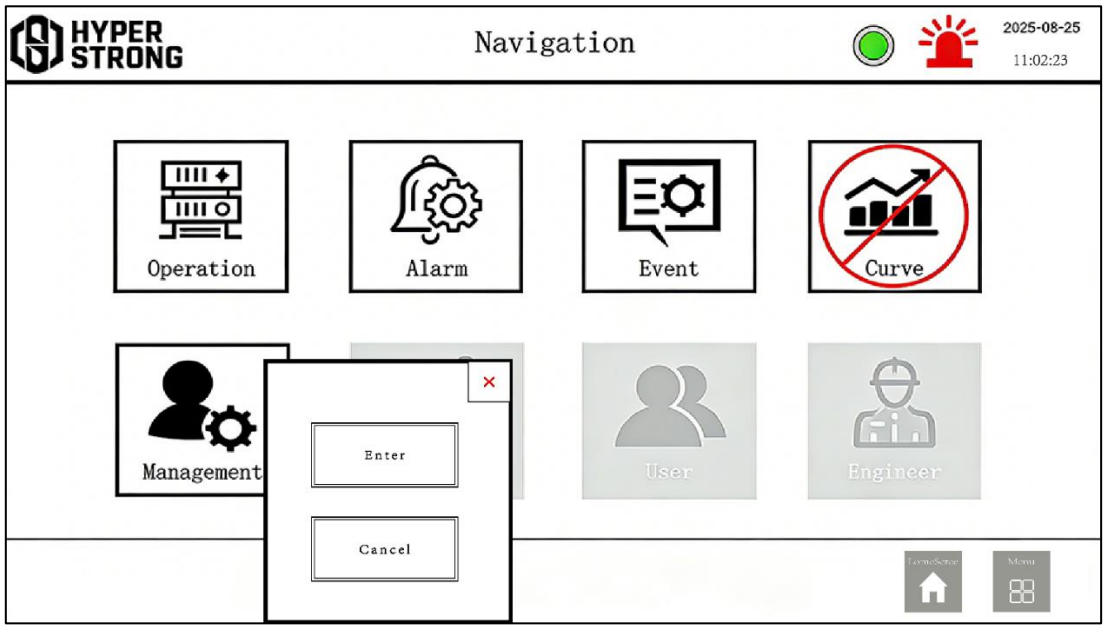


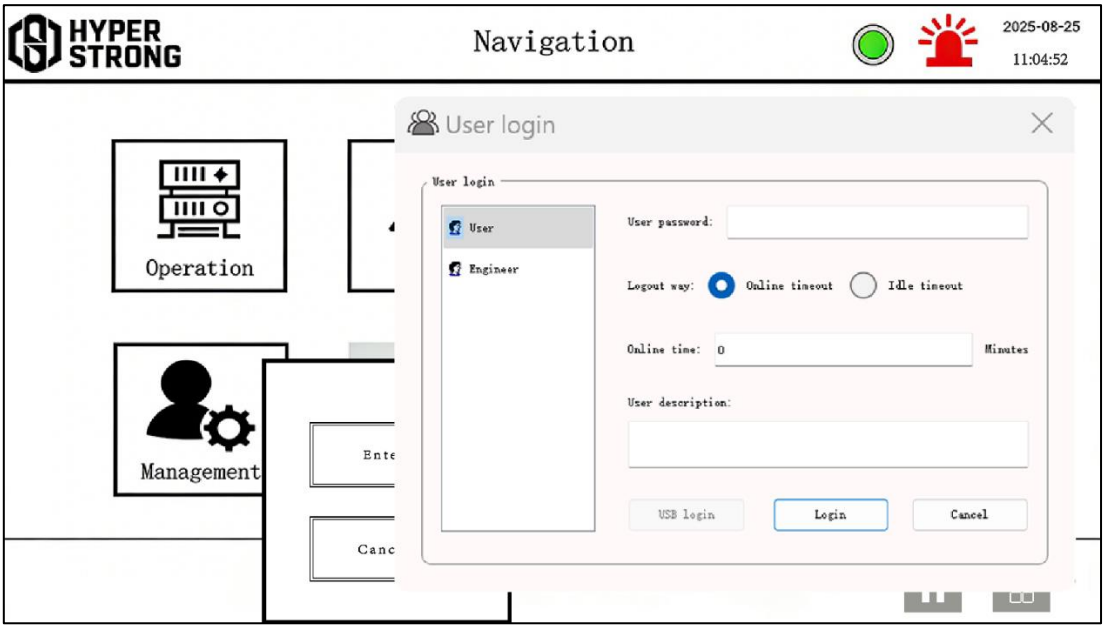
 Copyright © Beijing HyperStrong Technology Co., Ltd.

User Management Interface

Click on 

and enter the Menu, click “User Management,” and the “Login” interface will pop up in the main interface or other interfaces. Select “User” or “Engineer”, enter the password, and click “Login” to complete the corresponding operation. If you log in as an engineer, the “Engineer Operation” button in the menu interface will be enabled.

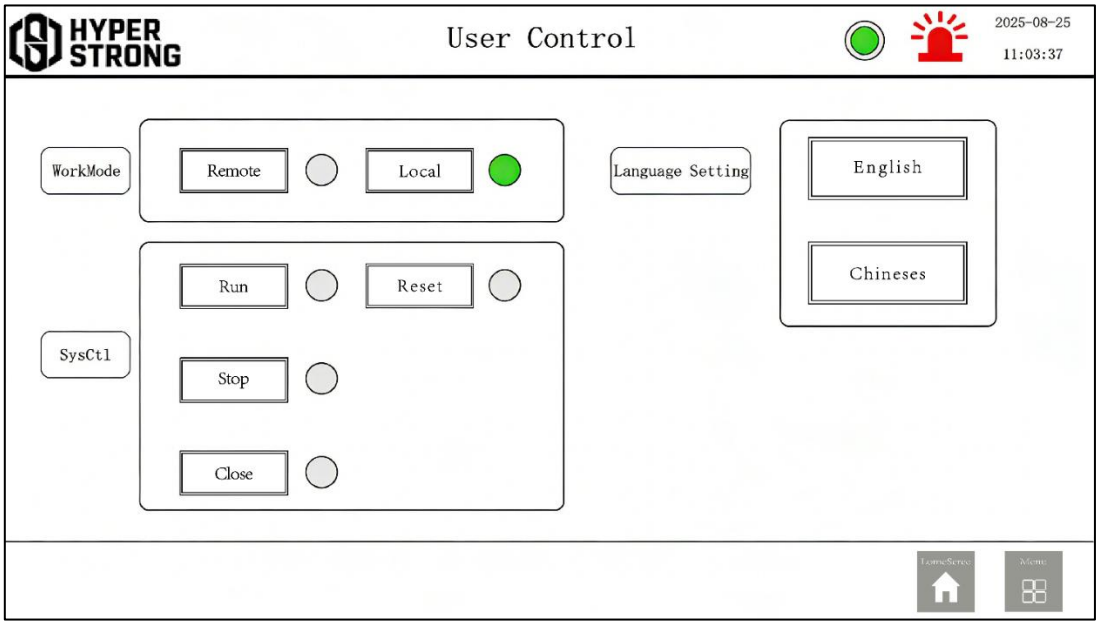




Click on “User” to enter User Control. This interface's main functions include subsystem run/shutdown/reset control, remote/local mode switching, and

 Copyright © Beijing HyperStrong Technology Co., Ltd.

subsystem selection/operation. (Contact the operation and maintenance personnel of HyperStrong to get the password).



The user can select the mode based on the situation and click on Run depending on the operation condition; then, the corresponding subsystem will turn to the running state. Click on Stop, the corresponding subsystem will be shut down. Click on Close, the corresponding subsystem will be turned off. If a subsystem enters a fault-recoverable state and needs to reset, click on Reset, and the system reset instruction will be successfully issued.

8.5 Communications

The HMI has a network cable interface and is connected to BAMU-PLC via a network cable. Modbus TCP/IP transmission protocol is used for data transmission.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

9. Emergency Operation and Accident Handling

The faults of the ESS primarily include battery faults, control system faults, equipment faults and physical damage. For common faults, please refer to the maintenance manual. This chapter will address the emergency faults, such as fire and electric leakage.

If there is any uncertainty about identification or handling of abnormal conditions, please contact the relevant technical or after-sales service department of the Company for professional technical support.

When contacting us, please provide the following information, which will help you root the cause and clear the fault quickly and accurately:

- Equipment serial number, production date and software version
- Fault information and brief description
- Photos of faulty parts
- Information about the manufacturer, model, and configuration of the external devices connected to the energy storage system (e.g., transformers, upstream circuit breakers, and cable models).

9.1 Emergency Handling

If any of the following conditions (including but not limited to) occur, immediately stop using the ESS container and the adjacent battery container to prevent electric shock, fire, or explosion.

In Case of Fire:

In case of electrical equipment catches fire, please immediately disconnect the power supply and use dry powder or carbon dioxide fire extinguishers to extinguish the fire.

If there is an open flame outside the ESS container, please immediately evacuate from the ESS system and contact the local fire emergency services for assistance.

After emergency troubleshooting, stand and observe for a specific period. If no abnormality occurs, proceed to the next step of troubleshooting.

Typical ESS Emergencies and Anomalies:

- Abnormal sound, spark, odor or other irregularities occurs during the operation or maintenance.
- Smoke or fire in the container.
- Expansion, rupture, melting, deformation of the battery pack.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

- External or internal short circuit.
- Extreme weather conditions, such as rainstorms and flash floods.
- Thermal runaway occurs during use, indicated by a battery temperature exceeding 85°C.
- Structural damage that has not yet caused a severe accident.
- The coolant of the chiller leaks.
- The auxiliary power supply is abnormal or the communication is interrupted.
- Damage to the main structure of the battery container or IP protection of the battery system.
- Loose or burnt marks on electrical connections.
- Insulation fault alarm. If triggered, stop using immediately and verify whether the alarm is false.

If there are any of the above situations occur (include but not limited to the above case), which may result in the ESS dysfunction, please contact professional maintenance team or technical support immediately of the Company.

If the situation is critical, please activate the contingency plan (refer to the maintenance manual), and notify first responders or fire services.

⚠ CAUTION

- Before troubleshooting faults, replacing spare parts, or upgrading software, ensure that the DC and AC power supplies are disconnected. Use a multimeter to check that all devices are powered off. Ensure that the devices under maintenance cannot unexpectedly power on.

9.2 Emergency Operations

Power-off Process

1. Power off via remote control

   When the system is in remote mode, the EMS can issue a shutdown command to the BAMU via communication. Upon receiving the shutdown command, the BAMU forwards it to each ESS. After receiving the shutdown instruction, the BCMU in each ESS will open the contactor and disconnect the semi-fused switch, thereby eliminating high voltage in the main circuit. The RUN indicator on the cabinet door turns off after the system is powered off.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

2. Power off via local control

   When the system is in local mode, the user can issue a stop command via the HMI. The remaining steps of the shutdown process are identical to those followed during a remote stop.

Emergency shutdown operation process

1. If the emergency stop button on the SCDC front door is pressed, the BAMU will issue an emergency stop command to each ESS. Each ESS will then turn on its contactor and semi-fused switch, cutting off the high-voltage live circuit. If circuit breakers are configured for each branch within the SCDC, pressing the emergency stop button will also simultaneously trigger these circuit breakers to open, further ensuring the requirement for rapid power-down.
2. When the emergency stop button is pressed, the BAMU will disconnect the safety interlock with the external PCS. Upon receiving this signal, the PCS will execute an emergency shutdown, cutting off the power circuit.

⚠ CAUTION

- Manual emergency stop shall not be performed in event of fire.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

10. Product Labels

The figures below are for reference only. Some are examples for illustration purposes. Refer to the actual product for details.



Figure 10-1 ES Label

Remark: A circular green reflector sign with a minimum diameter of 100 mm, featuring the letters “ES” on



Figure 10-2 Restricted Access Sign



Figure 10-3 Typical Restricted Access Sign

 Copyright © Beijing HyperStrong Technology Co., Ltd.



Figure 10-4 Explosion Hazard Label

Remark: Minimum dimensions of 175mm*175mm



Figure 10-5 Toxic Fumes Hazard Label



Figure 10-6 Shutdown Procedure Notice Label

Note:

Disconnection devices shall be labelled with an identification name or number as specified in the BESS wiring diagram, consistent with the shutdown procedure.

The minimum size of this sign shall ensure readability from a distance of 1m.

All switches shall clearly and reliably indicate the isolating position of the device. The symbols “0" (off) and “I"(on) fulfil this requirement.

The isolation switch model: MVS 25 DA1 A4F2.

 Copyright © Beijing HyperStrong Technology Co., Ltd.

11. Contact Information

Headquarter

Company Name:	Beijing HyperStrong Technology Co., Ltd.
Adress:	Building 2C, No. 9 Fenghao East Road, Haidian District, Beijing, China
Email:	support@hyperstrong.com

Australia

Company Name:	HyperStrong International (Australia) Pty Ltd
Adress:	Level 32, 367 Collins St, Melbourne, 3000，Victoria, Australia
Email:	support@hyperstrong.com

EMEA

Company Name:	HyperStrong International (Germany) GmbH
Adress:	Max-Holder-Str. 19 60437 Frankfurt am Main Germany
Email:	support@hyperstrong.com

North America

Company Name:	HyperStrong International USA Corp.
Adress:	2033 Gateway Place #500, San Jose, CA95110
Email:	Service_HSNA@hyperstrong.com

 Copyright © Beijing HyperStrong Technology Co., Ltd.

Website

http://www.hyperstrong.com

For Business Request

global@hyperstrong.com

For Media Request

marcomm@hyperstrong.com